Inequality, Markups, and the Phillips Curve

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Abstract

Micro evidence suggests that poorer households are more price elastic than richer ones. We build this fact into a two-agent New Keynesian model with a closed-form Phillips curve. Higher inequality raises desired markups by tilting demand toward the less elastic rich. Relative to a common-elasticity model with the same steady-state markup, the Phillips curve is flatter and cost-push shocks raise inflation by more. Fiscal transfers to poorer households lower desired markups and thereby shape the Phillips curve. Empirically, oil shocks raise the inverse labor share by more where inequality is higher, a weak pattern that the common-elasticity model also predicts.

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1 Introduction

How does the income distribution shape the response of inflation and output to aggregate shocks? The answer matters for monetary policy: if the distribution affects how prices respond to output, it also affects the central bank's trade-off between inflation and output. The question became more pressing after 2021, when cost shocks raised inflation (Gagliardone and Gertler, 2023) and renewed the debate on markups and cost pass-through (Conlon et al., 2023; Bilbiie and Känzig, 2024).

Most of the literature answers this question through aggregate demand alone. In heterogeneous-agent and two-agent New Keynesian models, households differ in their marginal propensity to consume (Kaplan et al., 2018; Auclert, 2019), but firms face the same price elasticity of demand from all households, so the distribution does not affect how they set prices. Micro evidence suggests that this assumption is restrictive: price sensitivity declines with income. Using Swiss household scanner data and the 2015 Swiss franc appreciation, Auer et al. (2024) estimate a price elasticity of 6.6 at an annual income of 20,000 Swiss francs and 3.0 at 120,000. The composition of demand then shapes firms' desired markups, so the distribution also matters on the supply side.

We develop and quantify this channel in a two-elasticity version of the Two-Agent New Keynesian (TANK) framework of Debortoli and Galí (2018, 2024), which we call TE-TANK. Fixing one elasticity per household type gives the Phillips curve in closed form. Its slope is the product of three terms: the slope without distributional feedback, a Ricardian wage channel present in any TANK model, and a desired-markup correction present only when elasticities differ (Proposition 2). This correction flattens the curve, and for a given demand block a flatter curve leaves more of a cost-push shock in inflation (Proposition 4).

We measure the channel against a TANK model whose single elasticity matches TE-TANK's steady-state markup, so that the two economies differ only in the desired-markup correction. When the high elasticity goes to hand-to-mouth households and the low one to Ricardian households, elasticity dispersion lowers the slope by 4.3 percent and raises cost-push inflation pass-through by 3.1 percent. The sign holds whenever profits are not fully rebated to hand-to-mouth households. If elasticities also vary with consumption within type, it holds only when the poor's elasticity does not fall too steeply with consumption relative to the rich's. Against a TANK model with the population-mean elasticity, pass-through is 15.7 percent higher. This larger gap also contains a markup-level effect, since the mean elasticity implies a lower markup, and it depends on how the shock is normalized.

The paper makes two contributions. First, with fixed elasticities by household type, we derive a distribution-dependent supply block in closed form. Second, we sign and

quantify the effect of elasticity dispersion on cost-push inflation pass-through. The empirical patterns are descriptive consistency checks on the model.

Two papers that also let elasticities differ across households are closest. Mongey and Waugh (2025) study how the composition of demand shifts markups when elasticities vary with wealth, with oligopolistic firms but without nominal rigidities, monetary policy, or supply shocks. Olivi et al. (2024) let baskets and elasticities vary across households in a multi-sector New Keynesian model with productivity shocks and study optimal monetary policy in a cost-of-living crisis. In their model the Phillips-curve slope depends on the expenditure distribution, and redistribution moves markups through an endogenous wedge. Their elasticities differ because non-CES preferences common to all households make them fall with spending. We instead fix one CES elasticity per household type in a one-sector model. The slope then factors in closed form into the slope of a common-elasticity TANK model and a desired-markup correction, and we sign the effect of elasticity dispersion on cost-push pass-through against a markup-matched TANK model. A separate literature studies the flattening of the Phillips curve (Blanchard, 2016; Hooper et al., 2020; Hazell et al., 2022) and links it to population aging (Ambrocio, 2026) and market concentration (Andrés et al., 2026). We study how the slope shapes shock transmission, not its historical decline.

Section 2 develops the model, in which demand composition moves desired markups. A positive demand shock compresses inequality, shifts demand toward the price-sensitive poor, and lowers desired markups, which damps inflation. A cost-push shock runs the mechanism in reverse: inequality rises, demand tilts toward the less elastic rich, and desired markups rise on top of the initial cost pressure. Redistribution therefore works on the supply side. Section 3.4 shows that, under the calibrated Taylor rule, moving from full profit rebating to none raises cost-push pass-through by 17.2 percent in TE-TANK but lowers it in both common-elasticity benchmarks.

The evidence that price elasticities differ by income comes from several sources. Households with more time for shopping search more and pay lower prices for the same goods (Aguiar and Hurst, 2007; Kaplan and Menzio, 2016), in line with the law of diminishing elasticity of demand of Harrod (1936). Spatial estimates are less clear: some find only small differences across income groups (Argente and Lee, 2021; Faber and Fally, 2022), while Handbury (2021) finds differences of similar size once preferences for quality are controlled for.

More broadly, desired markups that move with the income and wealth distribution are documented and modelled elsewhere:[4] retail markups rise with household income and with local income dispersion (Sangani, 2024).[5] The distribution also reaches

[4] Enders (2020) also obtains countercyclical markups from consumption-weighted elasticities, but under flexible prices, with elasticities that move as monetary injections reach segmented asset markets.

[5] The cross-sectional evidence that firms charge higher markups to less price-sensitive customers is

prices through other margins. Non-homothetic preferences shift demand and earnings across sectors (Andreolli et al., 2024). In Kaplan and Menzio (2016), buyers who search less weaken competition, but the search margin operates under flexible prices and never reaches a nominal rigidity. In Cravino et al. (2020), the goods bought by richer households have stickier prices, but realistic changes in inequality leave the aggregate price and output responses essentially unchanged. On the demand side, Bilbiie et al. (2025) likewise suggest that household heterogeneity alone leaves aggregate dynamics largely unchanged. Variable markups also flatten the Phillips curve through firm entry (Fujiwara and Matsuyama, 2026). Jaravel (2019) supplies the micro evidence for the entry margin, where demand growth in the segments bought by high-income households draws in entrants who erode incumbents' market shares and markups, though there income groups differ in the varieties they buy rather than in their elasticity of substitution. We place the composition term inside the Phillips curve.

Section 3 compares RANK, TANK, and TE-TANK under a common calibration and quantifies the demand-composition feedback shock by shock. Against the markup-matched benchmark, elasticity dispersion deepens the output contraction after a cost-push shock by the same percentage as it raises inflation pass-through. The two gaps are equal by construction: matching the markup also equates the demand blocks of the two economies, so the output contraction is the same fixed multiple of inflation in both. For demand shocks, the markup-matched benchmark shares the IS amplification generated by hand-to-mouth households, so there the damping of inflation runs entirely through the slope. Against the mean-matched benchmark, the elasticity gap also slightly lowers that amplification (Proposition 3). Technology shocks make the channel visible: they raise inequality, shift demand toward the less price-elastic rich, and dampen the fall in inflation.

The cost-push amplification also bears on the post-pandemic debate over markups and inflation. In our mechanism, the markup increase creates its own distributional consequence: higher markups temporarily raise profits, and the fiscal scheme rebates only part of the profit flow to hand-to-mouth households, so relative income shifts toward the rich and desired markups rise further. The model does not derive the distributional incidence of oil shocks. External evidence nevertheless motivates the em-

stronger than the cyclical evidence the mechanism ultimately requires. Stroebel and Vavra (2019) find that local house-price increases raise retail prices through markups rather than costs, concentrated in the high-homeownership areas where the shock makes buyers less price-sensitive. Döpper et al. (2025) estimate that a ten percent fall in a product's consumer price sensitivity accompanies a 7.3 percent higher markup within brand, category, market and retailer. Faber and Fally (2022) show that rich and poor households buy from different parts of the firm-size distribution within the same product group. What remains open is the business-cycle frequency: this evidence is regional, cross-sectional, or secular, and none of it identifies the aggregate demand-composition channel. Near-uniform pricing across the stores of a chain (DellaVigna and Gentzkow, 2019) also weakens local tests of the mechanism without disabling the aggregate channel.

pirical application: Broer et al. (2025) use 45 years of German administrative data and the oil-supply news shock of Känzig (2021) to show that a 10 percent oil-price increase lowers the earnings growth of low-income workers by about two percentage points after two years, while top incomes are largely unaffected. Their mechanism is different from ours, but the income shift they document is directionally consistent with the redistribution that raises desired markups in our model.

Finally, Section 4 reports descriptive patterns in US quarterly data and a cross-country panel. Countries with higher long-run income concentration have higher average markup levels, the cross-country counterpart of the steady-state relation between inequality and the desired markup. Markups rise after the expansionary technology shock and the contractionary oil supply-news shock of Känzig (2021), and consumption inequality rises after the technology shock. Markups also rise after an expansionary monetary shock, contrary to the shared sticky-price core (Nekarda and Ramey, 2020). These signs assess that core, not the elasticity-gap channel.

2 A Two-Agent New Keynesian Model with Heterogeneous Elasticities

The model extends the two-agent New Keynesian (TANK) framework of Debortoli and Galí (2018, 2024) with one new assumption: hand-to-mouth households have a higher price elasticity of demand than households with asset market access, in line with the micro evidence that price sensitivity declines with income (Berry et al., 2004; Aguiar and Hurst, 2007; Auer et al., 2024). A fraction λ of households are hand-to-mouth (poor, denoted P); the remaining $1-\lambda$ have full asset market access (rich, denoted R); and $\varepsilon_P > \varepsilon_R$. This single change makes firms' desired markups depend on the composition of demand: when spending shifts toward rich households, firms face a less elastic customer base and raise desired markups.

The mechanism operates through the division of income between wages and profits. When the output gap is positive, markups and profit income fall relative to labor income, demand shifts toward the more price-sensitive poor households, and desired markups fall further. We call this loop from the output gap to the distribution and back into price setting the demand-composition feedback. Its strength depends on fiscal policy: δ determines how much profit income the government collects and τ how it is redistributed across the two types. Redistribution therefore has supply-side effects, because it changes the elasticity of demand firms face. With a common elasticity ε, by contrast, the desired markup is the constant $\varepsilon/(\varepsilon-1)$ whatever the income distribution, so standard TANK cannot connect inequality to markups.

Maintained assumptions. Throughout, $\sigma > 0$, $\varphi \geq 0$, $\alpha \in [0,1)$, $\zeta > 0$, $\lambda \in (0,1)$, $\varepsilon_P \geq \varepsilon_R > 1$, and $(\delta, \tau) \in [0,1]^2$, so that $R \equiv \delta(1-\tau) \in [0,1]$. We also assume that, at every steady state considered, the symmetric price is each firm's global profit maximizer when all other firms charge it. With an elasticity gap this is not automatic, because a higher price loses mostly the elastic poor customers. Appendix D.1 states which restrictions bind, gives a sufficient condition in primitives, and verifies global optimality at every calibration reported in the paper.

2.1 Households

Households have unit mass, a fraction λ hand-to-mouth (poor) and $1-\lambda$ Ricardian (rich), and both types share the same preferences over consumption and leisure,

$$U_t^h = \left(\frac{(C_t^h)^{1-\sigma}}{1-\sigma} - \frac{(N_t^h)^{1+\varphi}}{1+\varphi} \right) Z_t, \tag{1}$$

where C_t^h is consumption, N_t^h is labor supply, σ is the inverse intertemporal elasticity of substitution, φ is the inverse Frisch elasticity, and Z_t is a preference shock with $\log Z_t = \rho_z \log Z_{t-1} + \varepsilon_t^z$. Each type aggregates consumption over varieties with its own CES elasticity of substitution ε_h, which implies the type-specific demand schedule

$$C_t^h(i) = \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} C_t^h, \tag{2}$$

where P_t^h is the price index of type h. Appendix D.3 derives this demand function. We set $\varepsilon_P = 6.6$ and $\varepsilon_R = 3.0$, the baseline quantification of Auer et al. (2024) that Section 3 discusses.

For rich households, utility maximization yields the familiar Euler equation, labor-supply condition, and equity-pricing equation, which Appendix D.2 derives. Poor households cannot save or borrow and consume all labor income plus transfers each period. Hours are common across types, $N_t^P = N_t^R \equiv N_t$, at a wage that satisfies the Ricardian intratemporal condition.

2.2 Firms, Demand Composition, and Markups

A continuum of monopolistically competitive firms produces differentiated goods using the technology

$$Y_t(i) = A_t N_t(i)^{1-\alpha}, \tag{3}$$

where A_t follows $\log A_t = \rho_a \log A_{t-1} + \varepsilon_t^a$ and $\alpha \in (0,1)$ governs decreasing returns to labor. Firms face Rotemberg quadratic price adjustment costs: changing the price from $P_{t-1}(i)$ to $P_t(i)$ costs $\frac{\zeta}{2}\left(\frac{P_t(i)}{P_{t-1}(i)} - 1\right)^2 Y_t$. These costs are paid in the same composites of varieties that households consume.

Aggregate demand for variety i reflects the two household types:

$$Y_t(i) = \sum_h s_h \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h, \tag{4}$$

where $s_h \in \{1-\lambda, \lambda\}$ is the population share of type h. Define the rich household's demand relative to the average as $H_t \equiv Y_t^R / Y_t = C_t^R / C_t$, so that rich households account for the expenditure share $(1-\lambda)H_t$ and, by market clearing, $H_t^P = [1-(1-\lambda)H_t]/\lambda$. The effective (expenditure-weighted) elasticity faced by a firm in a symmetric allocation is then

$$\varepsilon_{\text{eff},t} = (1-\lambda)H_t\varepsilon_R + \lambda H_t^P \varepsilon_P = H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P.$$

This object carries demand composition into price setting: a higher H_t shifts demand toward the less price-sensitive rich, so the elasticity firms face falls. The firm chooses $P_t(i)$ to maximize the present value of profits subject to (4). Appendix D.5 states the full problem. In a symmetric equilibrium all firms choose the same price, and because firms do not price discriminate across household types, both types face the same price level, $P_t^R = P_t^P = P_t$.

We treat ε_R and ε_P as fixed within type and calibrate their between-type gap using Auer et al. (2024). The fixed gap stands in for a demand system in which the elasticity is derived from preferences, such as Mongey and Waugh (2025), where elasticities vary endogenously with wealth; it keeps $\varepsilon_{\text{eff},t}$ moving only with the composition of demand H_t.[6]

Optimal markup. Under flexible prices ($\zeta = 0$), the first-order condition yields the desired markup

$$\mathcal{M}_t = \frac{\varepsilon_{\text{eff},t}}{\varepsilon_{\text{eff},t} - 1} = \frac{H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P}{H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1}. \tag{5}$$

[6] If elasticities also vary with consumption within type, the channel scales by $1-\gamma$, where γ is the cross-type difference in elasticity-consumption gradients relative to the between-type gap. Against a markup-matched benchmark with the same average within-type gradient, the channel keeps its sign for $\gamma < 1$ and reverses for $\gamma > 1$. Appendix D.13 derives the scaling and reports the pass-through gap under assumed values of γ, which the evidence does not identify.

When rich households consume more relative to the average (higher H_t) and are less price-sensitive ($\varepsilon_R < \varepsilon_P$), firms optimally charge higher markups. Appendix D.5.1 gives the derivation.

2.3 Fiscal and Monetary Policy

Fiscal policy. The government collects a fraction δ of firm profits and redistributes the revenue across household types according to the transfer-tilt parameter $\tau \in [0, 1]$, with $\tau = 0$ the most progressive scheme, equal per-capita transfers. The government budget balances each period, and Appendix D.2 gives the details. Only one combination of the two fiscal parameters matters for equilibrium outcomes.

Lemma 1 (Fiscal Sufficient Statistic). *Fix the non-fiscal parameters $(\beta, \alpha, \varphi, \sigma, \lambda, \zeta, \varepsilon_R, \varepsilon_P)$ and the monetary-policy rule (ϕ_π, ϕ_y). After imposing symmetric equilibrium and eliminating transfers and equity values from the household budget constraints, the redistribution share δ and the transfer tilt τ enter every steady-state allocation object and every coefficient of the first-order equilibrium system for $(\tilde{y}_t, \pi_t, \hat{h}_t, \hat{\imath}_t)$ only through the scalar*

$$R \equiv \delta(1 - \tau), \qquad 1 + \delta(\tau - 1) = 1 - R. \tag{6}$$

In particular, $\left(\bar{H}, \bar{\mathcal{M}}, \delta_2, \kappa_y, \kappa_h, \Gamma, \chi_y, \kappa\right)$, which Sections 2.4 and 2.5 define, are all functions of R alone given the non-fiscal parameters. Hence fiscal pairs with the same R generate identical steady-state allocations and first-order dynamics.

Appendix D.10.1 gives the proof.

The parameter R measures how much profit income is rebated according to population shares. At $R = 0$, poor households receive none; at $R = 1$, profits are fully collected and returned as equal lump-sum transfers, which reproduces the RANK allocation. The baseline calibration $(\delta, \tau) = (1, 0.7)$ gives $R = 0.30$.

Monetary policy. The central bank follows a Taylor rule:

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \hat{y}_t + v_t, \tag{7}$$

where ρ is the steady-state real rate, ϕ_π and ϕ_y are the policy response coefficients, and v_t is a monetary policy shock. We impose the Taylor principle, $\phi_\pi > 1$, a non-negative output response, $\phi_y \geq 0$, and a positive effective IS elasticity in (11) below, $\chi_y < 1$. A sufficient condition in primitives is $\lambda(1 - R)(\varphi + \alpha) \leq 1 - \lambda$, which holds on the whole fiscal branch at the baseline calibration. Under these three conditions the equilibrium is determinate; the demand-composition feedback cannot overturn determinacy, because the slope stays strictly positive (Proposition 2). Appendix D.10.5 gives the argument.

2.4 Steady State

The steady state has a recursive structure. Define the composite parameters

$$\mathsf{A} = (1-\lambda)(\varepsilon_R - \varepsilon_P), \qquad \mathsf{B} = \varepsilon_P, \qquad \mathsf{C} = \frac{1-\lambda R}{1-\lambda}, \qquad \mathsf{D} = \frac{(1-\alpha)\lambda(1-R)}{1-\lambda}.$$

The heterogeneity index and the flex-price markup are jointly determined by

$$\bar{H} = \mathsf{C} - \frac{\mathsf{D}}{\bar{\mathcal{M}}}, \tag{8}$$

$$\bar{\mathcal{M}} = \frac{\mathsf{A}\bar{H} + \mathsf{B}}{\mathsf{A}\bar{H} + \mathsf{B} - 1}. \tag{9}$$

The first equation follows from the rich household's budget constraint: C captures the concentration of profit-related income among rich households, and $\mathsf{D}/\bar{\mathcal{M}}$ the equalizing effect of common labor income. The second is (5) in steady state. The elasticity gap enters only through A, and fiscal policy only through R, via C and D.

The fixed point is a feedback between redistribution, inequality, and market power. A decline in R shifts profit income toward rich households and raises $\bar{H}$. Because rich households are less price-sensitive, the effective demand elasticity falls and the desired markup rises. The higher markup shifts income from wages toward profits and raises inequality further. Substituting (9) into (8) reduces the fixed point to the quadratic (D.22) in $\bar{H}$, and output then follows recursively; Appendices D.4 and D.6 give the algebra.

Proposition 1 (Existence, Uniqueness, and Fiscal Comparative Statics)**.**

(i) ***Existence and uniqueness.*** *For every fiscal pair $(\delta, \tau) \in [0,1]^2$, the quadratic (D.22) has exactly one root consistent with strictly positive consumption for both household types. Under the maintained pricing assumption, for which (D.1) is sufficient, the symmetric price supporting this root is each firm's global profit maximizer, so the root is the unique symmetric steady state. If $R < 1$, this root satisfies $\bar{H} \in (1, \frac{1}{1-\lambda})$ and $\bar{\mathcal{M}} > 1$; if $R = 1$, it satisfies $\bar{H} = 1$.*

(ii) ***Fiscal comparative statics.*** *Along this branch, $\bar{H}$ is strictly decreasing in R. Because $\varepsilon_P > \varepsilon_R$, the steady-state markup is strictly increasing in $\bar{H}$ and therefore strictly decreasing in R. A higher transfer share to poor households thus lowers both steady-state inequality and the desired markup.*

Appendix D.10.2 gives the proof.

The model nests both benchmark economies. When $\varepsilon_R = \varepsilon_P = \varepsilon$, it reduces to standard TANK with fixed desired markup $\varepsilon/(\varepsilon - 1)$; that economy can retain consumption inequality when $R < 1$, but its markup does not respond to it. At $R = 1$, $\bar{H} = 1$,

and TE-TANK coincides in steady state and in all first-order dynamics with RANK at the population-weighted elasticity $\bar{\varepsilon} = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$. Appendix D.10.2 proves this equivalence. Elasticity dispersion therefore has aggregate effects only when consumption is unequal.

2.5 Log-Linear Equilibrium

Lowercase letters denote logarithms, $\hat{x}_t$ denotes the percentage deviation of x_t from its steady-state value, and $\tilde{y}_t \equiv \log(Y_t/Y_t^n)$ is the output gap relative to flexible-price output Y_t^n. The log-linear equilibrium consists of a distributional relation, a dynamic IS curve, a New Keynesian Phillips curve, and the Taylor rule (7). The distributional relation ties the heterogeneity index to the output gap period by period, so substituting it out returns the familiar three-equation system.

Distribution and aggregate demand. Log-linearizing the rich household's budget constraint, normalized by aggregate output, gives

$$\hat{h}_t = -\chi_y\,\tilde{y}_t, \qquad \chi_y = \frac{\Gamma\left(\dfrac{\varphi+\alpha}{1-\alpha}+\sigma\right)}{1+\Gamma\sigma}, \qquad \Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}}. \tag{10}$$

Appendix D.4 gives the derivation. When the output gap is positive, labor income rises relative to profit income and inequality falls. The composite Γ scales the demand-composition feedback and vanishes in the RANK limit $R = 1$; χ_y turns it into the loading of inequality on the output gap. At the baseline calibration, $\Gamma = 0.188$ and $\chi_y = 0.4746$: a one percent positive output gap lowers the heterogeneity index by about half a percent.

Since $\operatorname{sign}(\chi_y) = \operatorname{sign}(1-R)$, inequality moves opposite to the output gap for any disturbance whenever $R < 1$. Demand shocks move output and the gap in the same direction, so inequality falls in a demand-driven boom; a cost-push shock lowers both, so inequality rises as output falls. A positive technology shock raises potential output, and under the calibrated rule actual output rises by less, so the gap turns negative and inequality rises with output.

Dynamic IS curve. The log-linearized Euler equation of rich households, combined with market clearing and the hand-to-mouth constraint, gives

$$\tilde{y}_t = \mathbb{E}_t[\tilde{y}_{t+1}] - \frac{1}{\sigma(1-\chi_y)}\left(\hat{i}_t - \mathbb{E}_t[\pi_{t+1}] - r_t^n\right), \tag{11}$$

where $r_t^n = (1-\rho_z)z_t + \sigma \mathbb{E}_t[\Delta \hat{y}_{t+1}^n]$ is the natural real rate. As in standard TANK (Campbell and Mankiw, 1989; Bilbiie, 2008), hand-to-mouth households create a New Keynesian cross (Bilbiie, 2025): a lower real rate raises Ricardian consumption and labor demand, hand-to-mouth households spend the extra wage income, and the IS semi-elasticity rises from $1/\sigma$ to $1/[\sigma(1-\chi_y)]$. In the RANK limit, $\chi_y = 0$ and (11) is the textbook IS curve. Appendix D.7 gives the derivation.

New Keynesian Phillips curve. Appendix D.8 log-linearizes the firm's first-order condition around the zero-inflation steady state, which yields

$$\pi_t = \beta \, \mathbb{E}_t[\pi_{t+1}] + \kappa_h \, \hat{h}_t + \kappa_y \, \tilde{y}_t + u_t, \tag{12}$$

with

$$\begin{aligned}
\kappa_y &= \delta_2 G, \qquad G \equiv \frac{\varphi + \alpha}{1-\alpha} + \sigma, \\
\kappa_h &= \frac{(1-\lambda)\bar{H}}{\zeta}\left(1 - \overline{MC}\right)(\varepsilon_P - \varepsilon_R) + \delta_2 \sigma, \\
\delta_2 &= \frac{1}{\zeta}[\bar{H}(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1] = \frac{\varepsilon_{\text{eff}}(\bar{H}) - 1}{\zeta},
\end{aligned}$$

where $\overline{MC} = 1/\bar{\mathcal{M}}$ is steady-state real marginal cost, δ_2 is the marginal-cost loading, and u_t is an exogenous cost-push disturbance, which Section 2.6 specifies. The coefficient κ_y is the output-gap loading with the demand-composition feedback switched off: the conventional marginal-cost block, evaluated at the TE-TANK steady state. The coefficient κ_h carries two channels.

The first is a Ricardian wage channel, $\delta_2 \sigma$, present even when both types share one elasticity. Hand-to-mouth households are rationed at the common hours, so the real wage equals the rich household's marginal rate of substitution, $(C_t^R)^\sigma N_t^\varphi$, and real marginal cost is $\widehat{mc}_t = \sigma \hat{h}_t + G\tilde{y}_t$. The term $G\tilde{y}_t$ is the usual RANK loading; $\sigma \hat{h}_t$ is the TANK piece. A boom compresses inequality, so rich consumption and the wage rise by less, and inflation responds less to output.[7]

The second is a desired-markup channel, present only when $\varepsilon_P \neq \varepsilon_R$. A boom tilts spending toward the more price-elastic poor, so $\varepsilon_{\text{eff},t}$ rises and the desired markup falls. The first channel changes what firms pay, through who sets the wage; the second changes what firms charge, through who buys the goods. Since $1 - \overline{MC} = 1/\varepsilon_{\text{eff}}(\bar{H})$,

[7] The same wage-setting object appears in a worker–capitalist economy with flexible wages, where marginal cost tracks workers' consumption rather than aggregate output (Broer et al., 2020). Appendix D.8 derives the channel.

the second channel is itself an elasticity:

$$\eta_\mu \equiv \left.\frac{\partial \log \mathcal{M}_t}{\partial \log H_t}\right|_{\bar{H}} = \frac{(1-\lambda)\bar{H}(\varepsilon_P - \varepsilon_R)}{\varepsilon_{\text{eff}}(\bar{H})\left[\varepsilon_{\text{eff}}(\bar{H}) - 1\right]} \geq 0, \qquad \kappa_h = \delta_2(\sigma + \eta_\mu), \tag{13}$$

where η_μ is the steady-state elasticity of the desired markup with respect to the heterogeneity index, strictly positive if and only if $\varepsilon_P > \varepsilon_R$.

Reduced-form slope. Substituting the distributional relation (10) gives the reduced-form NKPC:

$$\pi_t = \beta\, \mathbb{E}_t[\pi_{t+1}] + \underbrace{(\kappa_y - \kappa_h \chi_y)}_{\kappa}\, \tilde{y}_t + u_t. \tag{14}$$

A positive output gap compresses inequality, which lowers the wage that clears the rich household's labor supply and, when $\varepsilon_P \neq \varepsilon_R$, the desired markup, and so puts downward pressure on inflation. Substituting $\chi_y = \Gamma G/(1+\Gamma\sigma)$ and $\kappa_h = \delta_2(\sigma + \eta_\mu)$ cancels the terms in $\Gamma\sigma$, and the slope factors exactly:

$$\kappa = \kappa_y - \kappa_h \chi_y = \underbrace{\delta_2 G}_{\substack{\text{feedback-off} \\ \text{slope } \kappa_y}} \cdot \underbrace{\frac{1}{1+\Gamma\sigma}}_{\substack{\text{Ricardian wage channel,} \\ \text{present in any TANK}}} \cdot \underbrace{(1-\Gamma\eta_\mu)}_{\substack{\text{desired-markup correction,} \\ \text{absent under a common } \varepsilon}} \tag{15}$$

Together the two factors are the demand-composition feedback; setting $\hat{h}_t \equiv 0$ shuts both off and returns the slope to κ_y. The wage factor is a supply-side object and differs from the New Keynesian cross $1/[\sigma(1-\chi_y)]$ in (11).

Proposition 2 (Endogenous Flattening of the Phillips Curve)**.** *Under the maintained assumptions, the reduced-form slope κ in (14) satisfies the following.*

(i) ***Factorization.*** *The slope factors as in (15), into the feedback-off slope $\kappa_y = \delta_2 G > 0$, the Ricardian wage channel $1/(1+\Gamma\sigma)$, and the desired-markup correction $1-\Gamma\eta_\mu$.*

(ii) ***Positive slope.*** $0 < 1 - \Gamma\eta_\mu \leq 1$, *hence*

$$\kappa \in (0, \kappa_y].$$

The reduced-form slope is strictly positive everywhere on the fiscal square $(\delta, \tau) \in [0,1]^2$.

(iii) ***Flattening.*** $\kappa_h > 0$ *and*

$$\kappa < \kappa_y \iff \chi_y > 0 \iff R < 1.$$

Given $R < 1$, the desired-markup factor satisfies $1 - \Gamma\eta_\mu < 1$ if and only if $\varepsilon_P > \varepsilon_R$.

The demand-composition feedback therefore flattens the Phillips curve relative to the same steady state with $\hat{h}_t \equiv 0$, without ever reversing its sign.

Both factors are at most one, so each works toward the flattening. This comparison holds the steady state fixed and removes only the demand-composition feedback; it is not a comparison with a homogeneous-elasticity economy. Appendix D.10.3 gives the proof.

2.6 Benchmark Comparisons and Cost-Push Transmission

Three benchmarks isolate different parts of the mechanism. The *feedback-off* benchmark is TE-TANK itself with $\hat{h}_t \equiv 0$: it keeps $(\bar{H}, \bar{\mathcal{M}}, \delta_2)$ and has slope κ_y. The *mean-matched* TANK replaces the two elasticities by their population mean $\bar{\varepsilon}$. Its steady-state markup is lower, so a comparison with it mixes dispersion with a markup-level effect. The *markup-matched* TANK(ε^*) uses the single elasticity whose markup $\varepsilon^*/(\varepsilon^* - 1)$ equals $\bar{\mathcal{M}}$. It then also shares $\bar{H}$, δ_2, Γ, and χ_y, has $\eta_\mu = 0$, and differs from TE-TANK only in κ_h. By (15),

$$\kappa^{\text{TE}} = \kappa^{\text{TANK}^*}\left(1 - \Gamma\eta_\mu\right),$$

so the factor $1 - \Gamma\eta_\mu$ is the entire dispersion-pure difference in slopes.

The two homogeneous benchmarks give different magnitudes for the same mechanism. At the baseline calibration, cost-push inflation pass-through in TE-TANK is 3.1 percent higher than in markup-matched TANK($\varepsilon^* = 3.93$) and 15.7 percent higher than in mean-matched TANK($\bar{\varepsilon} = 4.44$). The larger number adds a markup-level effect. At a given mean elasticity, the rich account for more than their population share of spending, so dispersion lowers the effective elasticity and raises the steady-state markup; the higher markup flattens the curve through δ_2, and this step alone raises pass-through by 12.2 percent. The markup-level step depends on how the cost-push shock is normalized. Appendix C.3 considers alternative normalizations, under which the mean-matched gap becomes -1.3 percent. The markup-matched gap is invariant to the normalization, because the two economies share δ_2, and its sign follows from Propositions 2 and 4. We therefore quote it as the headline magnitude.

IS amplification. Against the mean-matched benchmark, dispersion also changes the IS amplification $1/(1 - \chi_y)$ through the steady state.

Proposition 3 (Elasticity Dispersion and IS Amplification)**.** *Fix the population-weighted mean elasticity $\bar{\varepsilon} \equiv (1 - \lambda)\varepsilon_R + \lambda\varepsilon_P$ and fiscal parameters with $R < 1$ such that $\chi_y^{\text{TANK}} < 1$. Relative to homogeneous-elasticity TANK with $\varepsilon = \bar{\varepsilon}$, TE-TANK has*

$$\bar{H}^{\text{TE}} > \bar{H}^{\text{TANK}}, \qquad \bar{\mathcal{M}}^{\text{TE}} > \bar{\mathcal{M}}^{\text{TANK}}.$$

Consequently,

$$\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}, \qquad 1 < \frac{1}{1-\chi_y^{\text{TE}}} < \frac{1}{1-\chi_y^{\text{TANK}}}.$$

Thus TE-TANK's IS curve is more rate-sensitive than RANK's but less rate-sensitive than that of mean-matched TANK.

The ordering is a markup-level effect: dispersion raises $\bar{\mathcal{M}}\bar{H}$ and so lowers χ_y. Markup-matched TANK(ε^*) shares $\bar{\mathcal{M}}\bar{H}$ and hence χ_y, so against it the elasticity gap acts only through the slope of the Phillips curve. Appendix D.10.4 gives the proof.

Cost-push transmission. The disturbance u_t in (12) is an AR(1), $u_t = \rho_u u_{t-1} + \varepsilon_t^u$: a reduced-form wedge that shifts marginal cost independently of the output gap.[8] For the cost-push results below, maintain $\beta \in (0,1)$, $\sigma > 0$, $\kappa > 0$, $\phi_\pi > 1$, $\phi_y \geq 0$, $\rho_u \in [0,1)$, and $0 \leq \chi_y < 1$, in both economies when two are compared. Then $\tilde{\sigma} \equiv \sigma(1-\chi_y) > 0$, the reciprocal of the IS semi-elasticity in (11).

Proposition 4 (Cost-Push Pass-Through)**.** *Consider the cost-push wedge u_t in isolation, so $z_t = a_t = v_t = 0$, $r_t^n = 0$, and $\hat{y}_t = \tilde{y}_t$. Under the maintained cost-push conditions in the preceding paragraph, the unique bounded solution is $\pi_t = \Psi_\pi u_t$ and $\tilde{y}_t = \Psi_y u_t$, where*

$$\Psi_\pi = \frac{1}{D_\pi} > 0, \qquad \Psi_y = -\Theta\Psi_\pi < 0,$$

with

$$D_\pi = (1-\beta\rho_u) + \kappa\Theta > 0, \qquad \Theta = \frac{\phi_\pi - \rho_u}{\tilde{\sigma}(1-\rho_u) + \phi_y} > 0.$$

Pass-through satisfies

$$\frac{\partial\Psi_\pi}{\partial\kappa} = -\Theta\Psi_\pi^2 < 0, \qquad \frac{\partial\Psi_\pi}{\partial\tilde{\sigma}} = \frac{(1-\rho_u)\kappa(\phi_\pi-\rho_u)}{\left[\tilde{\sigma}(1-\rho_u)+\phi_y\right]^2}\Psi_\pi^2 > 0.$$

Holding all non-elasticity primitives, policy coefficients, and the wedge process fixed, TE-TANK has greater pass-through than homogeneous-elasticity TANK if and only if

$$\frac{\Psi_\pi^{\text{TE}}}{\Psi_\pi^{\text{TANK}}} > 1 \iff \kappa^{\text{TE}}\Theta^{\text{TE}} < \kappa^{\text{TANK}}\Theta^{\text{TANK}}.$$

Appendix D.10.5 gives the proof.

A sufficient condition for greater pass-through is $\kappa^{\text{TE}} \leq \kappa^{\text{TANK}}$ and $\chi_y^{\text{TE}} \leq \chi_y^{\text{TANK}}$, with at least one inequality strict. The condition holds structurally against the *markup-matched* benchmark whenever $R < 1$: the two economies share χ_y, and $\kappa^{\text{TE}} =$

[8] The calibration sets $\rho_u = 0.9$, $\sigma_u = 0.05$; the paper's headline results are TE-TANK/TANK ratios of impact responses to the same unit shock, which are invariant to σ_u.

$\kappa^{\text{TANK}^*}(1-\Gamma\eta_\mu) < \kappa^{\text{TANK}^*}$. Against the *mean-matched* benchmark, Proposition 3 delivers $\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}$, and Section 3 verifies the slope ranking at the baseline calibration. Since $\Psi_y = -\Theta\Psi_\pi$, a positive wedge also generates stagflation, and the output contraction is larger in TE-TANK whenever the pass-through amplification dominates any offsetting difference in Θ.

The economics is the demand-composition feedback in reverse. A cost-push recession opens a negative output gap and raises inequality, so demand shifts toward the less price-sensitive rich and firms raise desired markups on top of the initial cost pressure. A flatter Phillips curve means that a given output contraction buys less disinflation, so more of the wedge stays in prices. When inflation is driven by the output gap instead, as under demand and technology shocks, a flatter curve shrinks the inflation response.

A cross-sectional prediction. Across steady states, indexed by inequality $\bar{H}$, which rises as the profit rebate R falls, the impact markup response to the cost-push wedge, $\partial\hat{\mu}_t/\partial u_t$, is strictly increasing in $\bar{H}$ under a policy condition that the calibrated Taylor rule satisfies. Appendix D.11 proves this result as Corollary 5(iii). Facing the same shock, a more unequal economy raises markups by more. Section 4.2 takes this prediction to the data. Its sign is shared with homogeneous-elasticity TANK, so the elasticity gap contributes magnitude, not direction.

Divine coincidence. Absent the wedge ($u_t = 0$), setting $\pi_t = \tilde{y}_t = 0$ satisfies (14) at all times. Demand and technology shocks leave the flexible-price H_t, and hence the flexible-price desired markup, at its steady-state value, so heterogeneous elasticities create no additional cost-push wedge; they change only the pass-through of u_t, through κ.[9]

3 Quantitative Analysis

3.1 Calibration

Table 1 reports the quarterly calibration, and Appendices C.1 and C.2 give the sources. Its upper panel lists the parameters common to the three economies and its lower panel the model-specific ones. In the TE-TANK column, the demand elasticities follow the rich–poor gradient that Auer et al. (2024) estimate from Swiss household scanner data; their level is a normalization, so the gradient, not the level, disciplines the mechanism.

[9] The divine coincidence is a flexible-wage property. Under nominal wage rigidity it breaks for disturbances that move the flexible-wage marginal rate of substitution, in this model's shock set only the technology shock.

The TANK and RANK columns use the markup-matched $\varepsilon^* = 3.93$, which reproduces the TE-TANK markup $\bar{\mathcal{M}} = 1.34$. The model has less consumption dispersion than the CEX rich-to-poor ratio, a held-out diagnostic; since CEX groups are ranked by consumption, not asset access, this does not show that the model understates the channel. Table C.5 shows that, with the estimated elasticity difference held fixed, the dispersion-pure pass-through gap stays below 7.0 percent over all admissible (λ, R).

TABLE 1: Baseline Calibration

Description	Symbol	TE-TANK	TANK	RANK
Common parameters				
Discount factor	β		0.995	
Risk aversion	σ		1.00	
Inverse Frisch elasticity	φ		1.00	
Capital share	α		1/3	
Price adjustment cost	ζ		105.63	
Model-specific parameters				
Share of hand-to-mouth	λ	0.40	0.40	0.00
Share of dividends	δ	1.00	1.00	1.00
Transfer tilt toward rich	τ	0.70	0.70	0.00
Rich elasticity	ε_R	3.00	3.93	3.93
Poor elasticity	ε_P	6.60	3.93	3.93

Notes: Common parameters are shared across all three economies. TANK and RANK use the markup-matched elasticity $\varepsilon^* = 3.93$, at which $\varepsilon/(\varepsilon - 1)$ equals TE-TANK's steady-state markup $\bar{\mathcal{M}} = 1.34$. RANK is the limit $\lambda = 0$; by Lemma 1 and Proposition 1, the parameterization with $\lambda = 0.40$ and $R = 1$ is the same economy. Appendix C.1 reports the full parameterization, including the Taylor rule and shock persistence.

3.2 The Flattening of the Phillips-Curve Slope

Table 2 shows that elasticity dispersion flattens the Phillips curve by 4.34 percent at the baseline calibration, holding the steady state fixed. Its rows apply the factors of (15) in turn. Row 1 is the *feedback-off* slope $\kappa_y = 0.0833$, which is also the slope of RANK(ε^*). In row 2, the Ricardian wage channel, common to any TANK, lowers it by 15.82 percent and lands exactly on *markup-matched* TANK(ε^*), with slope 0.0701. In row 3, the desired-markup correction $1 - \Gamma\eta_\mu$ lowers the slope by a further 4.34 percent, to 0.0671 in TE-TANK. In total the demand-composition feedback lowers the

slope by 19.48 percent, of which the wage channel accounts for 15.82 percentage points and the desired-markup channel for 3.66.

TABLE 2: The Phillips-Curve Slope, Factor by Factor

	Economy	κ	vs. κ_y
Feedback off, $\kappa_y = \delta_2 G$	—	0.0833	—
+ Ricardian wage channel	TANK(ε^*)	0.0701	−15.82%
+ desired-markup route	TE-TANK	0.0671	−19.48%

Notes: The two factors of Proposition 2(i), applied in turn at the baseline calibration of Table 1. Row 1 is the feedback-off slope $\kappa_y = \delta_2 G$, which is also the slope of RANK(ε^*). Row 2 multiplies by the Ricardian wage channel $1/(1+\Gamma\sigma) = 1/1.188$. Row 3 multiplies by the desired-markup correction $1 - \Gamma\eta_\mu$. The final column is the cumulative reduction relative to κ_y. Only the step from row 2 to row 3 requires $\varepsilon_P \neq \varepsilon_R$.

The last step is the paper's headline object. It is the only step that vanishes under a common elasticity, and by Proposition 4 it turns a 4.3 percent flatter Phillips curve into 3.1 percent higher cost-push inflation pass-through. Table C.6 shows that TE-TANK also has a flatter slope than mean-matched TANK($\bar{\varepsilon}$), and by Proposition 3 it has a smaller χ_y, so it satisfies the sufficient condition that Section 2.6 derives. The pass-through gap is then 15.7 percent, which also contains the markup-level effect.

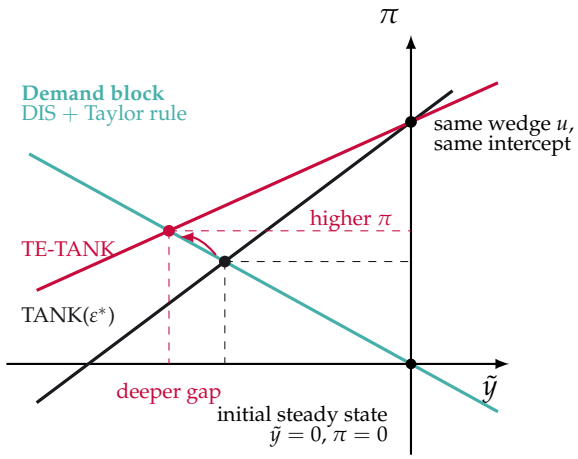
3.3 Impulse Response Functions

We solve the log-linearized model by first-order perturbation in Dynare. Figure D.2 in Appendix D.12 shows that a positive technology shock raises inequality and desired markups, so the elasticity gap shrinks the fall in inflation; Figure D.3 shows that after an expansionary monetary policy shock the gap damps inflation instead.

Cost-push shock. Heterogeneous elasticities raise inflation and deepen the output loss after a cost-push shock. On impact, the wedge raises annualized TE-TANK inflation by 0.60 percentage points and lowers output by 0.51 percent. Relative to TANK(ε^*), inflation pass-through is 3.1 percent higher and the output contraction 3.1 percent deeper: Ψ_π rises from 2.928 to 3.019 and $|\Psi_y|$ from 9.896 to 10.204. The two gaps coincide because the two economies share $\bar{H}$, $\bar{\mathcal{M}}$, and χ_y, so the entire first-order difference runs through the slope of the Phillips curve.

Figure 1 shows why the flatter curve leaves more of the wedge in inflation. The horizontal axis is the output gap and the vertical axis inflation, with the steady state at the origin. The teal line is the common demand block. The same wedge shifts the red TE-TANK and the black TANK(ε^*) Phillips curves to a common intercept, and the flatter red curve meets the demand line at higher inflation and a deeper output gap.

FIGURE 1: Why a Flatter Phillips Curve Raises the Inflationary Cost of a Cost-Push Shock

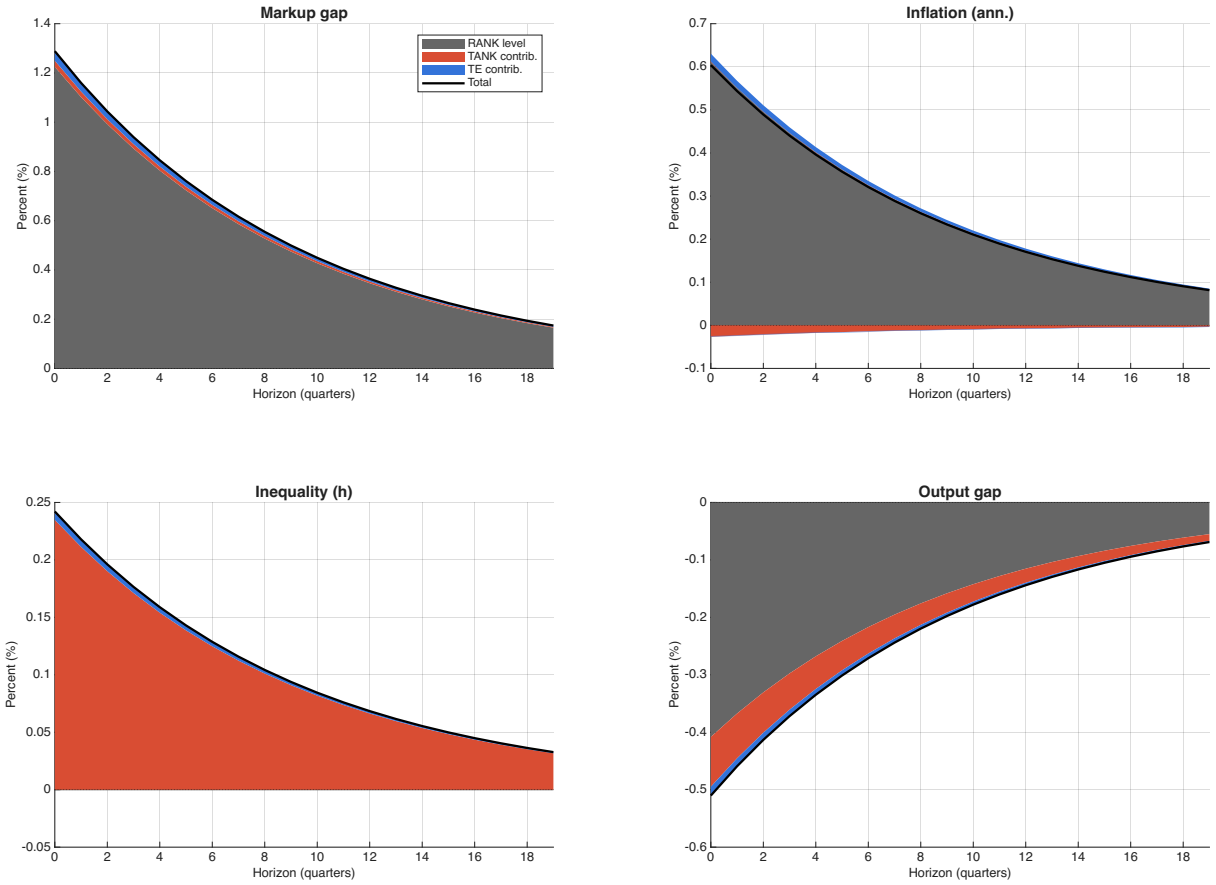


Notes: Stylized, with the slope difference exaggerated for legibility. The origin is the initial steady state; each economy slides back down the demand locus to the origin as the AR(1) wedge decays. The downward-sloping locus is the demand block, (11) and (7) solved forward, with slope $-1/\Theta$; it is common to the two economies because matching the markup gives them the same χ_y. The upward-sloping curves are (14) solved forward, $\pi = [\kappa/(1-\beta\rho_u)]\tilde{y} + u/(1-\beta\rho_u)$, so only κ differs. Calibrated, $\kappa^{\text{TE}} = 0.0671$ against $\kappa^{\text{TANK}^*} = 0.0701$ delivers $\Psi_\pi = 3.019$ against 2.928. RANK has $\tilde{\sigma} = 1$ and so does not share the locus.

Figure 2 shows that the elasticity gap raises the markup gap and inflation and deepens the fall in the output gap. The top panels plot the markup gap and annualized inflation, the bottom panels inequality $\hat{h}_t$ and the output gap, each in percent against quarters after the shock. The black line is TE-TANK, built from three stacked bands: gray is the RANK response, red adds hand-to-mouth aggregation, and blue adds the elasticity gap. The blue band is thin, in line with the 3.1 percent gaps above. It reflects the desired-markup channel in reverse: inequality rises with the negative output gap, demand tilts toward the less price-elastic rich, and firms add inflation on top of the wedge. Markups and profits thus rise during a cost-driven inflation with no change in conduct, the pattern at the center of the post-2021 debate on "sellers' inflation" that Bilbiie and Känzig (2024) evaluate; here it follows from static profit maximization under a shift in customer composition.

RANK has the highest pass-through, $\Psi_\pi = 3.061$. Hand-to-mouth aggregation makes the IS curve more rate-sensitive, $\tilde{\sigma} = 0.53$ against 1 in RANK, so the same Taylor rule buys more disinflation, and the red band in Figure 2 is negative for inflation.

FIGURE 2: Cost-Push Shock: Sequential Channel Decomposition at the Markup-Matched Benchmark



Notes: Impulse responses to a one-standard-deviation cost-push shock, the AR(1) wedge u_t in equation (12) ($\rho_u = 0.9$, $\sigma_u = 0.05$). The **solid black line** is TE-TANK ($\varepsilon_R = 3.0$, $\varepsilon_P = 6.6$). Three additive bands build it along the path RANK $\rightarrow$ TANK $\rightarrow$ TE-TANK: **gray** is the RANK response, **red** the contribution of hand-to-mouth aggregation (TANK minus RANK), and **blue** the contribution of the elasticity gap (TE-TANK minus TANK). The accounting is sequential, not a partition into independent channels. Both homogeneous corners use the markup-matched $\varepsilon^* = 3.93$, so the blue band is the dispersion-pure contribution. Inflation is annualized; other variables are quarterly percentage deviations from steady state. Figure C.2 in Appendix C.3 plots TE-TANK directly against TANK(ε^*).

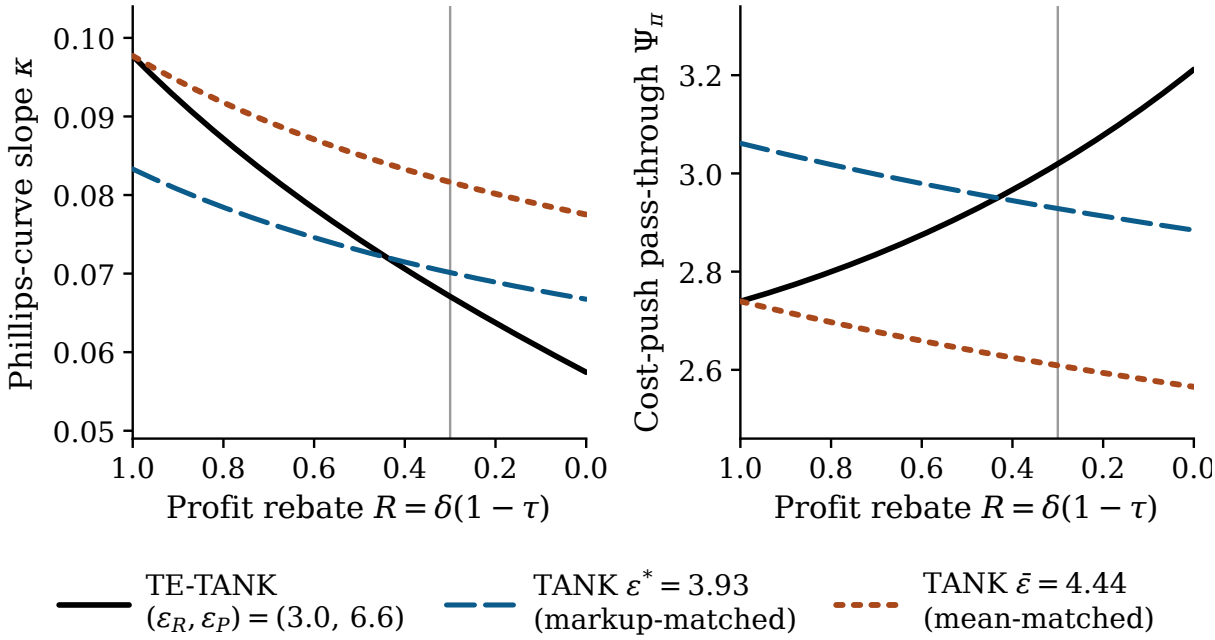
3.4 Fiscal Policy as a Supply-Side Instrument

Profit redistribution moves the Phillips curve in this environment, not only the demand block. Index steady states by the profit rebate $R = \delta(1 - \tau)$, from full rebating ($R = 1$, no steady-state inequality) to none ($R = 0$), holding all other primitives and the policy rule at baseline. Figure 3 shows that along this branch the curve flattens in all three economies, while cost-push pass-through rises only in TE-TANK. In both panels the horizontal axis is R, falling from left to right, and the vertical rule marks the baseline. The left panel plots the slope κ and the right panel impact pass-through Ψ_π. The solid black line is TE-TANK, the dashed blue line *markup-matched* TANK, and the dotted orange line *mean-matched* TANK. Pass-through rises in TE-TANK and falls

in the benchmarks along the whole branch, not only between its endpoints.

Table 3 reports the change between the two endpoints. Its first column shows that the slope falls by 41.2 percent in TE-TANK, 19.9 percent in *markup-matched* TANK($\varepsilon^* = 3.93$), and 20.7 percent in *mean-matched* TANK($\bar{\varepsilon} = 4.44$). Its last column shows that cost-push pass-through nevertheless rises by 17.2 percent in TE-TANK and falls by 5.8 and 6.4 percent in the two benchmarks. Corollary 5 in Appendix D.11 proves all signs except that of TE-TANK pass-through, which is numerical.

FIGURE 3: The Phillips-Curve Slope and Cost-Push Pass-Through Along the Fiscal Branch



Notes: The reduced-form Phillips-curve slope κ (left) and impact cost-push inflation pass-through Ψ_π (right) along the admissible fiscal branch, with the profit rebate $R = \delta(1-\tau)$ falling from full rebating ($R = 1$, no steady-state inequality) to none ($R = 0$). Non-fiscal primitives and the policy rule are held at the baseline calibration. Table 3 reports the change in each line over the whole branch. The vertical rule marks the baseline calibration, $R = 0.30$ ($\delta = 1$ and $\tau = 0.7$). TE-TANK and *mean-matched* TANK coincide at $R = 1$, where the effective elasticity equals the population mean.

Two forces damp inflation after a cost shock, and pass-through depends on them only through their product: the slope κ, the disinflation per unit of lost output, and Θ, the output contraction the policy rule produces per unit of inflation. A lower rebate raises inequality, which flattens the curve but also makes demand more rate-sensitive, so Θ rises. The Ricardian wage channel enters κ as $1/(1+\Gamma\sigma)$ and Θ as $1+\Gamma\sigma$, so it cancels from the product. In the homogeneous benchmarks this channel is the whole flattening, and the small demand-side residual lowers their pass-through at the calibrated rule. TE-TANK flattens for a second, supply-side reason: as inequality rises, more spending comes from the price-insensitive rich, the demand firms face becomes less elastic, and firms move prices less for a given change in cost. Nothing in Θ offsets

TABLE 3: Fiscal Redistribution and Cost-Push Transmission Along the Rebate Branch

	% change along the branch, $R{=}1 \to R{=}0$			
Economy	κ	$\partial\hat{\mu}/\partial u$	$\lvert\Psi_y\rvert$	Ψ_π
TE-TANK $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$	-41.2	$+27.5$	$+58.3$	$+17.2$
TANK$(\varepsilon^* = 3.93)$, markup-matched	-19.9	$+2.7$	$+28.2$	-5.8
TANK$(\bar{\varepsilon} = 4.44)$, mean-matched	-20.7	$+2.5$	$+29.3$	-6.4

Notes: Percentage change in each object between the endpoints of the admissible fiscal branch, from full profit rebating ($R = \delta(1-\tau) = 1$) to none ($R = 0$), holding all non-fiscal primitives and the policy rule at baseline. κ is the reduced-form Phillips-curve slope, $\partial\hat{\mu}/\partial u$ the impact markup response to a cost-push wedge (Corollary 5(iii)), $|\Psi_y|$ the impact output contraction, and Ψ_π impact inflation pass-through. Every entry is monotone along the branch on a 2001-point grid in R.

this force, which is about 3.7 times the demand-side residual, so the TE-TANK line in the right panel of Figure 3 rises.

The markup and pass-through comparative statics carry different information. The second column of Table 3 shows that the impact markup response to a cost-push wedge rises in all three economies, by 27.5 percent in TE-TANK against 2.7 percent in the *markup-matched* benchmark: the same sign, an order of magnitude apart. Only pass-through separates the two economies by sign, and only under the calibrated rule: a rule with output coefficient ϕ_y above $(1-\rho_u)(\varphi+\alpha)/(1-\alpha) = 0.200$ reverses the homogeneous sign, and Appendix D.11 derives this threshold and the elasticity gap, 1.19, below which the net change in TE-TANK pass-through over the branch turns negative.

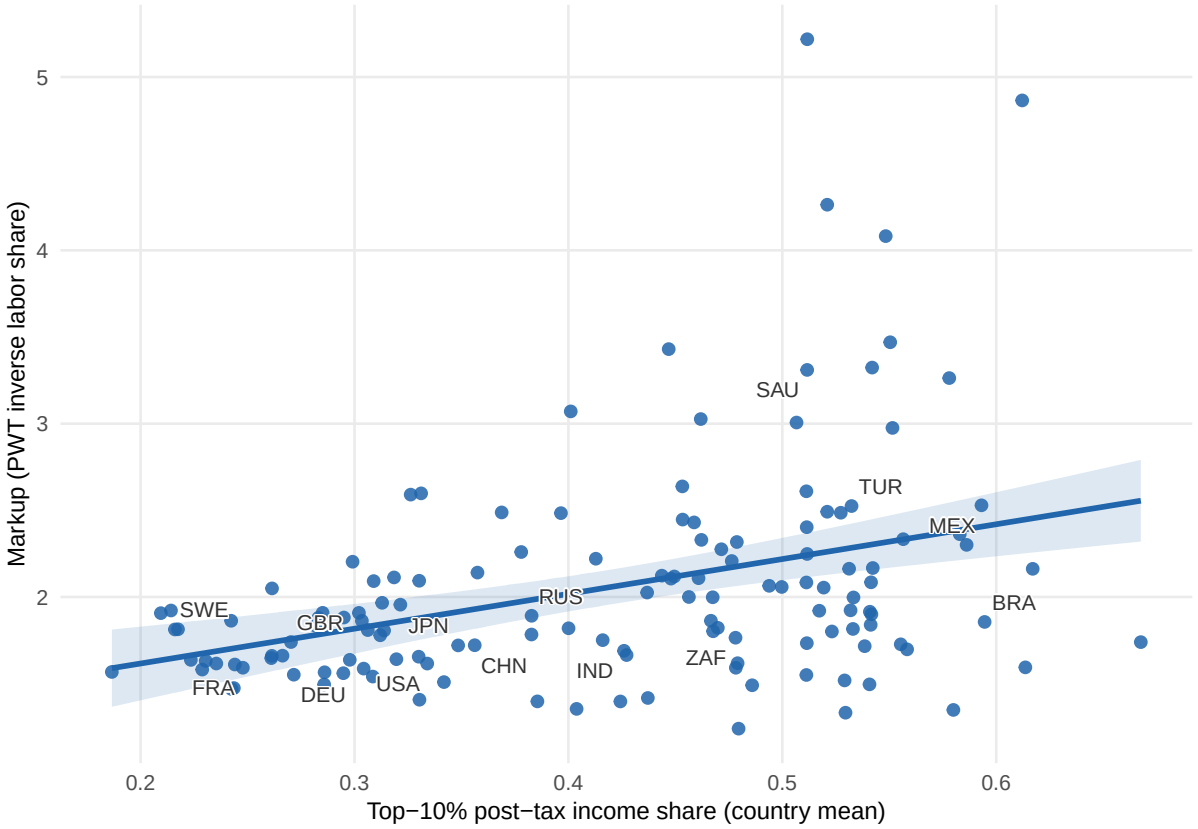
4 Empirical Patterns

The patterns below are descriptive consistency checks, not tests of the elasticity gap. Most of the model's aggregate implications, including the markup responses to supply shocks, follow from the sticky-price core shared with RANK and homogeneous-elasticity TANK. The one implication that separates TE-TANK from *markup-matched* TANK, a policy-dependent sign difference in inflation pass-through, is not detected in the aggregate data. Appendix B states the local-projection specification, identified shocks, and markup proxies, and Table B.6 there collects the full sign record.

4.1 Markup Levels and Inequality

Figure 4 shows that economies with higher long-run income concentration have higher average markup levels. Each point is one country. The horizontal axis is the country-

FIGURE 4: Markup Levels and Inequality Levels Across Countries

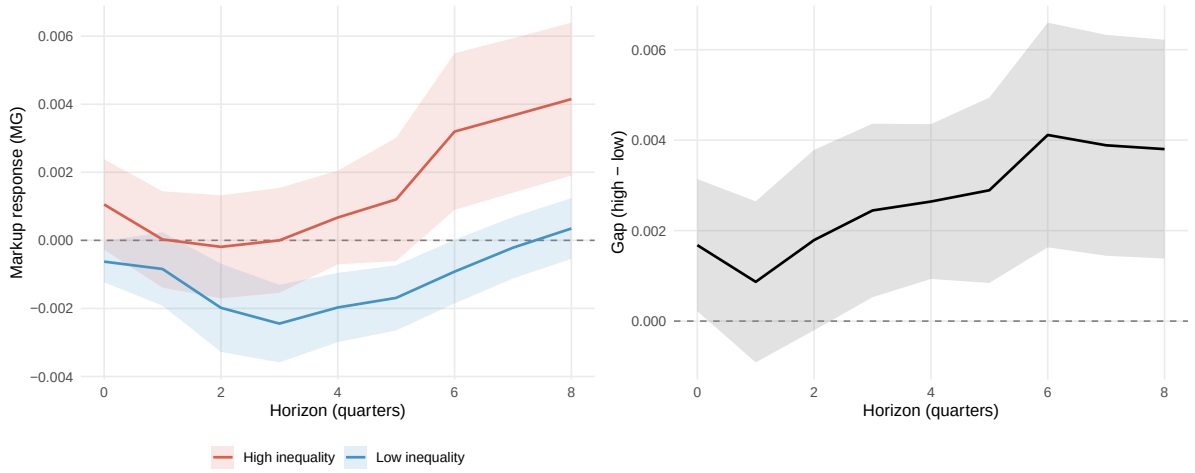


Notes: Each point is one country. The vertical axis is the markup level measured as the inverse labor share, the geometric time mean $\exp\left(\overline{\log(1/\mathrm{LS})}\right)$ of the labor share LS; the horizontal axis is the time-mean WID post-tax top-10% income share. Both means are taken over each country's own overlap window with WID. The labor share is from the Penn World Table, adjusted for the labor income of the self-employed, for 138 countries (5480 country-years, 1951 to 2019). The line is the OLS fit with a 95% confidence band based on conventional standard errors. Labels mark selected large economies.
Sources: Penn World Table 10.01, ILOSTAT, WID.world, and De Loecker and Eeckhout (2018).

mean WID post-tax top-10% income share, and the vertical axis is the markup proxy, the inverse of the Penn World Table labor share; the line is the OLS fit with its 95% band. With the income share measured as a fraction, the slope of the fitted line is 2.009 (SE 0.450, $N = 138$); with the ILO labor share it is 2.245 (SE 0.347, $N = 185$). The inverse labor share has a long right tail, and without the 19 countries with a mean above 2.5, among them oil and gas exporters and offshore financial centers, the log slope falls from 0.829 to 0.351 (SE 0.110): the sign survives, but the magnitude falls to 0.42 of its full-sample value. Table A.1 in Appendix A reports the log specification for all three markup sources and the within-country estimates.

The relationship is descriptive and does not identify the model's steady-state link from inequality to the desired markup. Cross-country differences in market structure, regulation, technology, sectoral composition, and measurement may drive both variables, and causality can also run from markups to inequality, as in Boar and Midri-

FIGURE 5: Cross-Country Markup Response by Inequality Group (Mean Group, Oil Shock)



Notes: Mean Group markup responses to a +1 SD Känzig (2021) oil supply-news shock, high- versus low-inequality countries (35 OECD and other European economies, split at the median WID post-tax top-10% share, without the 5 net oil exporters that Appendix B.4 lists). Right panel: the high-minus-low gap. Shaded bands are 90% intervals from the cross-country dispersion of the country coefficients; Appendix B.6 reports the common-shock-robust bootstrap. Inference on the gap uses the standard error of the difference, so the overlap of the left panel's bands is not informative about it.

gan (2025), where higher markups raise the income of concentrated firm owners. The within-country estimates are weaker and in one case wrong-signed, so the evidence speaks to long-run levels, not to changes within countries.

4.2 Shock-Conditional Evidence

US markups rise after the technology and oil supply shocks but also after an expansionary monetary policy shock, and consumption inequality rises after the technology shock; Appendix B.3 and Table B.6 report these signs.

Figure 5 shows that, across countries, the inverse-labor-share markup response to the common oil supply-news shock of Känzig (2021) is larger in more unequal economies, a sign that TE-TANK and homogeneous TANK share. Appendix D.11 derives the TANK sign. The shock is an empirical analogue of a contractionary supply disturbance, not a structural implementation of u_t. We estimate country-specific local projections for 35 economies, average them with the Mean Group estimator of Pesaran and Smith (1995), and split countries at the median WID post-tax top-10% income share. In both panels, the horizontal axis is the horizon in quarters and the vertical axis the cumulative change in the log inverse labor share after a one-standard-deviation shock. The left panel shows the high-inequality group in red and the low-inequality group in blue, the right panel their difference, and the shaded areas are 90% intervals. The high-minus-low gap is positive at every horizon, $+0.0026$ at $h = 4$ and $+0.0038$

at $h = 8$. A moving-block bootstrap that resamples the shared quarterly axis gives $p = 0.226$ and $p = 0.057$, so the common-shock-robust evidence is insignificant at four quarters and significant at the ten percent level at eight. The positive estimates cannot confirm the elasticity-gap channel, because homogeneous TANK predicts the same sign and the estimated gap is several times the calibrated one. Power against the calibrated amplification is only 6.1 percent at a nominal size of five percent, so a failure to reject could not refute the channel either. Appendices B.4–B.9 give the details.

The discriminating inflation prediction is not detected. On the same shock and split, the high-minus-low cumulative price-level gap carries the homogeneous-TANK sign at both horizons, is not significant, and could be resolved only if it were much larger than the model implies. Appendix B.5 reports this test.

5 Conclusion

Heterogeneous price elasticities place the consumption distribution inside the Phillips curve. In a two-agent New Keynesian model calibrated to Swiss micro evidence (Auer et al., 2024), desired markups depend on the composition of demand across household types, so they move whenever a shock moves inequality. The reduced-form slope factors in closed form into a feedback-off slope, a Ricardian wage channel, and a desired-markup correction; only the last requires an elasticity gap.

The flattening has a definite sign; its size depends on the benchmark. Holding the steady state fixed, elasticity dispersion lowers the slope by 4.3 percent, and cost-push inflation is 3.1 percent higher than in markup-matched TANK; against mean-matched TANK, which also has a lower steady-state markup, the gap is 15.7 percent.

Fiscal redistribution operates on the supply side. Rebating profits to hand-to-mouth households lowers steady-state inequality in both TE-TANK and homogeneous-elasticity TANK, but by Proposition 1 it lowers the steady-state desired markup only when elasticities differ. Under the baseline calibration and Taylor rule, less redistribution raises cost-push pass-through in TE-TANK and lowers it in homogeneous TANK.

The empirical evidence is suggestive. Markups are higher in more unequal countries and rise after identified supply shocks, but also after an expansionary monetary shock (Nekarda and Ramey, 2020), and the inflation gap that separates TE-TANK from homogeneous TANK is not significant. The main contribution is the mechanism and its sign: when households differ in price sensitivity, a shift in spending toward the rich leaves firms with less price-sensitive customers and raises desired markups.

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Appendix

Inequality, Markups, and the Phillips Curve

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A Markup and Inequality: Additional Evidence

Section 4.1 documents a cross-country association between long-run markup and inequality levels. This appendix tests its robustness to the markup proxy and the estimator, and to the tax base of the income share.

Table A.1 shows that the between-country slope is positive for all three markup measures, but that the production-approach slope rests on Latin America. Its rows are three log markup measures: the Penn World Table and ILO inverse labor shares and the production-approach markup of De Loecker and Eeckhout (2018). Each is regressed on the WID post-tax top-10% share. The four columns are the between-country estimator, the same estimator without Latin America, the within-country estimator with fixed effects, and the long difference. Without the 7 Latin American countries, the production-approach slope falls from 0.475 to -0.014, while the two inverse labor-share slopes become larger.

TABLE A.1: The Between-Country Markup-Inequality Slope: Proxy and Estimator Robustness

	Between	Between, excl. Latin America	Within FE	Long-diff.
Inverse LS: PWT `labsh`	0.829***	1.018***	0.307**	0.346*
N_{cty}=138	(0.172)	(0.242)	(0.149)	(0.183)
Inverse LS: ILO labour income share	0.999***	1.288***	0.101	−0.348
N_{cty}=185	(0.141)	(0.148)	(0.173)	(0.236)
De Loecker–Eeckhout production approach	0.475***	−0.014	0.236	0.442
N_{cty}=41	(0.160)	(0.209)	(0.314)	(0.670)

Notes: Dependent variable: log markup; regressor: WID post-tax top-10% national income share; heteroskedasticity-robust standard errors in parentheses, clustered by country in the within column. Between: OLS on country means; within FE: country and year fixed effects; long-diff.: change from each country's first to last year. N_{cty} is the number of countries, which is the sample size of the between and long-difference columns. The second column repeats the between-country regression without the Latin American and Caribbean countries (World Bank region), on 112 countries for PWT, 155 for ILO and 34 for De Loecker–Eeckhout. The within-country fixed-effects column is estimated on 5,451 country-years for PWT, 3,876 for ILO and 1,411 for De Loecker–Eeckhout. In the PWT sample, the within estimator drops 29 US country-years (1951–1979) that form singleton year cells. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The three samples differ in coverage. The Penn World Table has the longest time span and the most country-years. The production-approach markups of De Loecker and Eeckhout (2018) exist only for countries and years with detailed firm financial-statement data, and the ILO labor-share series has its own reporting coverage. The intersection of the three sources is too thin for a separate common-sample estimate, so we report the coverage difference rather than compute one.

The last two columns show that the level association is better identified than the

corresponding time-series one. The within-country fixed-effects and long-difference estimates are significant only in the Penn World Table sample, at 0.307 (SE 0.149, $p = 0.0410$) and 0.346 (SE 0.183, $p = 0.0611$), the long difference only at the ten percent level. They have the wrong sign in the ILO long difference and are positive but imprecise in the production-approach sample. The data therefore speak mainly to differences in long-run levels across countries. The level association is consistent with the model's steady-state channel from $\bar{H}$ to $\bar{\mathcal{M}}$ but does not identify it. The shock-conditional evidence in Section 4 is likewise descriptive and does not isolate heterogeneous demand elasticities from other distributional mechanisms.

Pre-tax versus post-tax income shares. Table A.2 shows that the between-country association does not depend on the tax base. It re-estimates the level specification in Panel A and the log specification in Panel B on the WID pre-tax share, next to the post-tax share used everywhere else in the paper. All twelve slopes, for three markup measures, two tax bases, and the level and the log, are positive and significant at the 1% level. In logs, the post-tax slope is slightly below the pre-tax one for two of the three measures. For the production-approach markups it is 0.475 against 0.564, a ratio of 0.84. For the Penn World Table it is 0.829 against 0.935, and the ratio is 0.90 once the pre-tax slope is measured on the country-years that both bases cover, where it is 0.921. For the ILO labor income share the post-tax slope is slightly above the pre-tax one, 0.999 against 0.964, a ratio of 1.04.

Coverage changes with the tax base only in the Penn World Table panel, where WID reports a post-tax share for 5480 of the 6183 country-years that carry a pre-tax share. The gap between the two columns comes from the tax base and not from the sample. Re-running the pre-tax specification on the post-tax country-years gives a log slope of 0.921 (SE 0.198), against 0.935 (SE 0.197) on the full pre-tax panel, so almost all of the move to 0.829 survives when the country-years are held fixed. The production-approach and ILO panels contain identical country-years on both bases, 1411 and 3876, so their columns differ in the tax base alone.

Post-tax is the paper's baseline because the model's inequality object is post-transfer. The heterogeneity index $H_t = C_t^R / C_t$ is consumption per rich household relative to population-average consumption, so the rich households' aggregate consumption share is $(1 - \lambda) H_t$, which they finance out of income after taxes and transfers. The WID post-tax national income share (`sdiinc`) is the closest available counterpart to post-transfer resources, and hence to $\bar{H}$. We report the pre-tax column because it is the series most of this literature uses, not because one base gives the more favorable result.

TABLE A.2: The Between-Country Markup-Inequality Slope: Pre-Tax versus Post-Tax Income Shares

	Pre-tax share	Post-tax share
Panel A. Markup level		
De Loecker–Eeckhout production approach	0.779***	0.662***
N_{cty}=41	(0.272)	(0.231)
Inverse LS: PWT `labsh`	2.248***	2.009***
N_{cty}=138	(0.512)	(0.450)
Inverse LS: ILO labour income share	2.171***	2.245***
N_{cty}=185	(0.448)	(0.347)
Panel B. Log markup		
De Loecker–Eeckhout production approach	0.564***	0.475***
N_{cty}=41	(0.189)	(0.160)
Inverse LS: PWT `labsh`	0.935***	0.829***
N_{cty}=138	(0.197)	(0.172)
Inverse LS: ILO labour income share	0.964***	0.999***
N_{cty}=185	(0.181)	(0.141)

Notes: Between-country regressions of the country-mean markup on the country-mean WID top-10% national income share, one observation per country, heteroskedasticity-robust standard errors in parentheses. Panel A puts the markup *level* on the left-hand side, the quantity the scatterplots draw; Panel B the log markup, the specification quoted in the text. The pre-tax share is WID `sptinc`, the post-tax share WID `sdiinc`; the post-tax column is the paper's headline specification and reproduces the between-country column of Table A.1. Every post-tax observation in WID has a pre-tax counterpart but not the reverse, so the two columns can differ in coverage as well as in tax base. De Loecker–Eeckhout: identical country-years on both bases (1,411); PWT: 6,183 country-years pre-tax against 5,480 post-tax, and re-running the pre-tax specification on the post-tax country-years gives 2.241 (0.520) in levels and 0.921 (0.198) in logs, so the gap between the columns is the tax base and not the coverage; ILO: identical country-years on both bases (3,876). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Local Projections

This appendix reports the estimation detail behind the local-projection evidence of Section 4. It covers the specification and the markup measures, the unconditional responses to technology, oil, and demand shocks, and the cross-country Mean Group design with its inflation pass-through test, common-shock inference, power calculation, structural-sorter second stage, and robustness checks. Markup cyclicality depends on

how the markup is measured (Nekarda and Ramey, 2020), so the unconditional responses are compared across several measures. The appendix closes with the signs by shock.

B.1 Specification

For each horizon $h = 0, 1, \ldots, 12$ we estimate the local projection of Jordà (2005),

$$\mu_{t+h} - \mu_{t-1} = a_h + \beta_h s_t + \sum_{l=1}^{L+1} \gamma_{l,h} s_{t-l} + \sum_{l=1}^{L+1} d_{l,h} \Delta\mu_{t-l} + e_{t+h}, \tag{B.1}$$

where μ_t is the log markup *level* and s_t an identified shock, so that β_h is the cumulative markup response at horizon h, measured from $t-1$ and normalized to a one-standard-deviation shock. The lag length L is chosen by AIC (at most eight lags) from an auxiliary VAR in the shock and $\Delta\mu_t$, is held fixed across horizons, and is common to the markup proxies of an exhibit. The controls include one lag more than L. With this lag augmentation, ordinary least squares with Eicker–Huber–White (HC3) standard errors gives valid inference without a HAC correction (Montiel Olea and Plagborg-Møller, 2021). HC3 is more conservative in small samples than the HC1 formula of the underlying Montiel Olea–Plagborg-Møller asymptotic theory, so inference is conservative where the per-regime sample is modest. The smooth local projection is the one exception; for it we report HC1 standard errors.

B.2 Markup Measures

Table B.1 lists the five markup measures, which span the main methodological choices in the literature: (i) the inverse labor share under Cobb–Douglas technology, the Nekarda and Ramey (2020) baseline, which we also use in the cross-country panel below; (ii) a Cobb–Douglas specification that adds overhead labor following Galí and Gertler (1999), counting only production and nonsupervisory workers as variable labor; (iii) a CES labor-margin measure; (iv) a CES capital-margin measure constructed under constant utilization; and (v) the price-over-unit-cost markup for non-financial corporations from Enders (2020).

TABLE B.1: Markup measures

Markup measure	Definition	Source / parameter
Inverse labor share (CD)	$\mu^{CD} = -\ln s_L$	BLS labor share s_L
CD + overhead labor	$\mu^{CD,OH} = -\ln s'_L,\ s'_L = \nu\, WS/Y$	ν = production-worker share of private employment; from 1964Q1
CES, labor margin	$\mu_L^{CES} = \mu^{CD} + \ln\left[\alpha_L\,(Y/ZL)^{1/\sigma-1}\right]$	BLS, $\sigma = 0.5$
CES, capital margin	$\mu_K^{CES} = \mu^{CD} + \ln\left[1-\alpha_K\,(Y/uK)^{1/\sigma-1}\right]$	BLS + BEA capital, const. u
Price / unit cost (Enders)	$\mu = \ln(P/UC)$	BLS NFC Prod. & Costs

Notes: The five markup measures used in the local-projection analysis. Measures (i)–(iv) are constructed from quarterly US national-accounts data; measure (v) is the price-over-unit-cost ratio for non-financial corporations. Measure (ii) divides business-sector wages and salaries WS, not compensation, by nominal output Y, multiplies this ratio by the variable-labor share ν of Nekarda and Ramey (2020), and omits the constant Cobb–Douglas output elasticity, which cancels in log changes. Section B.4 also uses measure (i) in the cross-country panel.

B.3 Unconditional Responses to Identified Shocks

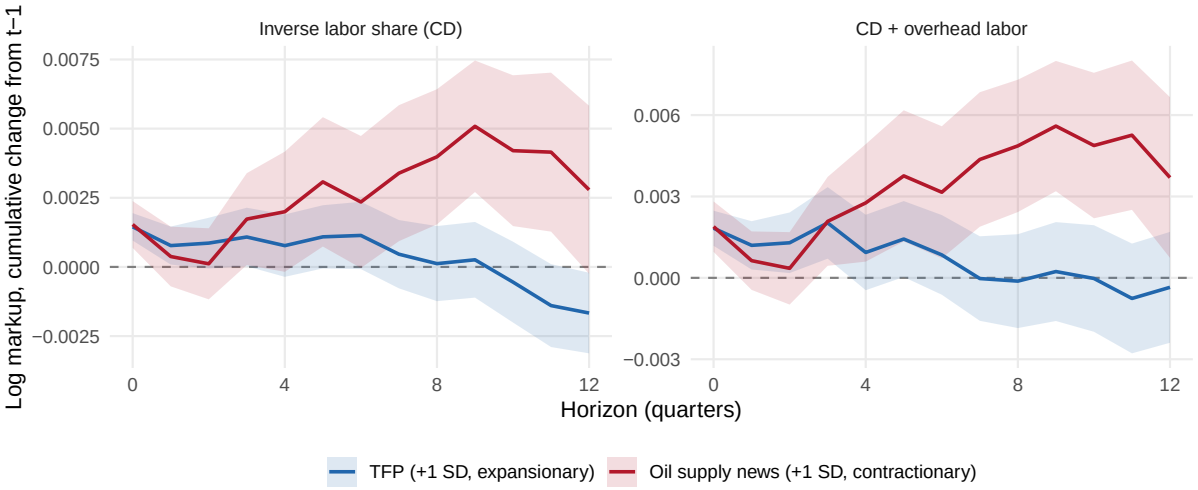
The unconditional signs are mixed and test only the core that TANK and TE-TANK share. Markups rise after the technology and oil supply shocks, but they also rise, on all four proxies, after an expansionary Jarociński and Karadi (2020) monetary policy shock, whereas the sticky-price class predicts a decline.[10] Consumption inequality rises after the technology shock and is an uninformative null after the oil and demand shocks.

Markups across both supply shocks: the headline picture. Figure B.1 shows that markups rise after both the technology and the oil shock, on two proxies of Nekarda and Ramey (2020) estimated from equation (B.1). The left panel uses the Cobb–Douglas inverse labor share and the right panel its overhead-labor variant. In each panel, the horizontal axis gives the horizon in quarters after the shock and the vertical axis the cumulative change in the log markup from $t-1$. The blue line is the response to an expansionary technology shock, the red line the response to a contractionary oil supply-news shock, and the shaded bands are ± 1 standard error. In both panels the two responses start positive on impact. The oil-shock response then builds over about two years, while the technology-shock response fades to about zero within two years and turns slightly negative thereafter. The technology shock is the clean test of the model's mechanism: marginal cost falls and, with sticky prices, the ratio of price to marginal cost rises. The response replicates Nekarda and Ramey (2020) and is consistent with New Keynesian pricing in general, not only with our extension. Two qualifications

[10] Matching this sign requires machinery that the model does not contain: a cost channel, under which a lower policy rate lowers financing costs and marginal cost; nominal wage rigidity (Erceg et al., 2000), which Bilbiie and Känzig (2024) show decides whether profits rise after a demand shock; or Kimball demand curvature and strategic complementarities, which damp the predicted decline (Kimball, 1995; Klenow and Willis, 2016). The composition of measured marginal cost also decides the sign of the estimate itself (Bils et al., 2018). Adding capital investment does not resolve the issue: in Bilbiie and Känzig (2024) it restores demand amplification, but procyclical profits still keep the inflation response muted.

apply. First, Figure B.2 shows that the US markup response to the technology shock is larger before 1975 than after, so the oil shock is the more reliable driver in modern data. Second, the inverse-labor-share proxy used here measures the labor margin, while energy costs enter marginal cost directly, so part of its rise after the contractionary oil shock could in principle be the mechanical wedge between the two.

FIGURE B.1: Markup Responses to Supply Shocks: Unconditional Local Projections



Notes: Cumulative log-markup responses from equation (B.1) on US quarterly data. Left panel: inverse labor share under Cobb–Douglas technology, measure (i) of Table B.1; right panel: the variant with overhead labor, measure (ii). Estimation samples are the merged shock–proxy windows and differ by shock: 1949Q1–2017Q4 ($N = 276$) for the technology shock, except in the overhead-labor panel, whose proxy begins in 1964Q1 ($N = 216$); and 1975Q1–2017Q4 ($N = 172$) for the oil shock in every panel, the oil series beginning in 1975. N is the number of quarters in each merged shock–proxy window. A horizon-h regression loses the first three quarters to the lagged controls and the last h quarters to the lead, so the regression samples at $h = 4$ and $h = 8$ are 269 and 265 for the technology shock (209 and 205 in the overhead-labor panel) and 165 and 161 for the oil shock. Blue: identified TFP shock, +1 SD expansionary (SVAR innovation). Red: Känzig (2021) oil supply news, +1 SD contractionary (published sign)—the two shocks move the economy in opposite directions by construction. Shaded bands are ±1 standard error (approximately 68% pointwise; EHW/HC3).

The technology shock $\varepsilon_t^{\text{TFP}}$ is identified in a *first stage*, exactly as in Nekarda and Ramey (2020). We estimate a quarterly five-variable Cholesky SVAR with four lags and a quadratic trend. Utilization-adjusted TFP (Fernald, 2012) in log levels is ordered first, followed by log real GDP per capita, the log GDP deflator, the three-month Treasury-bill rate, and the log markup. The reduced-form residual of the TFP equation enters equation (B.1) as the shock s_t for horizons $h = 0, \ldots, 12$. The identified shock changes little when the markup is dropped from the SVAR: its correlation with the baseline shock is 0.995, so the outcome entering the system does not drive the shock. Other specification choices move it more. The correlation is 0.988 with a linear trend, 0.960 with TFP ordered last, 0.930 with all variables in first differences, 0.910 under a bivariate long-run restriction on TFP and output growth (Galí, 1999), and 0.840 when the

SVAR is estimated on the post-1984 sample. The local projections and their bootstrap treat this identified shock as a given regressor; first-stage estimation uncertainty is not propagated into the bands. The correlations above limit but do not eliminate this concern. The cross-country oil design is unaffected, since the oil supply-news series is a published external instrument with no first stage estimated here.

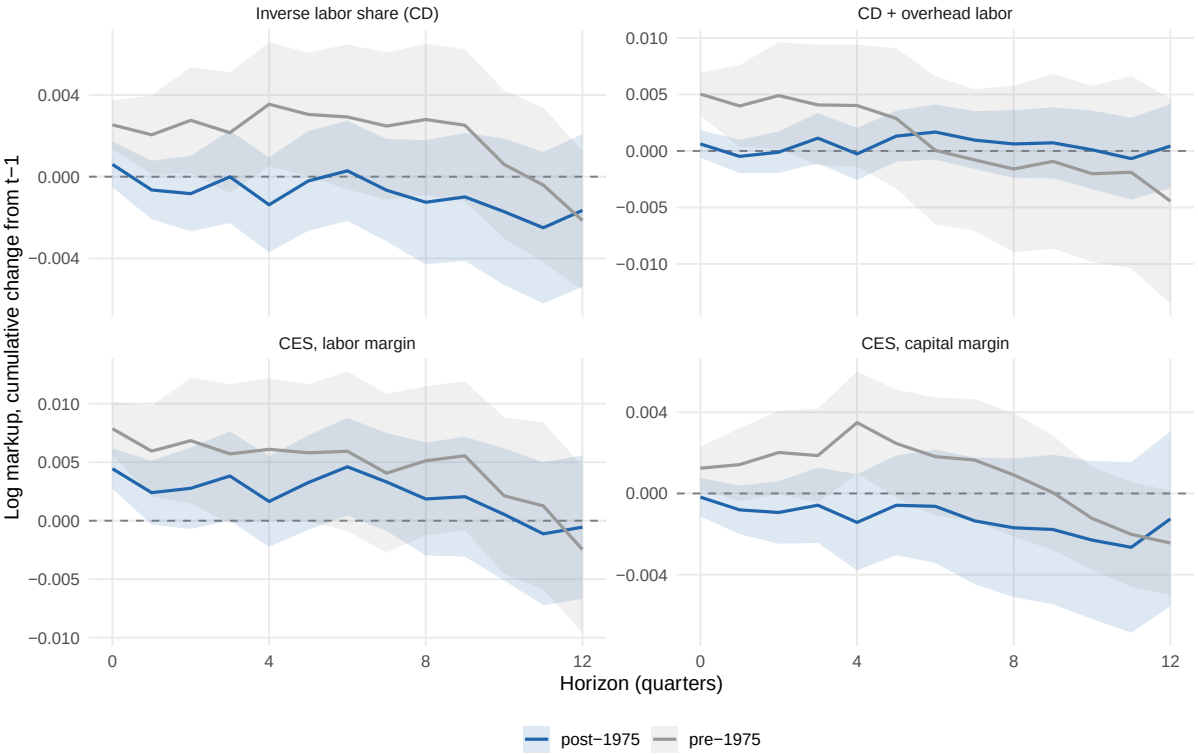
For the unconditional markup response to the technology shock, a joint bootstrap test across all horizons ($h = 0, \ldots, 12$) rejects the null that the entire path is zero ($p = 0.041$), and the sup-t simultaneous confidence band (Montiel Olea and Plagborg-Møller, 2019) excludes zero at the impact horizon. The impact response drives the rejection: the path is positive but imprecise thereafter, and the mild negative excursion beyond two years is indistinguishable from zero at every horizon, pointwise and jointly.

The impact rise is robust across measures. Four of the five measures are positive and significant at impact; only the CES capital-margin measure is insignificant. The response also holds on the Enders price-over-unit-cost measure, which does not reduce to the labor-share identity. Adding current and lagged hours growth as controls raises the impact response from 0.0014 (SE 0.0005) to 0.0022 (SE 0.0005) and leaves the response at $h = 4$ almost unchanged, at 0.0007 against 0.0008. Hours themselves respond to the shock, so conditioning on them changes the estimand; the check is a robustness test, not one that rules out labor hoarding. Smooth (Barnichon and Brownlees, 2019) and bias-corrected (Herbst and Johannsen, 2024) local projections leave the response intact: smoothing sharpens the joint test, and the estimated finite-sample bias is at most 2.1 percent of the standard error, at $h = 7$. The result matches Nekarda and Ramey (2020): the nature of the shock, not its sign, governs markup cyclicality.

Technology shock before and after 1975. Figure B.2 shows that, in the first year after the technology shock, the markup response is larger before 1975 on all four US proxies. The difference is significant on impact for the overhead-labor measure but not jointly across horizons for the baseline measure. The figure re-estimates equation (B.1) separately on the two windows, with the full-sample shock in the same units in both. Its four panels show measures (i)–(iv) of Table B.1: the Cobb–Douglas inverse labor share and its overhead-labor variant in the top row, and the CES labor-margin and capital-margin measures in the bottom row. The axes are those of Figure B.1. The grey line is the pre-1975 response, the blue line the post-1975 response, and the shaded bands are 90 percent pointwise intervals. On the baseline Cobb–Douglas measure the impact response is 0.0025 (SE 0.0007) before 1975 and 0.0006 (SE 0.0007) after. A pooled local projection, fully interacted with a post-1975 indicator, tests the difference directly. For the baseline measure it gives $p = 0.052$ on impact, $p = 0.034$ at $h = 4$, and $p = 0.165$ at

$h = 8$, and a sup-t bootstrap test over $h = 0, \ldots, 8$ gives $p = 0.157$. For the overhead-labor measure, whose pre-1975 window starts in 1964Q1, the impact difference has $p = 0.001$, or 0.013 after a Bonferroni correction over the nine horizons, while the differences at $h = 4$ and $h = 8$ are not significant. After 1975 the impact response remains significant on the CES labor-margin measure, at 0.0044 (SE 0.0011).

FIGURE B.2: Markup Responses to the Technology Shock before and after 1975



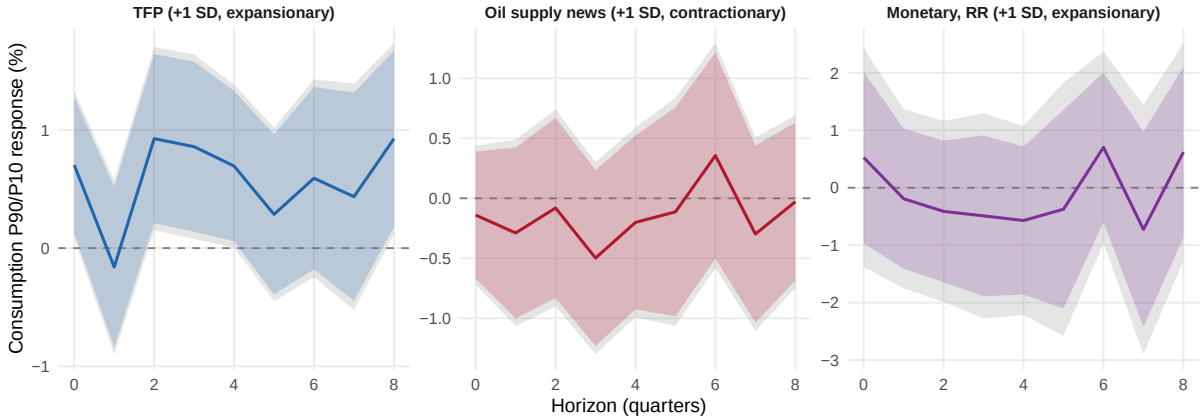
Notes: Cumulative log-markup responses from equation (B.1) to a +1 SD technology shock for measures (i)–(iv) of Table B.1, estimated separately on two windows. Grey: 1949Q1–1974Q4 ($N = 104$), except for the overhead-labor measure, whose proxy begins in 1964Q1 ($N = 44$). Blue: 1975Q1–2017Q4 ($N = 172$). N is the number of quarters in each merged shock–proxy window. The shock is the full-sample SVAR shock, not re-estimated by window, and is scaled by its full-sample standard deviation, so the two lines are comparable in size. Shaded bands are 90% pointwise intervals (EHW/HC3), wider than the ±1 standard-error bands of Figure B.1. Vertical axes differ across panels.

Oil shock: a broader cost base. The Enders (2020) markup, price over unit labor plus nonlabor cost (capital consumption, taxes on production, and net interest), also rises after the oil shock, so the rise survives a broader cost base. That cost base contains no energy, however, so the wedge between the labor margin and true marginal cost remains open.

Consumption inequality. Figure B.3 shows that consumption inequality rises after the technology shock and responds imprecisely after the oil and monetary shocks.

Consumption inequality is the Coibion et al. (2017) P90/P10 ratio from the Consumer Expenditure Survey; for this outcome the monetary shock is that of Romer and Romer (2004). The three panels show, from left to right, the responses to an expansionary technology shock, a contractionary oil supply-news shock, and an expansionary monetary shock. In each panel, the horizontal axis gives the horizon in quarters, from 0 to 8, and the vertical axis the cumulative percent response of the P90/P10 ratio. Colored bands are 90 percent pointwise intervals and grey bands 90 percent sup-t simultaneous bands. After the technology shock the ratio rises and the path is jointly significant, so inequality is procyclical conditional on technology, as Section 2.5 predicts. After the oil and monetary shocks the responses are imprecise in both directions. These nulls are uninformative rather than evidence against the model: Broer et al. (2025) detect distributional effects of the same oil shock in large monthly administrative earnings data, so a null in a short quarterly survey ratio is expected.

FIGURE B.3: Consumption Inequality after Three Identified Shocks



Notes: Cumulative percentage responses of the Coibion et al. (2017) consumption P90/P10 ratio (BLS CEX, quarterly), estimated by lag-augmented local projections with EHW/HC3 standard errors and normalized to a +1 SD shock. Shocks are labeled in the panel headers; the monetary shock is Romer and Romer (2004). Colored bands: pointwise 90% intervals; grey bands: 90% sup-t simultaneous bands. Samples and N refer to the $h = 0$ regression: TFP 1981Q4–2008Q4 ($N = 109$), oil 1981Q3–2008Q4 ($N = 110$), monetary 1981Q4–2008Q4 ($N = 109$). Joint p-values (sup-t, $h = 0$–8): TFP 0.018, oil 0.662, monetary 0.950.

B.4 Cross-Country Panel: Design Detail and Robustness

The cross-country panel asks whether markups rise more after the common Känzig (2021) oil shock in more unequal economies; Section 4.2 reports the headline Mean Group estimates. The sample contains 35 OECD and other European economies through 2019Q4 and excludes 5 net oil exporters by the rule set out below. The oil shock also redistributes income toward the top, so the exercise is a joint test of distribution-dependent pricing rather than a pure pass-through test.

In the closest model, Chan et al. (2024) bring an energy shock into a two-agent New Keynesian setting. There the heterogeneity is the standard borrowing constraint, and the energy price matters because energy–labor complementarity in production compresses the labor share, not because types differ in the elasticity of demand.

Using the same instrument, Broer et al. (2025) find textbook supply-shock responses of German inflation and activity, with a first-stage F-statistic near 10, which supports its identification outside the United States. In their counterfactual in which a monetary non-response is fully anticipated, the shock's aggregate and distributional effects change little, although repeated surprise non-responses do raise activity and low-income employment. The small effect of an anticipated non-response partly mitigates the concern that cross-country differences in policy reactions, rather than in demand composition, drive the heterogeneity documented below.

Mean Group design. For each country i we estimate a separate local projection on the common global oil shock,

$$\begin{aligned} \mu_{i,t+h} - \mu_{i,t-1} &= a_{i,h} + \beta_{i,h}\, s_t + \sum_{l=1}^{L+1} \rho_{i,l,h}\, \Delta\mu_{i,t-l} + \sum_{l=1}^{L+1} \omega_{i,l,h}\, s_{t-l} + e_{i,t+h}, \\ \hat{\beta}_h^{\mathrm{MG}} &= \frac{1}{N} \sum_i \hat{\beta}_{i,h}. \end{aligned} \tag{B.2}$$

The country coefficients are averaged with the Mean Group estimator of Pesaran and Smith (1995), with standard errors from the cross-country dispersion. Unlike a pooled estimator, this estimator is consistent under slope heterogeneity. The lag length L is common and chosen by a panel AIC rule. The shock s_t is the published oil supply-news series of Känzig (2021), identified from high-frequency oil-futures reactions to OPEC announcements, aggregated to the quarter and standardized with the published sign kept, so $+1$ SD is contractionary. It is common to all countries and is not re-estimated, so identification comes entirely from how the response $\hat{\beta}_{i,h}$ differs across countries. The outcome is the inverse labor share $\mu_{i,t} = \log(\text{GDP}/\text{compensation of employees})$, and countries are split at the median time-averaged WID top-10% share (Alvaredo et al., 2017), an availability sorter rather than the model-matched consumption-dispersion object. Table B.2 lists the 35 economies in the sample, sorted by the post-tax top-10% share, with each country's first and last markup quarter, its group, and the excluded oil exporters in italics. Ending the sample in 2019Q4 keeps the finding from resting on the post-2021 inflation surge that Section 1 uses as motivation.

Oil-exporter exclusion. The baseline excludes net oil exporters, because for them the oil supply shock is a positive terms-of-trade shock and oil rents lower the labor share,

which raises the markup proxy mechanically. Following the convention of Peersman and Van Robays (2012), a country counts as a net oil exporter if its annual net exports of petroleum and other liquids, production minus consumption in the EIA International Energy Statistics, are positive on average over its own window, from its first markup quarter, but not before 1975, to 2019. The rule excludes Norway, Mexico, Canada, Denmark, and the United Kingdom. Only Norway has positive net exports in every year. The share of window years with positive net exports is 0.69 for Canada, 0.76 for Denmark, and 0.53 for the United Kingdom, and mean net exports are 40, 37, and 5 percent of consumption, so the United Kingdom is a borderline case. Earlier drafts of this paper instead excluded a hand list of Norway, Mexico, and Canada.

The positive gap does not depend on the exclusion set, but its size does. With country-dispersion standard errors in parentheses, the gap at $h = 4$ and $h = 8$ is $+0.0026$ (0.0010) and $+0.0038$ (0.0015) under the rule, with 35 countries; $+0.0027$ (0.0010) and $+0.0032$ (0.0014) when the United Kingdom is kept (36); $+0.0025$ (0.0010) and $+0.0031$ (0.0014) under the earlier hand list (37); $+0.0023$ (0.0010) and $+0.0025$ (0.0014) when only Norway is excluded (39); and $+0.0018$ (0.0012) and $+0.0023$ (0.0014) when no country is excluded (40), with the median recomputed in each case. The current rule gives the largest gap at $h = 8$. Without any exclusion the gap is smallest and not significant at $h = 4$; when only Norway or no country is excluded, the gap at $h = 8$ is significant only at the ten percent level.

A pooled regression on the continuous sorter complements the Mean Group gap and allows checks for confounders:

$$\mu_{i,t+h} - \mu_{i,t-1} = a_i + \beta_h s_t + \gamma_h \left(s_t \times \tilde{H}_i\right) + \sum_k \theta_{k,h} \left(s_t \times X_{k,i}\right) + \text{lags} + u_{i,t+h}, \qquad \text{(B.3)}$$

where $\tilde{H}_i$ is the demeaned country-mean post-tax top-10% income share and $X_{k,i}$ contains time-invariant country controls. Time fixed effects are omitted because they would absorb the common shock. The pooled estimator imposes a common slope, so Mean Group is the primary estimator, while the pooled specification supplies Driscoll–Kraay inference (Driscoll and Kraay, 1998) and confounder checks.

Table B.3 shows that the pooled interaction $\hat{\gamma}_h$ stays positive and significant at the 5 percent level under Driscoll–Kraay standard errors when seven time-invariant country traits enter as shock interactions, one at a time or jointly. The traits are imports, capital-account openness, energy intensity, the capital share, employment protection, inflation targeting, and net energy imports. The columns report $\hat{\gamma}_h$ and the number of country-quarters at $h = 4$ and at $h = 8$. The first row is the baseline without confounder controls, the middle block adds one trait at a time, and the last two rows add the first four traits and all seven traits jointly. The coefficient is a pooled slope on the

continuous sorter, so it is a different estimand from the Mean Group gap, for which Section B.6 reports weaker evidence from the shared-time bootstrap.

TABLE B.2: Cross-Country Sample: Coverage and Inequality Split

ISO3	First	Last	N	H_i^{post} (%)	Group	ISO3	First	Last	N	H_i^{post} (%)	Group
DNK	1995Q1	2019Q4	100	18.6	*excl. (oil)*	POL	1995Q1	2019Q4	100	29.9	High
NOR	1995Q1	2019Q4	100	21.7	*excl. (oil)*	HRV	1995Q1	2019Q4	100	30.1	High
SWE	1993Q1	2019Q4	108	22.2	Low	PRT	1995Q1	2019Q4	100	31.2	High
FIN	1990Q1	2019Q4	120	22.6	Low	GRC	1995Q1	2019Q4	100	31.7	High
NLD	1995Q1	2019Q4	100	23.4	Low	LUX	1995Q1	2019Q4	100	31.9	High
SVK	1995Q1	2019Q4	100	23.5	Low	LVA	1995Q1	2019Q4	100	32.7	High
CZE	1995Q1	2019Q4	100	23.8	Low	BIH	2000Q1	2019Q4	80	32.9	High
CHE	1990Q1	2019Q4	120	24.0	Low	BGR	1995Q1	2019Q4	100	33.6	High
BEL	1995Q1	2019Q4	100	24.3	Low	MNE	2006Q1	2019Q4	56	33.7	High
FRA	1980Q1	2019Q4	160	24.8	Low	SRB	1995Q1	2019Q4	100	34.9	High
ITA	1995Q1	2019Q4	100	25.2	Low	LTU	1995Q1	2019Q4	100	35.0	High
SVN	1995Q1	2019Q4	100	25.8	Low	JPN	1994Q1	2019Q4	104	36.1	High
AUT	1995Q1	2019Q4	100	26.0	Low	EST	1995Q1	2019Q4	100	36.3	High
IRL	1995Q1	2019Q4	100	26.2	Low	ROU	1995Q1	2019Q4	100	36.5	High
HUN	1995Q1	2019Q4	100	26.2	Low	USA	1947Q1	2019Q4	292	36.6	High
AUS	1959Q3	2019Q4	242	27.1	Low	ISR	1995Q1	2019Q4	100	43.2	High
DEU	1991Q1	2019Q4	116	28.1	Low	TUR	1995Q1	2019Q4	100	51.8	High
GBR	1955Q1	2019Q4	260	28.2	*excl. (oil)*	*MEX*	2008Q1	2019Q4	48	59.5	*excl. (oil)*
CYP	1995Q1	2019Q4	100	28.9	Low						
ESP	1995Q1	2019Q4	100	29.2	Low						
CAN	1981Q1	2019Q4	156	29.5	*excl. (oil)*						
MLT	1995Q1	2019Q4	100	29.7	Low						

Notes: All 40 countries for which the first stage delivers a coefficient, sorted by the WID post-tax top-ten-percent income share H_i^{post}, the simple mean of the annual share over 1990–2019; missing years are skipped, and at least 5 annual observations are required. "First" and "Last" are the first and last quarters with a non-missing markup observation at or before 2019Q4; N counts those quarters and measures panel coverage. The inclusion rule counts differently: a country needs at least 40 usable quarters, that is, quarters with both a markup growth rate and a shock value after the two series are merged, counted before lags and horizon leads are formed. The regression samples are therefore smaller, most visibly at $h = 8$ for countries with short windows. The balanced subsample of Table B.5, Panel D, thresholds the horizon-maximum estimation sample, and 2 countries fall below 80 on that measure while 1 does on coverage. Group assigns each country to the low or high half at the median H_i^{post} of the baseline sample, giving 17 high and 18 low. The 5 net oil exporters are set in italics and marked "excl. (oil)"; they enter the exclusion-set comparison in the text, the US technology-shock row of Table B.5, and the registered structural-sorter families. The inflation outcome of Appendix B.5 is available for all 35 baseline countries. It is the all-items consumer price index, monthly and not seasonally adjusted, taken at the last month of each quarter, with one source per country: the OECD for most countries, the Eurostat HICP for Cyprus, Malta, and Romania, and the IMF for Bosnia and Herzegovina, Montenegro, and Serbia; Australia enters with its quarterly OECD index. The inflation sample can start later than the markup coverage shown here, as for Bosnia and Herzegovina, whose index starts in 2005Q2.

TABLE B.3: Confounder Controls for the Pooled Interaction

	$h = 4$		$h = 8$	
	$\hat{\gamma}_h$ (DK se; t)	n	$\hat{\gamma}_h$ (DK se; t)	n
Baseline (no confounder controls)	0.0251^{**} (0.0120); $t = +2.10$	3446	0.0284^{**} (0.0121); $t = +2.35$	3306
Individual confounders (added one at a time)				
+ Imports / GDP	0.0244^{**} (0.0121); $t = +2.01$	3446	0.0293^{***} (0.0111); $t = +2.64$	3306
+ Capital-account openness (Chinn–Ito)	0.0281^{**} (0.0122); $t = +2.30$	3122	0.0314^{**} (0.0153); $t = +2.05$	2998
+ Energy intensity of production	0.0251^{**} (0.0119); $t = +2.11$	3446	0.0283^{**} (0.0121); $t = +2.34$	3306
+ Capital share	0.0262^{**} (0.0105); $t = +2.50$	3398	0.0262^{**} (0.0110); $t = +2.39$	3262
+ Employment protection legislation (coverage 31/35)	0.0223^{**} (0.0104); $t = +2.14$	3078	0.0229^{**} (0.0111); $t = +2.06$	2954
+ Inflation-targeting regime share (coverage 35/35)	0.0242^{**} (0.0100); $t = +2.43$	3446	0.0242^{**} (0.0108); $t = +2.25$	3306
+ Net energy imports (coverage 35/35)	0.0253^{**} (0.0121); $t = +2.08$	3446	0.0286^{**} (0.0124); $t = +2.32$	3306
Joint specifications				
Joint — first four confounders	0.0277^{**} (0.0124); $t = +2.23$	3122	0.0354^{**} (0.0147); $t = +2.40$	2998
Joint — all seven confounders	0.0225^{**} (0.0104); $t = +2.16$	2846	0.0266^{**} (0.0131); $t = +2.03$	2734

Notes: $\hat{\gamma}_h$ is the coefficient on the interaction $s_t \times \tilde{H}_i$ in equation (B.3), estimated on the pooled panel of the 35 baseline countries, with Driscoll–Kraay standard errors and the implied t-statistic in parentheses. It is a pooled slope on the demeaned sorter, not the Mean Group gap. Each row of the middle block adds one time-invariant country trait, interacted with the shock, to the baseline specification. "Joint — first four confounders" adds imports/GDP, capital-account openness (the Chinn–Ito index of Chinn and Ito 2006), energy intensity of production, and the capital share together; "Joint — all seven confounders" adds employment protection legislation, the inflation-targeting regime share, and net energy imports as well. The sample size n counts country-quarters and varies with trait coverage. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.5 Inflation Pass-Through by Steady-State Inequality

Corollary 5 and the homogeneous-elasticity benchmark of Appendix D.11 agree on the sign of the markup comparative static and disagree on the sign of the pass-through comparative static: along the calibrated fiscal branch, higher steady-state inequality raises inflation pass-through in TE-TANK by 17.2 percent and lowers it in markup-matched TANK by 5.8 percent. That disagreement, not the markup gradient, is what separates the two economies. The same cross-country design can take it to the data, but the test lacks the power to settle it.

Outcome, and the model object it maps to. The outcome is the cumulative price-level response, $\sum_{s=t}^{t+h} \pi_{i,s} = 100\,(\log P_{i,t+h} - \log P_{i,t-1})$ in percentage points, not an inflation rate: horizon $h = 4$ is the one-year-ahead and $h = 8$ the two-year-ahead cumulative response. The projection (B.2) is re-estimated per country with this outcome on

the same common oil supply-news shock, aggregated by the same Mean Group estimator, and split at the same median time-averaged WID top-10% income share; only the control-lag length is re-selected. The model object of Proposition 4 is instead the *impact* inflation response Ψ_π to a unit cost-push wedge, so the two are not directly comparable and the mapping has to be made rather than assumed. With a single AR(1) wedge and the linear policy block, the mapping is immediate: equilibrium inflation inherits the wedge's persistence exactly, $\pi_{t+s} = \Psi_\pi \rho_u^s u_t$, so the cumulative response through horizon h is

$$\sum_{s=0}^{h} \pi_{t+s} = \Psi_\pi \frac{1-\rho_u^{h+1}}{1-\rho_u} u_t. \tag{B.4}$$

The multiplier is a positive constant that depends only on ρ_u and h, both common across economies and across positions on the fiscal branch. The cumulative price-level response is therefore a fixed positive multiple of Ψ_π, and its comparative static in steady-state inequality carries the sign of $\mathrm{d}\Psi_\pi/\mathrm{d}\bar{H}$ at every horizon. The sign test is thus well posed on the empirical outcome: TE-TANK implies a positive high-minus-low gap, markup-matched TANK a negative one.

Sample. The inflation panel covers all 35 countries of the markup baseline, 17 in the high-inequality group and 18 in the low. Every country uses one price concept: the all-items consumer price index, monthly and not seasonally adjusted. The quarterly value is the index in the last month of the quarter, and inflation is 100 times the log difference between two consecutive quarters. Each country has exactly one source, so no series is spliced. The source is the OECD consumer price database for most countries, the Eurostat harmonized index (HICP) for Cyprus, Malta, and Romania, and the IMF consumer price series for Bosnia and Herzegovina, Montenegro, and Serbia. Australia publishes its index only quarterly, so it enters with the quarterly OECD index. A country's sample is its markup window intersected with the availability of its source. For Bosnia and Herzegovina the index starts in 2005Q2, so 21 quarters of its markup window have no inflation observation; the country is kept. Bulgaria's sample includes the hyperinflation of early 1997.

Lag length. The control-lag length is re-selected for the inflation series by the rule of the markup design: each country's AIC choice with a ceiling of eight lags, and the median across the 35 countries of the estimation sample. The rule gives the headline $L = 6$, against $L = 2$ in the markup design, so each country-horizon regression carries 14 lag controls and an intercept. The choice is not sharply determined. 14 of the 35 countries choose the ceiling, and raising the ceiling to twelve or sixteen lags gives $L = 5$ and $L = 6$. Taking the median over all countries with at least forty quarters of inflation

data, which adds four of the excluded oil exporters (Mexico has no inflation series), gives $L = 7$. At the ceiling of sixteen lags, Bosnia and Herzegovina and Montenegro have too few observations for the criterion and drop out of the selection. We therefore report the gap across a grid of lag lengths below.

Result and inference. At the headline $L = 6$, the high-minus-low gap is -0.0626 at $h = 4$ and -0.1284 at $h = 8$, with $t = -0.48$ and $t = -0.74$ under the cross-country dispersion standard errors. The common-time moving-block bootstrap of Section B.6, with the markup design's block length of 10 quarters and $B = 4999$ draws, gives $p = 0.737$ at $h = 4$ and $p = 0.600$ at $h = 8$ for the headline, with 90 percent intervals of $[-0.2831, 0.2390]$ and $[-0.5334, 0.3829]$. Block lengths of eight and twelve quarters also give large p-values. The point estimates have the sign of markup-matched TANK, not of TE-TANK, but they are not distinguishable from zero.

The gap varies in size, and in one cell in sign, across lag lengths. At $h = 4$ it is -0.1345 at $L = 8$, -0.0910 at the all-country choice $L = 7$, -0.0626 at the headline $L = 6$, -0.1611 at $L = 5$, and $+0.0648$ at the markup design's $L = 2$; at $h = 8$ the same cells are -0.2707, -0.1751, -0.1284, -0.1900, and -0.0155. Standard errors lie between 0.100 and 0.186, and the largest $|t|$ in the grid is 1.54, at $L = 8$ and $h = 8$. The size of the point estimate thus depends on the lag length, so the conclusion on pass-through rests on the power calculation below, not on the sign of any one cell.

Dropping Turkey moves the gap more than dropping any other country. Without Turkey, the gap is $+0.0372$ (SE 0.0943) at $h = 4$ and -0.0080 (SE 0.1151) at $h = 8$: the point estimate at $h = 4$ turns positive, and neither is significant. These values are the upper ends of the leave-one-out ranges; the lower ends are -0.0997 at $h = 4$ and -0.1717 at $h = 8$. Turkey is kept in the baseline.

Harmonized sample. The inflation and markup gaps use the same 35 countries and the same median split, so the harmonized markup gap is the baseline markup gap that Section 4.2 reports. On country-dispersion standard errors it is $+0.0026$ (SE 0.0010, $p = 0.011$) at $h = 4$ and $+0.0038$ (SE 0.0015, $p = 0.010$) at $h = 8$, so it is positive and pointwise significant at both horizons on these standard errors; the shared-time bootstrap of Section B.6 gives weaker evidence. On the same countries, the inflation gap is too imprecise to be distinguished from zero. The country windows of the two outcomes differ only where the price index starts after the markup series.

Power. The design has no more power here than on the markup side. With the convention that Section B.7 sets out, the minimum detectable effect at eighty percent power and five percent size is 0.4967 percentage points at $h = 4$ and 0.8653 at

$h = 8$. The model-implied gap is smaller by a factor of 23 at $h = 4$ and -55 at $h = 8$, where a negative factor reflects the sign of the unconditional response rather than a change in the prediction. The implied power is 5.2 and 5.0 percent against a nominal size of five percent, so the test has no power against the prediction. One caveat is sharper here than for the markup exercise. For the markup gap, Section B.7 maps the observed cross-country inequality spread into the model's fiscal branch with $A(g_{xc})$, but no inflation-specific model grid exists for this mapping. We therefore scale by the model's full branch contrast, the 17.2 percent rise in TE-TANK against the 5.8 percent fall in the benchmark. This contrast is the largest difference the fiscal block can generate and hence an upper bound on what any observed spread could imply. The calculation is illustrative rather than a calibrated interpolation, but its conclusion is not delicate. At $h = 8$ the mapped gap inherits a negative sign from the negative unconditional response, exactly as the markup mapping does at $h = 4$, so the comparison uses magnitudes, not signs.

Alternative sorters and split rules. Across six inequality sorters and three split rules, the inflation gap does not corroborate the markup pattern: the cells that are significant on country-dispersion standard errors carry both signs, and under Driscoll–Kraay inference neither the capital-share nor the market-Gini interaction is significant in any specification.

The policy-rule caveat. The discriminating sign is conditional on the systematic policy response. Equation (D.36) shows that the homogeneous-elasticity pass-through static has the sign of $\phi_y - (1 - \rho_u)b$, so a central bank responding to output more strongly than the threshold $(1 - \rho_u)b = 0.200$ reverses it, and at $\phi_y = 0$ both economies deliver a falling Ψ_π. Systematic rules differ across the countries in this panel, and the design holds none of that fixed, so cross-country dispersion in the inflation response to a common oil shock can reflect differences in reaction functions rather than in demand composition. The exercise is discriminating only conditional on comparable systematic policy responses. The markup design shares this assumption: Corollary 5(iii) signs the markup comparative static only under (D.34), and the homogeneous benchmark's markup static reverses above the same threshold. The sign mapping of equation (B.4) also assumes that the common oil shock reaches both groups as a domestic cost-push wedge of comparable size and persistence, which the design does not test. This limits the test, not the prediction; a design with country-specific policy rules would be the way to sharpen it.

B.6 Common-Shock Inference: Two Estimands, Three Standard Errors

The three inference methods apply to two estimands. Country-dispersion standard errors and the moving-block bootstrap use the Mean Group gap $\hat{\Delta}_h = \bar{\beta}_{\text{hi},h} - \bar{\beta}_{\text{lo},h}$, which averages heterogeneous country slopes before comparing groups and equals $+0.0026$ at $h = 4$ and $+0.0038$ at $h = 8$. Driscoll–Kraay standard errors use the pooled implied gap $\hat{\gamma}_h(\bar{H}_{\text{hi}} - \bar{H}_{\text{lo}})$ from equation (B.3), which fits one slope to the continuous sorter.

Country-dispersion standard errors treat the countries as independent, although all receive the same shock. The moving-block bootstrap (Künsch, 1989) resamples the common 1975Q1–2019Q4 time axis and re-estimates every country on the same quarters; with $B = 4999$ and blocks of 10 quarters, its percentile p-values are 0.226 at $h = 4$ and 0.057 at $h = 8$. Blocks of eight and twelve quarters give similar values, except that with eight-quarter blocks the p-value at $h = 8$ falls just below five percent. Driscoll–Kraay standard errors treat periods as the independent units; the pooled gap is somewhat smaller than the Mean Group gap and significant at the 5 percent level at both horizons, narrowly at $h = 4$. The bootstrap is the most conservative method, because it credits the sample with roughly the number of oil episodes rather than the number of countries or quarters.

B.7 Power and Model-Implied Magnitudes

The power calculation compares the smallest gap the design can detect with the gap the calibrated model predicts. The minimum detectable effect is $\text{MDE}_h = (z_{0.975} + z_{0.80})\,\hat{\sigma}_h = 2.80\,\hat{\sigma}_h$, the smallest gap that a two-sided test at five percent size detects with eighty percent power, where $\hat{\sigma}_h$ is the standard deviation across the $B = 4999$ draws of the moving-block bootstrap of Section B.6, with blocks of 10 quarters. The model-implied gap is $\Delta_h^{\text{model}} = A(g_{xc})\,\hat{\theta}_h$, where $\hat{\theta}_h$ is the estimated unconditional Mean Group response. The observed sorter gap g_{xc} is the difference between the mean H_i^{post} of the high and the low group, divided by the mean over all baseline countries; it equals 31.7 percent. The amplification $A(g)$ is the model's high-minus-low cost-push markup response, divided by the response of the baseline economy, for two steady states whose inequality differs by g. Scaling by the estimated response removes proportional measurement error in the labor-share proxy, which affects the gap and the average alike. On the calibrated fiscal branch the model is solved at four gaps up to 16.8 percent, just below the ceiling beyond which the fiscal block cannot raise the index further. The observed gap lies beyond this grid, so $A(g_{xc}) = 0.287$ is a linear extrapolation through the origin from the largest grid point; a quadratic fit through the origin

gives $A = 0.289$, and taking A at the grid maximum gives 0.152. The mapping also assumes a unit elasticity between the top-10% income share and the model's inequality index, although income and consumption dispersion are not interchangeable, as Appendix C.2 discusses.

Table B.4 shows that the design cannot detect the predicted gap. For each horizon, its columns give the estimated gap, its bootstrap standard deviation, the minimum detectable effect, the model-implied gap, their ratio, and the power. At $h = 8$ the minimum detectable effect is 0.00575, while the model predicts 0.000630, smaller by a factor of 9.1. The power of the test against the model's own prediction is 6.1 percent, against a nominal size of five percent. Without extrapolation, the predicted gap of 0.000334 lies below the detectable one by a factor of 17.2. Eighty percent power would require an effective sample about 83 times larger than the 45-year span, roughly 3800 years of quarterly data. This extrapolation holds the country sample, the effect size, the dependence across countries, and the information per time block fixed, and lets precision scale with the inverse square root of the sample length. Under a common shock the bootstrap credits the time dimension with only about 18 independent blocks, so this dependence limits what adding countries can gain; we have not evaluated by how much. At $h = 4$ the mapped gap is negative only because the estimated unconditional response is negative there: A is positive, and the model predicts a positive gap at every horizon, because the impact response signed in Corollary 5 decays geometrically without crossing zero.

TABLE B.4: Detectable Effects and Model-Implied Gaps: Cross-Country Design

	$\hat{\Delta}_h$	Bootstrap s.d.	MDE_h	$\Delta_h^{\mathrm{model}}$	$\Delta_h^{\mathrm{model}}/\mathrm{MDE}_h$	Power
h=4	0.0026	0.0027	0.00743	−0.000198	−0.027	0.051
h=8	0.0038	0.0021	0.00575	0.000630	0.109	0.061

Notes: $\hat{\Delta}_h$ is the Mean Group high-minus-low gap in the log inverse labor share after the oil supply-news shock; the bootstrap standard deviation is from the moving-block time bootstrap with $B = 4999$ and a block length of 10 quarters. $\mathrm{MDE}_h = 2.80\,\hat{\sigma}_h$ is the smallest gap detectable at a two-sided five percent size with eighty percent power. $\Delta_h^{\mathrm{model}}$ is the calibrated model's predicted gap at the observed cross-country inequality spread. Power is that of the two-sided five percent test evaluated at $\Delta_h^{\mathrm{model}}$; it is bounded below by the size of the test, so values near 0.05 mean the test carries essentially no information about the model's prediction.

B.8 Structural Sorters: A Second Stage Fixed in Advance

The structural-sorter test is a null. Its specification was fixed in an internal analysis plan dated 5 August 2026, before the result was computed; the plan is in the replica-

tion package as `PAP_structsort_2026-08-05.md`, frozen in git commit 528cd4f of that date. The first-stage country coefficients and earlier second-stage point estimates already existed, so the plan fixes how they are tested, not what they show. The second stage regresses each country's oil-shock coefficient $\hat{\beta}_{i,h}$, at $h = 4$ and $h = 8$, on five standardized traits: the steady-state markup, weak redistribution, their product, and an ordinal hand-to-mouth ranking, each with a predicted positive sign, plus the top-10% income share as an unsigned comparison. The markup trait is built from the inverse Penn World Table 10.01 labor share and the redistribution trait from the SWIID relative redistribution (Solt, 2020), both as country means over 1990–2019; the hand-to-mouth ranking is that of Almgren et al. (2022) for euro-area countries. Both stages are re-estimated in every shared-time bootstrap draw, with Holm adjustment (Holm, 1979). The plan's primary sample is all 40 countries, and its robustness family excludes Norway, Mexico, and Canada; both are kept as registered, so the robustness family uses the earlier hand list, not the baseline sample of this appendix. The code deviates from the plan in one respect: the plan registered the pre-tax WID top-10% share as the comparison trait, and since 8 October 2026 the code uses the post-tax share, the baseline sorter of this appendix. No cell survives the adjustment in either family. The markup, redistribution, and composite slopes have the predicted sign but are imprecise, and the hand-to-mouth trait is wrong-signed.

B.9 Cross-Country Robustness

Table B.5 shows that the robustness checks separate a stable sign from unstable magnitude and inference. The table starts from the baseline gap of $+0.0026$ at $h = 4$ and $+0.0038$ at $h = 8$. Its two gap columns report the high-minus-low gap at $h = 4$ and $h = 8$, and its last column the number of countries. Panels A to D vary, in turn, the identifying shock and instrument, the markup proxy, the inequality sorter, and the sample composition and inference. In Panel A, the alternative oil instrument, the Kilian (2024) oil supply-news series built from twelve-month futures (his m12 series), preserves the positive sign but removes significance. In Panel B, the GOS proxy gives $+0.0036$ and $+0.0030$, while the annual production-approach proxy is an imprecise null. In Panel C, top-income and top-wealth sorters give positive gaps, but the market Gini reverses the sign. In Panel D, every single-country omission preserves the positive gap, and dropping the low-labor-share countries leaves it significant at the 5 percent level. The sup-t simultaneous band excludes zero at both horizons, but it comes from a country bootstrap and, like the dispersion standard errors, treats countries as independent. One qualification applies to both factor-share-based cross-country outcomes: they move with wage incidence, employment composition, and energy costs while de-

sired markups are unchanged. The profit-side GOS proxy checks this without ruling it out.

TABLE B.5: Cross-Country Robustness of the High-Minus-Low Gap

	Gap (high−low)		N
	$h = 4$	$h = 8$	
Panel A: identifying shock and instrument			
Känzig (2021), headline (≤2019Q4)	0.0026** (0.0010)	0.0038*** (0.0015)	35
Kilian (2024) m12, matched window (1989Q2–2017Q2)	0.0012 (0.0015)	0.0016 (0.0016)	35
US TFP SVAR shock (secondary)	0.0012 (0.0019)	0.0022 (0.0029)	39
Panel B: markup proxy			
GOS profit-share proxy (μ_{gos})	0.0036*** (0.0009)	0.0030*** (0.0011)	34
Baseline proxy, restricted to μ_{gos} country-quarters	0.0028*** (0.0011)	0.0034** (0.0015)	34
De Loecker–Eeckhout annual production approach (annual frequency)	0.0075 (0.0112) at $h = 4$ years		17
Panel C: inequality sorter			
WID post-tax top-10% income share (baseline)	0.0026** (0.0010)	0.0038*** (0.0015)	35
WID pre-tax top-10% income share	0.0021* (0.0011)	0.0028* (0.0015)	35
WID top-10% wealth share	0.0014 (0.0011)	0.0031** (0.0015)	35
Capital share	0.0008 (0.0011)	0.0010 (0.0016)	34
SWIID market Gini	−0.0016 (0.0011)	−0.0017 (0.0015)	35
SWIID disposable Gini	0.0005 (0.0011)	0.0014 (0.0016)	35
Panel D: sample composition and inference			
Leave-one-out range (min–max across 35 single-country drops)	[+0.0023, +0.0032]	[+0.0026, +0.0040]	35
Sup-t 90% simultaneous band (country bootstrap, $B = 2000$)	[+0.0003, +0.0050]	[+0.0005, +0.0071]	35
Balanced subsample (≥80 usable quarters)	0.0035*** (0.0009)	0.0049*** (0.0014)	33
Excl. low labor-share countries (mean labor share <0.35)	0.0026** (0.0011)	0.0033** (0.0016)	32

Notes: High-minus-low Mean Group gap in the cumulative markup response at quarterly horizons, with country-dispersion standard errors in parentheses. The baseline sorter is the WID post-tax top-10% income share and the baseline sample has 35 countries; the pre-tax share appears as its own row in Panel C. The SWIID rows use the Gini point estimates of the SWIID summary file (Solt, 2020), release available in July 2026, as plain country means over 1990–2019 with no minimum number of observations, split at the median as for the baseline sorter. The De Loecker–Eeckhout row uses the annual production-approach markups of De Loecker and Eeckhout (2018). The Kilian row uses the m12 series of Kilian (2024), the oil supply-news proxy built from twelve-month futures. The GOS profit-share proxy is $\mu_{gos} = \log(\text{GDP}/(\text{GDP} - \text{GOS}))$, where GOS is gross operating surplus plus gross mixed income, in current prices and seasonally adjusted, from Eurostat and OECD quarterly national accounts. Country-quarters with a GOS-to-GDP ratio outside $(0.05, 0.85)$ are dropped. Japan has no quarterly GOS series, so the two Panel B rows on the μ_{gos} sample exclude it. The US TFP row retains Norway, Canada, Denmark, and the United Kingdom, because the oil-exporter exclusion applies to the oil shock only; Mexico drops out because its markup series starts in 2008Q1 and leaves fewer than forty usable quarters before the TFP series ends in 2017Q4. Leave-one-out and sup-t rows report ranges or bands and carry no stars. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Summary: Signs by Shock and the Overall Reading

Table B.6 shows that the estimated signs match the shared TANK core after the technology shock and for the markup after the oil shock, contradict it for the markup after the demand shock, and leave every TE-TANK increment untested or inconclusive. For each shock, the rows give TANK and the incremental TE-TANK effect; the columns

give the model prediction, the empirical estimate, and whether the two align, each for the markup and for consumption inequality. Conditional on the technology shock, both signs match: markups and consumption inequality rise, as the sticky-price mechanism and the cyclicality relation of Section 2.5 predict. The markup response holds across measures and when hours are controlled for, but it is larger before 1975, while the inequality response is estimated on 1981Q4–2008Q4, so the two legs hold on disjoint samples. Conditional on the demand shock, the markup sign contradicts the sticky-price core, and the inequality response is an uninformative null. Conditional on the oil shock, markups rise, also over a cost base that adds nonlabor to labor cost; that base contains no energy, so the energy wedge remains open, and the inequality response is again an uninformative null.

The test of the TE amplification prediction, the cross-country markup gap after the oil shock, is inconclusive. The gap is positive in the headline, on the GOS profit-share proxy, in the balanced subsample, under every single-country omission, and under every oil-exporter exclusion set. Its inference depends on how the common shock is treated: it is significant at both horizons on the dispersion standard errors and on the country-bootstrap simultaneous band, but the time bootstrap gives $p = 0.226$ and $p = 0.057$. Significance also requires a top-concentration sorter while the market Gini reverses the sign, and the annual production-approach proxy is uninformative. The inflation pass-through test, which separates TE-TANK from markup-matched TANK, has power of about 5.0 percent at $h = 8$ against a size of five percent, so it cannot discriminate between the two.

TABLE B.6: Model Predictions versus Empirical Estimates: Signs by Shock

	Model prediction		Empirics		Aligned?	
	Markup	Inequality	Markup	Inequality	Markup	Inequality
TFP (expansionary)						
TANK	$\uparrow$	$\uparrow$	$\uparrow^{a}$	$\uparrow^{d}$	consistent	consistent^{d}
TE-TANK	$\Uparrow$	$\Uparrow$	+, imprecise^{b}	—	tested–inconcl.^{b}	not sep. tested
Demand (expansionary)^{g}						
TANK	$\downarrow$	$\downarrow$	$\uparrow^{c}$	0^{d}	inconsistent	tested–inconcl.^{d}
TE-TANK	≈ 0	≈ 0	—	—^{i}	not identified^{h}	not sep. tested^{i}
Cost-push (oil)						
TANK	$\uparrow$	$\uparrow$	$\uparrow^{e}$	0^{d}	consistent^{e}	tested–inconcl.^{d}
TE-TANK	$\Uparrow$	$\Uparrow$	+, fragile^{f}	—	tested–inconcl.^{f}	not sep. tested

Notes: $\uparrow$ / $\downarrow$ gives the response sign. TE-TANK rows report the incremental effect relative to TANK: $\Uparrow$ / $\Downarrow$ denotes amplification relative to homogeneous-elasticity TANK, and ≈ 0 a negligible increment; "—" means no dedicated estimate. "Consistent" and "inconsistent" require an informative estimate with the predicted or opposite sign. "Tested–inconcl." denotes an imprecise or design-sensitive estimate, "not sep. tested" no estimate of the incremental TE effect, and "not identified" a design that cannot isolate that effect. [a] The US TFP response is larger before 1975, as Figure B.2 shows. [b] The cross-country high-minus-low split on the US-identified technology shock, reported in Table B.5, is positive but imprecise. [c] The demand-shock markup rises, against the sticky-price prediction. [d] Consumption inequality is the CEX P90/P10 ratio; joint p-values are 0.018 (TFP), 0.662 (oil), and 0.950 (demand). [e] A markup over unit labor plus non-labor cost also rises after the oil shock; that cost base contains no energy, so the energy wedge is not ruled out. [f] The oil-shock markup gap is positive; its country-bootstrap sup-t simultaneous band excludes zero at both horizons. The time bootstrap gives $p = 0.226$ at $h = 4$ and 0.057 at $h = 8$, and the result depends on the sorter and markup proxy. [g] The markup and inequality outcomes use different monetary shocks. [h] The design cannot separate κ_h from κ_y. [i] A level null cannot identify the incremental TE effect.

C Data, Calibration, and Compact Benchmarks

This appendix reports the full model calibration, the discipline of the distributional block, and the homogeneous-elasticity benchmarks. The replication code and data document series-construction detail beyond what the tables below need, such as raw source files and vintage handling. They are available from the authors and will be deposited with the journal on acceptance.

C.1 Calibration

Table C.1 reports the parameters common to all three economies, which take standard New Keynesian values: $\beta = 0.995$ (a 2% annual real rate), $\sigma = \varphi = 1$, $\alpha = 1/3$, $\zeta = 105.63$, a determinacy-consistent Taylor rule (Debortoli and Galí, 2018, 2024), and the persistence of the four shocks. The adjustment cost reproduces the marginal-cost slope of a Calvo economy with decreasing returns at a price-stickiness parameter of 0.753, corresponding to an average price duration of four quarters.[11] Table C.2 reports the parameters that distinguish TE-TANK, TANK, and RANK: the hand-to-mouth share

[11] The mapping equates the Rotemberg slope $(\varepsilon^* - 1)/\zeta$ with the Calvo slope $\frac{(1-\theta)(1-\beta\theta)}{\theta} \cdot \frac{1-\alpha}{1-\alpha+\alpha\varepsilon^*}$, whose second factor corrects for decreasing returns; at $\varepsilon^* = 3.93$ and $\zeta = 105.63$ it holds at $\theta = 0.753$.

$\lambda = 0.40$, the dividend and transfer parameters δ and τ, and the demand elasticities $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$ from Auer et al. (2024). We impose full dividend taxation, $\delta = 1$, in all three economies. RANK has no hand-to-mouth households, and TANK and RANK set both elasticities to the markup-matched $\varepsilon^* = 3.93$.

Demand elasticities. Following the baseline quantification of Auer et al. (2024), we set $\varepsilon_P = 6.6$ for poor households and $\varepsilon_R = 3.0$ for rich households, corresponding to annual incomes of 20,000 and 120,000 Swiss francs; hence $\varepsilon_P/\varepsilon_R = 2.2$.[12] Their first approach identifies expenditure switching between imported and domestic goods. Applying these estimates to substitution across domestic varieties assumes that the rich–poor gradient in price sensitivity carries over from one margin to the other. The assumption is weaker for their second approach, which estimates substitution between any pair of products within a sector regardless of import status, on a sample that includes domestic goods, and recovers essentially the same gradient as the first; restricting that sample to imports leaves the estimates largely unchanged. Three caveats remain: the data cover groceries, the identifying price variation is an import-price shock in both approaches, and their demand system is nested, with substitution estimated within sectors, so one elasticity applies to all products in a sector. Appendix D.11 reports how the Phillips-curve slope and the steady-state markup move with the elasticity level along a fixed-ratio path.

[12] Auer et al. (2024) estimate the slope of the elasticity in log income at 2.19 (SE 0.55) in their first approach and 1.93 (SE 0.87) in their second, and calibrate it to 2 elasticity points per log income point, which gives a gap of 3.6 points between annual incomes of 20,000 and 120,000 francs. Neither baseline design identifies the level; they choose the elasticity at 120,000 francs to be 3, so the poor elasticity is 6.6. They also recover the level under stronger assumptions and obtain 4.92 and 1.76 at 20,000 francs in the two approaches; the level is therefore much less tightly determined than the gradient, which is the object the baseline uses. Their own sensitivity exercise varies the top-income elasticity from 1.5 to 5 while holding the elasticity differences across households fixed, which implies a ratio between 1.72 and 3.39, computed from the unrounded gap $2 \ln 6$ rather than the rounded value.

TABLE C.1: Common Model Parameters

Description	Symbol	Value
Discount factor	β	0.995
Risk aversion	σ	1.0
Inverse Frisch elasticity	φ	1.0
Capital share	α	1/3
Price adjustment cost	ζ	105.63
Taylor rule on inflation	ϕ_π	1.5
Taylor rule on output	ϕ_y	0.125
AR(1) preference shock	ρ_z	0.5
AR(1) technology shock	ρ_a	0.9
AR(1) monetary shock	ρ_ν	0.5
Cost-push persistence	ρ_u	0.9

TABLE C.2: Model Calibration Differences

Description	Symbol	TE-TANK	TANK	RANK
Share of hand-to-mouth	λ	0.40	0.40	0.00
Dividend tax	δ	1.00	1.00	1.00
Transfer share	τ	0.70	0.70	0.00
Rich elasticity	ϵ_R	3.00	3.933	3.933
Poor elasticity	ϵ_P	6.60	3.933	3.933

Steady states. Table C.3 reports the steady states of the three economies, which share the markup but not the Phillips-curve slope. Rich households consume about 23 percent more than the average household in both TANK specifications, where $\bar{H} = 1.2347$. The markup is $\bar{\mathcal{M}} = 1.34$ in all three economies, because the homogeneous economies use $\varepsilon^* = 3.93$, which reproduces TE-TANK's markup. The composite Phillips-curve slope is 0.0671 in TE-TANK, 0.0701 in TANK, and 0.0833 in RANK. At this benchmark, TANK and TE-TANK differ only in the inequality loading κ_h within the analytical first-order policy block.

TABLE C.3: Steady-State Values

Variable	Symbol	TE-TANK	TANK	RANK
Heterogeneity Index	H	1.2347	1.2347	1.0000
Markup	M	1.3409	1.3409	1.3409
Output	Y	0.7384	0.7384	0.7922
Marginal Cost	MC	0.7458	0.7458	0.7458
NKPC Slope wrt Output	κ_y	0.0833	0.0833	0.0833
NKPC Slope wrt Inequality	κ_h	0.0342	0.0278	0.0278
DIS GE-term	χ_y	0.4746	0.4746	0.0000
Composite NKPC Slope	κ	0.0671	0.0701	0.0833

C.2 Disciplining the Distributional Calibration

Table C.4 shows that every input of the distributional block is set externally, that the steady-state markup is an implied outcome, and that the CEX consumption ratio is held out as a validation moment rather than targeted. The hand-to-mouth share $\lambda = 0.40$ is at the upper end of evidence that places it near one-third (Kaplan et al., 2014). The fiscal parameter $R \equiv \delta(1-\tau) = 0.30$ is externally selected rather than targeted at any moment. By Lemma 1 fiscal policy enters only through R. For any $R < 1$ the sign of the mechanism is unchanged and R scales only its magnitude; Appendix D.11 reports the comparative statics along the fiscal branch from $R = 1$ to $R = 0$. The demand elasticities $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$ come from Auer et al. (2024). These inputs imply $\bar{H} = 1.2347$ and a rich-to-poor consumption ratio of 1.905.

TABLE C.4: Calibration Ledger for the Distributional Block

Object	Value	Status
Hand-to-mouth share	$\lambda = 0.40$	Externally calibrated (hand-to-mouth share)
Demand elasticities	$(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$	Externally calibrated (Auer et al.: gradient estimated, top-income level normalized; import-margin to variety-CES mapping assumed)
Steady-state markup	$\bar{\mathcal{M}} = 1.34$	Implied outcome, untargeted; equals the production-approach formula $(1-\alpha)/s_L$ in model data
Fiscal parameter	$R = 0.30$ ($\delta = 1$, $\tau = 0.70$)	Externally selected, not targeted; enters only through R (Lemma 1; CEX-implied value -0.042 infeasible)
CEX two-type ratio	2.93	Non-targeted validation; model reaches at most 2.759 (at $R = 0$), baseline value 1.905

Notes: The CEX two-type ratio is the mean consumption of the top 60 percent of households relative to the bottom 40 percent. It is computed each quarter from the cross-sectional standard deviation of log consumption in Coibion et al. (2017), which is seasonally adjusted, includes durables, and is not equivalized, under the assumption that log consumption is normally distributed. The quarterly ratios are averaged over 1980Q1–2008Q4 (116 quarters; mean standard deviation 0.64).

The CEX moment exposes a limitation of the calibration. The moments in Coibion et al. (2017) imply a ratio of 2.93 at $\lambda = 0.40$ under a lognormal approximation. The model reaches at most 2.759 at the feasible boundary $R = 0$, so matching 2.93 would require $R = -0.042$. The baseline therefore understates consumption dispersion, and the fiscal block cannot close the gap. The raw CEX ratio includes durables and is not equivalized, but it would have to fall below 1.905 to reverse this conclusion.

The comparison is a diagnostic of overall dispersion, not a test of the model's types. The CEX groups are ranked by consumption, whereas the model's types differ by asset access. Equal group sizes do not imply equal membership, and dispersion within types also raises the CEX ratio. The direction of the resulting bias can be signed only within the model. As steady-state inequality $\bar{H}$ rises along the fiscal branch, the feedback strength Γ, the loading χ_y, and the markup sensitivity η_μ rise, the slope κ falls, and the markup response to a cost-push shock rises (Corollary 5). The dispersion-pure pass-through gap is strictly decreasing in R as well. If the CEX gap reflected too little dispersion between the model's two types, the calibration would understate the demand-composition feedback. The data do not establish that mapping, and even under it the size of the understatement is unknown, because the model cannot reach the CEX ratio.

The model's distributional object is consumption dispersion, so the income and wealth sorters of the empirical sections do not discipline the consumption block. Section 4.2 uses the top-10% income share only as the cross-country sorter, because it is widely available. The top-10% wealth share is a robustness sorter. These objects are not interchangeable.

The steady-state markup $\bar{\mathcal{M}} = 1.34$ is an implied outcome rather than a target because the demand elasticities are calibrated externally to Auer et al. (2024) and because markup estimates measure different objects. The elasticity level is a free calibration parameter in Auer et al. (2024), who sweep it themselves, and here it pins the steady-state markup $\varepsilon/(\varepsilon - 1)$. The sensitivity exercise therefore sweeps three one-dimensional paths through the plane of the mean elasticity and the elasticity gap. The first holds the elasticity gap fixed and varies the level, the second holds the elasticity ratio fixed and varies the level, and the third holds the mean fixed and varies the gap alone. The first is primary because the gap comes from the estimated gradient; the other two show what the level or the ratio would have to do to change the answer.

Table C.5 shows that the dispersion-pure cost-push gap stays positive across the published range of markups, and that how much it moves depends on which elasticity object is held fixed. Each row gives the implied markup, the two elasticities, their ratio and gap, and the dispersion-pure gap in percent. Panel A is the fixed-ratio path, Panel B the fixed-gap path, and Panel C reports three bounds; the starred base-

line row lies on all three paths. Along the fixed-gap path, which holds $\Delta\varepsilon = 3.6$, the dispersion-pure Taylor-rule gap moves from $+1.6$ to $+5.7$ percent around the baseline $+3.1$ percent. That range is taken over the implied markup span 1.21–1.61, the range of published estimates (Edmond et al., 2023; Burstein et al., 2025), along which the elasticity ratio drifts from 1.75 to 3.01, away from the estimated gradient. Holding the ratio at $\varepsilon_P/\varepsilon_R = 2.2$ instead narrows the range to $+2.4$ to $+3.8$ percent over the same markup span; the ratio depends on the chosen level, the gap does not. The cost-push result is therefore not tied to one markup level, but broader reparameterizations produce more variation.

TABLE C.5: Sensitivity of the Dispersion-Pure Cost-Push Gap along Three Elasticity Paths

	$\bar{\mathcal{M}}$	ε_R	ε_P	$\varepsilon_P/\varepsilon_R$	$\Delta\varepsilon$	Gap (%)
Panel A: fixed-ratio path ($\varepsilon_P/\varepsilon_R = 2.2$ held, level varies)						
DLEU 1980 aggregate / French firm median	1.21	4.34	9.54	2.20	5.20	+2.36
EMX cost-weighted aggregate	1.25	3.78	8.31	2.20	4.53	+2.63
Baseline (implied, untargeted)*	1.34	3.00	6.60	2.20	3.60	+3.11
French firm-level mean	1.39	2.73	6.01	2.20	3.28	+3.30
DLEU 2016 aggregate / EMX sales-weighted	1.61	2.06	4.53	2.20	2.47	+3.82
Panel B: fixed-gap path ($\Delta\varepsilon = 3.6$ held, level varies)						
Low end of markup span	1.21	4.77	8.37	1.75	3.60	+1.62
High end of markup span	1.61	1.79	5.39	3.01	3.60	+5.67
Panel C: bounds						
(a) 10% gap, fixed-mean path	1.45	1.23	9.26	7.54	8.03	+10.00
(b) Ceiling at ratio 2.2	1.93	1.65	3.63	2.20	1.98	+3.99
(c) Supremum over λ, $R^\dagger$	1.31	3.00	6.60	2.20	3.60	+7.03

Notes: Each row reports the dispersion-pure cost-push gap of TE-TANK against its own markup-matched TANK(ε^*), $\Psi_\pi/\Psi_\pi^* - 1$ in percent, at the baseline $\lambda = 0.40$ and $R = 0.30$ unless stated otherwise. Panel A holds the elasticity ratio at its calibrated $\varepsilon_P/\varepsilon_R = 2.2$ and moves the level, so the implied markup varies and $\Delta\varepsilon$ adjusts with it. Panel B holds the elasticity gap at its calibrated $\Delta\varepsilon = 3.6$ and moves the level, so the ratio instead drifts from 1.75 to 3.01, away from the estimated gradient. The markup span 1.21–1.61 common to both panels is the range of published aggregate estimates (Edmond et al., 2023; Burstein et al., 2025); DLEU and EMX denote De Loecker et al. (2020) and Edmond et al. (2023). The starred row ($*$) is the baseline calibration $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$, whose markup is an untargeted outcome; all three paths intersect there, so it is common to both panels and to row (a). Panel C reports three bounds. Row (a) holds the *mean* elasticity at its calibrated $\bar{\varepsilon} = 4.44$ and varies $\Delta\varepsilon$ alone: it is a third path, and the parameterization it reports lies on neither the fixed-ratio path of Panel A nor the fixed-gap path of Panel B. Row (b) is the maximum over all elasticity levels at the fixed ratio 2.2, and it is attained at $\bar{\mathcal{M}} = 1.93$, above the published markup span; within that span the largest attainable value is the $+3.82$ of the last row of Panel A. The daggered row ($\dagger$) maximizes over the distributional parameters at the baseline elasticities: the gap is strictly decreasing in R, so $R = 0.0$ is attained, but it is strictly increasing in λ and is bounded only by the strict admissibility constraint $\chi_y < 1$. The entry is therefore a supremum, approached as $\lambda \to 0.656$ and $\chi_y \to 1.000$ and attained at no admissible point; every admissible λ delivers strictly less than $+7.03$. At $\alpha = 1/3$ and $\sigma = \varphi = 1$ that boundary is exact in rationals: $\chi_y = 1$ forces $\Gamma = 1/2$, hence $(1-\lambda)\bar{H} = 2/3$ independently of λ, so $\varepsilon_{\text{eff}} = 21/5$, $\bar{\mathcal{M}} = 21/16 = 1.3125$, and $\lambda^* = 21/32 = 0.65625$; the numerical root agrees with the closed form to 2.2×10^{-16}.

Panel C of Table C.5 reports how far the gap can move away from the baseline. Row (a) runs along the fixed-mean path, which holds the mean elasticity at its calibrated $\bar{\varepsilon} = 4.44$ and varies $\Delta\varepsilon$ alone. At the baseline λ and R, a 10 percent dispersion-pure gap requires $\Delta\varepsilon \approx 8.0$, or $(\varepsilon_R, \varepsilon_P) = (1.23, 9.26)$ and a ratio of 7.5. The implied markup there is 1.45, inside the 1.21 to 1.61 span, so what rules out this parameterization is not the markup but $\varepsilon_R = 1.23$, near-unit-elastic demand from rich households, and a ratio more than three times the calibrated one. This parameterization also violates the maintained pricing assumption that Section 2 states. Condition (D.1) fails there, and although the local condition of Appendix D.5.2 holds at the symmetric price, profit has a second, higher peak at a price high enough to sell almost only to rich households, so the symmetric price is not each firm's profit maximum.

Rows (b) and (c) bound what the elasticity level and the distributional parameters can add. Holding $\varepsilon_P/\varepsilon_R = 2.2$, the gap cannot exceed 4.0 percent at any elasticity level, but that ceiling is reached at $\bar{\mathcal{M}} = 1.93$, above the published markup span, so the binding value within the span is the 3.8 percent in the last row of Panel A. The distributional parameters cannot substitute either. At $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$ the gap is strictly decreasing in R and strictly increasing in λ. Over $\lambda \in (0, 1)$ and $R \in [0, 1]$, subject to the strict admissibility constraint $\chi_y < 1$, it therefore has a supremum of 7.0 percent, approached at $R = 0.0$ as $\lambda \to 0.656$ and $\chi_y \to 1$ but attained at no admissible point.

The estimated rich–poor gradient therefore fixes the sign and order of magnitude of the mechanism. The elasticity level and the distributional block scale it within the bounds of Table C.5: between +1.6 and +5.7 percent across the published markup span at the estimated gap, and below +7.0 percent over all admissible (λ, R) at the baseline elasticities.

C.3 Homogeneous-Elasticity Benchmarks

Table C.6 shows that the dispersion-pure gap against the markup-matched benchmark does not depend on how the shock is normalized, while gaps against benchmarks with a different steady-state markup do. The table has three panels. Panel A compares TE-TANK with four homogeneous TANK benchmarks; for each it reports the benchmark's own κ and χ_y and the TE-TANK/TANK ratios of Taylor-rule pass-through and of the output contraction. Panel B splits the gap against the population-mean-preserving benchmark into a dispersion-pure and a markup-level component. Panel C reports impact-inflation ratios under three shock normalizations.

The primary homogeneous benchmark is TANK($\varepsilon^* = 3.93$), whose steady-state markup equals TE-TANK's $\bar{\mathcal{M}} = 1.34$. The two economies then share $\bar{H}$, $\bar{\mathcal{M}}$, Γ, and χ_y

and differ only in κ_h within the first-order policy block. In the starred row of Panel A, the pass-through and output-contraction ratios are both 1.031, so the dispersion-pure gap is 3.1 percent.

The population-mean-preserving benchmark $\bar{\varepsilon} = 4.44$ also includes the markup-level effect of introducing elasticity dispersion. Relative to this benchmark, pass-through rises by 15.74 percent and the output contraction by 14.53 percent. Panel B separates each total into the dispersion-pure component through ε^*, 3.11 percent for pass-through and 3.11 percent for output, and the remaining markup-level component, 12.25 and 11.08 percent.

Panel C checks whether the comparison depends on what is held fixed across economies. For an equal normalized NKPC wedge, the impact-inflation ratio is 1.031 against the markup-matched benchmark. An equal underlying real marginal-cost shifter instead enters as $u_t = \delta_2 \hat{u}_t$, while an equal log desired-markup shifter enters as $u_t = \delta_2 q_t$; the two structural-shifter columns are therefore identical. Because markup matching also matches δ_2, all three ratios remain 1.031 in the primary row. Against population-mean-preserving TANK, by contrast, the equal-wedge ratio is 1.157 while either structural-shifter ratio is 0.987. The primary comparison is thus invariant to shock normalization; alternatives that change the steady-state markup are not.

TABLE C.6: Benchmark Dependence of the TE-TANK/TANK Comparison

Panel A: Slopes and TE-TANK/TANK ratios by homogeneous benchmark

	κ	χ_y	TE-TANK / TANK ratio	
TANK benchmark $\bar{\varepsilon}$			Pass-through	Output
3.933 (ε^*, markup-matched)*	0.0701	0.4746	1.031	1.031
4.44 (population-mean-preserving)	0.0816	0.4930	1.157	1.145
4.00 (standard TANK)	0.0716	0.4773	1.048	1.046
4.80 (pair mean)	0.0898	0.5038	1.247	1.227
TE-TANK $(\varepsilon_R, \varepsilon_P)=(3, 6.6)$	0.0671	0.4746	1.000	1.000
RANK (ε^*, ref.)	0.0833	0.0000	–	–

Panel B: Decomposition of the TE-TANK/TANK(4.44) gap: dispersion-pure × markup-level

Object	Total TE / TANK(4.44)	Dispersion-pure TE / interm.	Markup-level interm. / TANK(4.44)
Taylor pass-through Ψ_π	+15.74%	+3.11%	+12.25%
Output contraction $\lvert\Psi_y\rvert$	+14.53%	+3.11%	+11.08%

Panel C: TE-TANK/TANK impact-inflation ratio by shock normalization

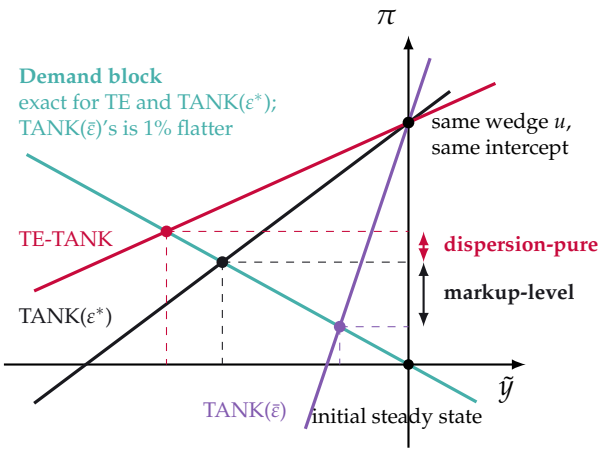
TANK benchmark	Equal NKPC wedge	Equal real-MC shifter	Equal desired-markup shifter
3.933 (ε^*, markup-matched)*	1.031	1.031	1.031
4.44 (population-mean-preserving)	1.157	0.987	0.987
4.00 (standard TANK)	1.048	1.024	1.024
4.80 (pair mean)	1.247	0.963	0.963

Notes: Each upper-panel row is a homogeneous-elasticity TANK benchmark. The starred markup-matched row is the main-text benchmark. The next rows preserve the population mean, use the standard TANK elasticity, and use the unweighted pair mean. The κ and χ_y columns report each benchmark's own coefficients; the last two columns report TE-TANK relative to that benchmark. TE-TANK and markup-matched RANK are shown for reference. The second panel decomposes the total against the population-mean-preserving benchmark through the exactly markup-matched ε^* intermediate. The final panel reports impact-inflation ratios for equal normalized NKPC wedges, equal real marginal-cost shifters, and equal log desired-markup shifters. The last two coincide because both structural shifters enter the pricing gap with loading δ_2.

Figure C.1 draws Panel B as two successive rotations of the Phillips curve about a common intercept. It uses the space of Figure 1: the horizontal axis is the output gap $\tilde{y}$ and the vertical axis is inflation π. The downward-sloping teal line is the demand block. The three upward-sloping lines are the Phillips curves of TANK($\bar{\varepsilon}$) in purple, TANK(ε^*) in black, and TE-TANK in red, from steepest to flattest. All three economies face the same wedge, so their Phillips curves share an intercept, and the decomposition is a two-step fan about it. The two arrows next to the inflation axis mark the steps between the three equilibria: the markup-level step from the population-mean benchmark to the markup-matched one, and the dispersion-pure step from there to TE-TANK. The slopes are exaggerated and the spacing is not to scale, so the figure shows the order of the steps, not their size.

The demand block is almost stationary across the two homogeneous benchmarks. Almost all of the 12.2 percent markup-level step therefore comes from the flatter Phillips curve, since κ falls with the steady-state markup through δ_2. The demand block contributes only a small part, and the two effects compound rather than add.

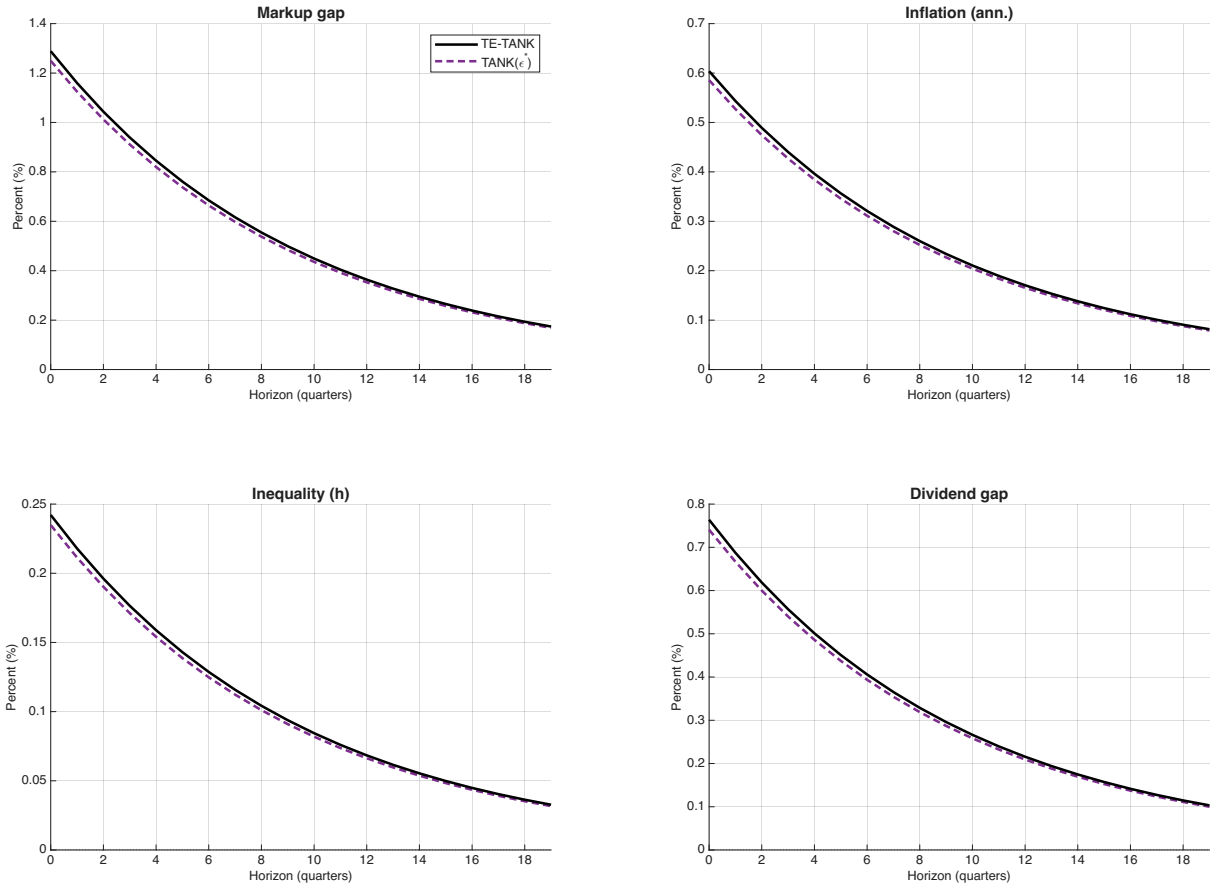
FIGURE C.1: Decomposing the Population-Mean Gap: Markup Level and Elasticity Dispersion



Notes: Panel B of Table C.6 drawn in the space of Figure 1; stylized, and not to scale in either dimension. Reading up the inflation axis, the markup-level step is TANK($\bar{\varepsilon}$) to TANK(ε^*) and the dispersion-pure step is TANK(ε^*) to TE-TANK, and the two steps compound as growth factors, $1 +$ total $\approx$ $(1 +$ markup-level step$)(1 +$ dispersion-pure step$)$, with each change expressed as a fraction. For Ψ_π the total is 15.74 percent, the markup-level step 12.25 percent, and the dispersion-pure step 3.11 percent. For the output contraction the same three numbers are 14.53, 11.08, and 3.11 percent, so one pair of numbers does not describe both axes. The demand locus is fixed only for the markup-matched pair: TE-TANK and TANK(ε^*) share $\chi_y = 0.4746$ exactly, so they share $\tilde{\sigma}$ and Θ. TANK($\bar{\varepsilon}$) has $\chi_y = 0.4930$ and hence a locus 1 percent flatter; one line is drawn because that difference is smaller than the line width, at the cost of placing its equilibrium with a 0.76 percent error. The slopes are exaggerated. **The shared intercept is a normalization.** Because δ_2 differs across $\bar{\varepsilon}$ and ε^*, the common wedge holds under the equal-NKPC-wedge convention only; under the marginal-cost and desired-markup conventions the TANK($\bar{\varepsilon}$) intercept is 17 percent higher and the population-mean gap reverses to -1.32 percent (Panel C). The dispersion-pure step is invariant to all three, and remains the paper's headline. RANK is omitted: with $\chi_y = 0$ it does not share the locus even approximately.

Figure C.2 shows the same markup-matched comparison as impulse responses to the cost-push shock that Section 3.3 studies. Its four panels plot the markup gap, annualized inflation, inequality $\hat{h}_t$, and the dividend gap, in percent, against the horizon in quarters. The solid black line is TE-TANK and the dashed purple line is TANK(ε^*). Because the two economies differ only in κ_h, the vertical distance between the two lines is the dispersion-pure effect of the elasticity gap. The TE-TANK line lies slightly above the TANK line in every panel, and the distance is largest on impact, where TE-TANK inflation is 1.031 times its TANK value, the equal-wedge ratio in Panel C of Table C.6.

FIGURE C.2: Cost-Push Shock: TE-TANK vs. TANK(ε^*)



Notes: Impulse responses to a one-standard-deviation cost-push shock, the AR(1) wedge u_t in equation (12) ($\rho_u = 0.9$, $\sigma_u = 0.05$). The **solid black line** is TE-TANK ($\varepsilon_R = 3.0$, $\varepsilon_P = 6.6$); the **dashed purple line** is TANK($\varepsilon^* = 3.93$), the homogeneous economy at the elasticity that reproduces TE-TANK's steady-state markup. Panels as in Figure 2, with the dividend gap in place of the output gap. Section 3.3 gives the units. The sequential three-band decomposition of the same response, from a RANK baseline through hand-to-mouth aggregation to the elasticity gap, is Figure 2.

D Model Derivations and Core Robustness

This appendix derives the model's equilibrium objects and proves the results that Section 2 states. It also reports the targeted comparative statics and the transmission of the monetary policy shock.

D.1 Maintained Assumptions

The results of Section 2 hold under restrictions maintained throughout and not repeated in individual statements. Preferences and technology satisfy $\sigma > 0$, $\varphi \geq 0$, $\alpha \in [0,1)$, and $\zeta > 0$. The population split is interior, $\lambda \in (0,1)$, and the elasticities satisfy $\varepsilon_P \geq \varepsilon_R > 1$; the weak inequality lets the same statements cover the

homogeneous-elasticity benchmarks, for which $\varepsilon_P = \varepsilon_R$. The fiscal instruments lie in the unit square, $(\delta, \tau) \in [0,1]^2$, so that $R = \delta(1-\tau) \in [0,1]$. Hours are common across types, $N_t^P = N_t^R \equiv N_t$, at a wage that satisfies the Ricardian intratemporal condition, and all steady-state objects are evaluated at the admissible branch established in Proposition 1. Three of these bind. At $\lambda = 0$ the demand-composition feedback vanishes for every R; $\varepsilon_R > 1$ is what keeps the desired markup finite; and the common-hours closure rations hand-to-mouth households below the hours they would choose at the common wage, as Appendix D.2 makes precise.

Finally, at every steady state considered, the symmetric price is each firm's global profit maximizer when all other firms charge it, so that the root of the pricing first-order condition is an equilibrium. This is not automatic: an elasticity gap makes demand more convex in the price, because a higher price loses mostly the elastic poor customers. A sufficient condition in primitives bounds the elasticity gap relative to the rich elasticity,

$$(\varepsilon_P - \varepsilon_R)^2 < \frac{4\,\varepsilon_R(\varepsilon_R - 1)(1 - \alpha + \alpha\varepsilon_R)}{1-\alpha}. \tag{D.1}$$

Under (D.1) each firm's profit is single-peaked in its own price at any composition of demand, so the pricing first-order condition selects the profit maximum. At the baseline calibration the two sides of (D.1) are 12.96 and 60. Because (D.1) is sufficient but not necessary, we also verify global optimality numerically for every calibration reported in the paper; it holds at all of them except one extreme sensitivity point, which Appendix C.2 rules out. Appendix D.5.2 derives the condition and reports the check.

D.2 Household Problem

This subsection states the fiscal transfer rule, the household budget constraints, and the Ricardian first-order conditions that Section 2 summarizes; they are the starting point for the heterogeneity-index and Euler-equation derivations that follow.

Fiscal policy. The government collects a fraction δ of firm profits and redistributes them according to:

$$\begin{aligned} T_t^R &= \left(1 + \frac{\tau\lambda}{1-\lambda}\right)\delta D_t, \\ T_t^P &= (1-\tau)\,\delta D_t, \end{aligned} \tag{D.2}$$

where $\tau \in [0,1]$ governs the tilt of transfers toward rich households ($\tau = 0$ is the most progressive scheme, equal per-capita transfers). The government budget balances each period: $(1-\lambda)T_t^R + \lambda T_t^P = \delta D_t$.

Rich households. Rich households maximize expected lifetime utility (1) subject to a budget constraint:

$$C_t^R + \frac{B_t^R}{P_t^R} + Q_t F_t^R = \frac{B_{t-1}^R(1+i_{t-1})}{P_t^R} + \frac{W_t}{P_t^R} N_t^R + \big((1-\delta)D_t + Q_t\big) F_{t-1}^R + T_t^R, \tag{D.3}$$

where B_t^R are nominal bond holdings, F_t^R are equity shares, Q_t is the share price, D_t are real dividends, W_t is the nominal wage, and T_t^R are real transfers. A fraction δ of firm profits is collected by the government and redistributed as described above.

Rich household's optimization. The rich household maximizes $\mathbb{E}_0 \sum_t \beta^t \big[\frac{(C_t^R)^{1-\sigma}}{1-\sigma} Z_t - \frac{(N_t^R)^{1+\varphi}}{1+\varphi} Z_t\big]$ subject to the budget constraint (D.3) (multiplier ξ_t), with consumption expenditure $\frac{1}{P_t^R}\int_0^1 P_t(i) C_t^R(i)\, di$, labor income $\frac{W_t}{P_t^R} N_t^R$, bond and equity holdings, and transfers T_t^R. The first-order conditions of this standard Lagrangian problem, after using the CES demand aggregator of Appendix D.3 and the price-index identity to simplify the consumption FOC, are the marginal utility of wealth,

$$\xi_t = (C_t^R)^{-\sigma} Z_t, \tag{D.4}$$

the intratemporal labor-supply condition,

$$\frac{W_t}{P_t^R} = (C_t^R)^{\sigma} (N_t^R)^{\varphi}, \tag{D.5}$$

the consumption Euler equation,

$$1 = \mathbb{E}_t \left[\Lambda_{t+1,t} \frac{1+i_t}{\Pi_{t+1}^R} \right], \qquad \Lambda_{t+1,t} \equiv \beta \left(\frac{C_{t+1}^R}{C_t^R} \right)^{-\sigma} \frac{Z_{t+1}}{Z_t}, \quad \Pi_{t+1}^R \equiv \frac{P_{t+1}^R}{P_t^R}, \tag{D.6}$$

and the equity Euler equation,

$$Q_t = \mathbb{E}_t \left[\Lambda_{t+1,t} \left((1-\delta) D_{t+1} + Q_{t+1} \right) \right].$$

These are the Euler equation, labor-supply condition, and equity-pricing equation that Section 2 uses.

Poor households. Poor households cannot save or borrow, so they spend all labor income plus transfers each period:

$$C_t^P = \frac{W_t}{P_t^P} N_t^P + T_t^P. \tag{D.7}$$

Throughout we impose a common equilibrium employment level, $N_t^P = N_t^R \equiv N_t$, and price labor off the Ricardian condition (D.5); equation (D.7) then prices the common N_t at W_t/P_t^P and residually determines C_t^P. Free choice of hours at the common wage would instead imply $W_t/P_t^P = (C_t^P)^\sigma (N_t^P)^\varphi$ and hence $N_t^P > N_t^R$ whenever $C_t^P < C_t^R$, so imposing common hours rations hand-to-mouth households below their desired hours.

D.3 CES Aggregation

Each household type $h \in \{R, P\}$ minimizes expenditure $\int_0^1 P_t(i) C_t^h(i)\, di$ subject to the CES aggregator $C_t^h = \left(\int_0^1 [C_t^h(i)]^{1-1/\varepsilon_h} di\right)^{\varepsilon_h/(\varepsilon_h-1)}$. This is the standard Dixit–Stiglitz expenditure-minimization problem; standard steps (first-order conditions across varieties, substitution into the aggregator, and the price-index definition $P_t^h \equiv (\int_0^1 P_t(j)^{1-\varepsilon_h} dj)^{1/(1-\varepsilon_h)}$) yield the CES demand function

$$C_t^h(i) = \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} C_t^h. \tag{D.8}$$

From variety demand to aggregate output. Price adjustment absorbs goods. We adopt the standard Rotemberg convention that adjustment costs are paid in the same composites of varieties that households consume. Aggregate adjustment resources $X_t \equiv k_t Y_t$, with $Y_t \equiv \sum_h s_h Y_t^h$ and $k_t \equiv \frac{\zeta}{2} \int_0^1 \left(P_t(j)/P_{t-1}(j) - 1\right)^2 dj$, are the sum of the firm-level costs $\frac{\zeta}{2}\left(P_t(j)/P_{t-1}(j) - 1\right)^2 Y_t$ in the firm's problem of Appendix D.5. They are purchased as type-specific bundles in proportion to each type's output, $X_t^h = k_t Y_t^h$, so that $\sum_h s_h X_t^h = X_t$, and each bundle is spread across varieties with the same CES weights as type h's consumption, $X_t^h(i) = (P_t(i)/P_t^h)^{-\varepsilon_h} X_t^h$. Output of variety i absorbed by type h, $Y_t^h(i) = C_t^h(i) + X_t^h(i)$, is therefore

$$Y_t^h(i) = \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h, \qquad Y_t^h \equiv C_t^h + X_t^h,$$

for any configuration of prices, not only at symmetric prices. Because each firm has measure zero, its own price leaves k_t, X_t^h, Y_t^h, and P_t^h unchanged, so the adjustment cost of firm i enters its demand only through aggregates it takes as given. The same convention gives $C_t^h = (1 - k_t) Y_t^h$, which in the symmetric equilibrium is (D.10) below, and makes $C_t^R/C_t = Y_t^R/Y_t$ exact. Aggregate output of variety i, summed across

household types with population share s_h, is

$$Y_t(i) = \sum_{h \in \{R,P\}} s_h Y_t^h(i) = \sum_{h \in \{R,P\}} s_h \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} Y_t^h. \tag{D.9}$$

D.4 Heterogeneity Index

We derive the heterogeneity index $H_t^h = C_t^h / C_t = Y_t^h / Y_t$ for rich households, first in levels and then in log-deviations. By construction, consumption equals output minus price adjustment costs for each good i:

$$C_t^h(i) = Y_t^h(i) - X_t^h(i).$$

In terms of the type-h composites,

$$C_t^h = Y_t^h - X_t^h.$$

By the convention of Appendix D.3, $X_t^h = k_t Y_t^h$, and in the symmetric equilibrium $k_t = \frac{\zeta}{2}(\Pi_t - 1)^2$, so

$$X_t^h = Y_t^h \left[\frac{\zeta}{2} (\Pi_t - 1)^2 \right],$$

which allows us to write consumption as

$$C_t^h = Y_t^h \left[1 - \frac{\zeta}{2} (\Pi_t - 1)^2 \right] = Y_t^h \Delta^p(\Pi_t), \tag{D.10}$$

where $\Delta^p(\Pi_t) = 1 - \frac{\zeta}{2}(\Pi_t - 1)^2$ captures the resource loss from price adjustment.

Starting from the rich household's budget constraint (D.3) and applying the CES aggregator for consumption, then using that bonds are in zero net supply and rich households' equity holdings are constant at $F_t^R = 1/(1-\lambda)$ (total shares equal one and the rich population share $1 - \lambda$ is constant), the budget constraint simplifies to $C_t^R = \frac{W_t}{P_t^R} N_t^R + (1 - \delta) D_t F_{t-1}^R + T_t^R$. Substituting C_t^R from (D.10), normalizing by aggregate output Y_t, and using the real wage $\frac{W_t}{P_t^R} = \frac{(1-\alpha) A_t}{\mathcal{M}_t} N_t^{-\alpha}$, labor-market clearing $N_t = N_t^R = N_t^P$, the transfer rule $T_t^R = \left(1 + \frac{\tau\lambda}{1-\lambda}\right) \delta D_t$, and $Y_t = A_t N_t^{1-\alpha}$ gives, after combining the dividend terms,

$$H_t = \frac{1}{\Delta^p(\Pi_t)} \left[\frac{(1-\alpha)}{\mathcal{M}_t} + \frac{D_t}{Y_t} \left(\frac{1 + \delta\lambda(\tau - 1)}{1 - \lambda} \right) \right]. \tag{D.11}$$

The aggregate accounting identity $C_t = Y_t \Delta^p(\Pi_t) = W_t N_t / P_t + D_t$ states that output net of adjustment costs equals real labor income plus dividends; together with

$W_t N_t / P_t = \frac{1-\alpha}{\mathcal{M}_t} Y_t$, this gives $D_t/Y_t = \Delta^p(\Pi_t) - \frac{1-\alpha}{\mathcal{M}_t}$. Substituting into (D.11) and collecting terms yields

$$H_t = \frac{1+\delta\lambda(\tau-1)}{1-\lambda} - \frac{1}{\Delta^p(\Pi_t)} \frac{(1-\alpha)}{\mathcal{M}_t} \left(\frac{\lambda(1+\delta(\tau-1))}{1-\lambda} \right).$$

In the steady state, where $\Delta^p(\Pi) = 1$, this simplifies to

$$H = \frac{1+\delta\lambda(\tau-1)}{1-\lambda} - \frac{(1-\alpha)}{\mathcal{M}} \left(\frac{\lambda(1+\delta(\tau-1))}{1-\lambda} \right).$$

Using $R = \delta(1-\tau)$ and the composite parameters defined in Section 2.4 gives equation (8). In contrast to the standard TANK model, the markup also depends on heterogeneity:

$$\mathcal{M} = \frac{H(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P}{H(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1}.$$

This is equation (9). Substituting it into (8) gives the quadratic equation (D.22). Appendix D.10.2 characterizes its economically admissible root.

Log-Linearization of the Heterogeneity Index. Write $H_t = C_1 - C_2/[\Delta^p(\Pi_t)\mathcal{M}_t]$ with constants $C_1 = \frac{1+\delta\lambda(\tau-1)}{1-\lambda}$ and $C_2 = (1-\alpha)\frac{\lambda(1+\delta(\tau-1))}{1-\lambda}$. Since $1/\Delta^p(\Pi_t) \approx 1$ to first order, H_t varies only through $\mathcal{M}_t$; a first-order Taylor expansion around the steady state, using $\partial H_t/\partial \mathcal{M}_t|_{ss} = C_2/\mathcal{M}^2$, gives the proportional deviation $\hat{h}_t \equiv (H_t - H)/H \approx (C_2/\mathcal{M}H)\hat{\mu}_t$, i.e.,

$$\hat{h}_t = \frac{(1-\alpha)}{\mathcal{M}H} \left(\frac{\lambda(1+\delta(\tau-1))}{1-\lambda} \right) \hat{\mu}_t \equiv \Gamma\, \hat{\mu}_t, \qquad \Gamma \equiv \frac{(1-\alpha)}{\mathcal{M}H} \left(\frac{\lambda(1+\delta(\tau-1))}{1-\lambda} \right).$$

Substituting $\hat{\mu}_t = -\sigma\hat{h}_t - \left(\frac{\varphi+\alpha}{1-\alpha} + \sigma\right)\tilde{y}_t$ from (D.26), collecting $\hat{h}_t$ on the left-hand side, and solving yields

$$\hat{h}_t = -\underbrace{\frac{\Gamma\left(\frac{\varphi+\alpha}{1-\alpha} + \sigma\right)}{1+\Gamma\sigma}}_{\chi_y} \tilde{y}_t,$$

where χ_y is the composite elasticity of heterogeneity with respect to the output gap. The denominator $1 + \Gamma\sigma > 1$ reflects a dampening feedback: higher heterogeneity raises Ricardian consumption relative to aggregate consumption, which through the intratemporal condition raises the real wage and compresses the markup, partially offsetting the original change in heterogeneity.

When $\delta = 1$ (all profits are redistributed) and $\tau = 0$ (equal per-capita transfers), the term $\lambda(1+\delta(\tau-1))/(1-\lambda) = 0$, and heterogeneity becomes acyclical in the log-linearized system.

D.5 Firm Problem and Flexible-Price Markup

This subsection derives the firm's pricing first-order condition, from which both the Phillips curve and the flexible-price markup follow. The firm chooses $P_t(i)$ to maximize discounted real profits $\max_{P_t(i)} \mathbb{E}_t \sum_{s=t}^{\infty} \frac{\Lambda_{s,t}}{P_s} \left[P_s(i) Y_s(i) - P_s \mathcal{C}(Y_s(i)) - \frac{\zeta}{2} \left(\frac{P_s(i)}{P_{s-1}(i)} - 1 \right)^2 Y_s P_s \right]$, where $\Lambda_{s,t} \equiv \beta^{s-t} U'_{c,s} / U'_{c,t}$ is the stochastic discount factor, subject to the demand constraint (D.9), $Y_s(i) = \sum_h s_h (P_s(i)/P_s^h)^{-\varepsilon_h} Y_s^h$, which holds at any price of firm i under the convention of Appendix D.3. The firm takes P_s, P_s^h, Y_s, and Y_s^h, and hence aggregate adjustment resources, as given. It values its adjustment cost at P_s, which equals the cost of the adjustment composite because $P_s^R = P_s^P = P_s$ in the symmetric equilibrium. Differentiate with respect to $P_t(i)$: the revenue and cost terms enter at $s = t$, and the Rotemberg adjustment-cost term enters at both $s = t$ (current-period cost) and $s = t+1$ (next period's cost, since $P_t(i)$ enters $P_{t+1}(i)/P_t(i)$). Evaluating each term under the symmetric equilibrium $P_t(i) = P_t$ and denoting real marginal cost $MC_t \equiv \mathcal{C}'(Y_t(i))$ gives the first-order condition

$$\begin{aligned} &\sum_h s_h \frac{Y_t^h}{P_t} (1 - \varepsilon_h) \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} + MC_t \sum_h s_h \frac{\varepsilon_h}{P_t(i)} \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} Y_t^h \\ &- \zeta (\Pi_t - 1) \Pi_t \frac{Y_t}{P_t} + \mathbb{E}_t \left[\Lambda_{t+1,t} \zeta (\Pi_{t+1} - 1) \Pi_{t+1} \frac{Y_{t+1}}{P_t} \right] = 0. \end{aligned} \tag{D.12}$$

Imposing the symmetric equilibrium $P_t(i) = P_t^h = P_t$, dividing by $\zeta Y_t / P_t$, and defining $H_t^h \equiv Y_t^h / Y_t$ as type h's output relative to the aggregate average gives

$$\sum_h \frac{s_h}{\zeta} H_t^h (1 - \varepsilon_h) + MC_t \sum_h \frac{s_h}{\zeta} \varepsilon_h H_t^h + \mathbb{E}_t \left[\Lambda_{t+1,t} (\Pi_{t+1} - 1) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right] = (\Pi_t - 1) \Pi_t. \tag{D.13}$$

At the zero-inflation steady state ($\Pi_{t+1} = \Pi_t = \Pi = 1$) the forward-looking term and the right-hand side vanish, giving steady-state marginal cost

$$MC = \frac{\sum_h s_h H^h (\varepsilon_h - 1)}{\sum_h s_h \varepsilon_h H^h} = \frac{\sum_h s_h H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h} \right)}{\sum_h s_h \varepsilon_h H^h}. \tag{D.14}$$

D.5.1 Flexible-Price Markup

Flexible prices. With $\zeta = 0$, the symmetric-equilibrium first-order condition (D.12) reduces to $\sum_h s_h (1 - \varepsilon_h) H_t^h - \frac{MC_t^{nom}}{P_t} \sum_h s_h (-\varepsilon_h) H_t^h = 0$, where $MC_t^{nom} \equiv P_t MC_t$ is nominal marginal cost, so real marginal cost is $MC_t^{nom} / P_t = \sum_h s_h (\varepsilon_h - 1) H_t^h / \sum_h s_h \varepsilon_h H_t^h$,

and the flexible-price markup, its inverse, is

$$\mathcal{M}_t^n \equiv \frac{P_t}{MC_t^{nom}} = \frac{\sum_h s_h \varepsilon_h H_t^h}{\sum_h s_h \left(\varepsilon_h - 1\right) \cdot H_t^h}. \tag{D.15}$$

Relating heterogeneity shares. Goods-market clearing aggregates the output sold to each type with its population share $s_h \in \{1-\lambda, \lambda\}$:

$$Y_t = \sum_h s_h Y_t^h = (1-\lambda) Y_t^R + \lambda Y_t^P. \tag{D.16}$$

Dividing through by aggregate output Y_t and using the relative demand indices $H_t^h \equiv Y_t^h / Y_t$ defined above, each type's demand measured against the economy-wide average, turns this into the adding-up condition

$$1 = (1-\lambda) H_t^R + \lambda H_t^P. \tag{D.17}$$

Solving for the poor household's share then expresses it in terms of the rich household's:

$$H_t^P = \frac{1 - (1-\lambda) H_t^R}{\lambda}. \tag{D.18}$$

Effective elasticity. The demand curve (4) aggregates two CES schedules with different exponents, so the price elasticity of total demand for variety i is not a primitive parameter but an endogenous average of ε_R and ε_P. Define it as the (negative) elasticity of $Y_t(i)$ with respect to the firm's own price. Since each CES term satisfies $\partial (P_t(i)/P_t^h)^{-\varepsilon_h} / \partial \ln P_t(i) = -\varepsilon_h (P_t(i)/P_t^h)^{-\varepsilon_h}$, differentiating $Y_t(i) = \sum_h s_h (P_t(i)/P_t^h)^{-\varepsilon_h} Y_t^h$ gives

$$-\frac{\partial \ln Y_t(i)}{\partial \ln P_t(i)} = \frac{\sum_h s_h \, \varepsilon_h \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h}{\sum_h s_h \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h}.$$

Evaluating at the symmetric allocation ($P_t(i) = P_t^R = P_t^P = P_t$), every price ratio equals one and the denominator becomes $\sum_h s_h Y_t^h = Y_t$ by goods-market clearing, so

$$\varepsilon_{\text{eff},t} \equiv -\left.\frac{\partial \ln Y_t(i)}{\partial \ln P_t(i)}\right|_{P_t(i)=P_t^h=P_t} = \sum_h s_h \, \varepsilon_h \, \frac{Y_t^h}{Y_t} = \sum_h s_h \, \varepsilon_h H_t^h = (1-\lambda) H_t \, \varepsilon_R + \lambda H_t^P \varepsilon_P,$$

where the last equality uses $s_h \in \{1-\lambda, \lambda\}$ and $H_t^R = H_t$. Each type's elasticity is therefore weighted by its demand share $s_h H_t^h = s_h Y_t^h / Y_t$, not by its population share s_h alone. The demand share equals the expenditure share, since both types pay the same price in the symmetric allocation. Substituting the market-clearing identity

(D.18) eliminates H_t^P:

$$\begin{aligned}\varepsilon_{\text{eff},t} &= (1-\lambda)H_t\,\varepsilon_R + \lambda\,\frac{1-(1-\lambda)H_t}{\lambda}\,\varepsilon_P \\ &= (1-\lambda)H_t\,\varepsilon_R + \varepsilon_P - (1-\lambda)H_t\,\varepsilon_P \\ &= H_t(1-\lambda)\,(\varepsilon_R-\varepsilon_P) + \varepsilon_P. \end{aligned} \tag{D.19}$$

Markup in terms of the effective elasticity. The flexible-price markup (D.15) is a function of the same share-weighted sums: its numerator is $\sum_h s_h\varepsilon_h H_t^h = \varepsilon_{\text{eff},t}$, and its denominator is $\sum_h s_h(\varepsilon_h-1)H_t^h = \varepsilon_{\text{eff},t} - \sum_h s_h H_t^h = \varepsilon_{\text{eff},t} - 1$ by the adding-up condition (D.17). Hence $\mathcal{M}_t^n = \varepsilon_{\text{eff},t}/(\varepsilon_{\text{eff},t}-1)$, and substituting (D.19) yields the flexible-price markup as a function of H_t alone:

$$\mathcal{M}_t^n = \frac{H_t(1-\lambda)\,(\varepsilon_R-\varepsilon_P)+\varepsilon_P}{H_t(1-\lambda)\,(\varepsilon_R-\varepsilon_P)+\varepsilon_P-1}.$$

D.5.2 Pricing Second-Order Condition

The first-order condition (D.12) selects the optimal price only if profit is single-peaked around it. Consider the zero-inflation steady state, in which one firm sets the log relative price $x \equiv \ln(P(i)/P)$ while all other firms charge P. Let $\omega_h \equiv s_h\bar{H}^h$ denote the expenditure shares, so that the firm's relative demand is $D(x) = \sum_h \omega_h e^{-\varepsilon_h x}$, with $D(0)=1$ and $-D'(0) = \varepsilon_{\text{eff}}$. Since $Y(i) = AN(i)^{1-\alpha}$, the real wage bill per unit of aggregate output is $(1-\alpha)\overline{MC}\,D(x)^{1/(1-\alpha)}$, and real profit per unit of aggregate output is

$$\varpi(x) = e^x D(x) - (1-\alpha)\,\overline{MC}\,D(x)^{1/(1-\alpha)}. \tag{D.20}$$

The condition $\varpi'(0) = 0$ is the steady-state condition (D.14), $\overline{MC} = (\varepsilon_{\text{eff}}-1)/\varepsilon_{\text{eff}}$. Differentiating twice and using $D''(0) = \text{Var}_\omega(\varepsilon) + \varepsilon_{\text{eff}}^2$ gives

$$\varpi''(0) = \frac{1}{\varepsilon_{\text{eff}}}\left[\text{Var}_\omega(\varepsilon) - \frac{\varepsilon_{\text{eff}}(\varepsilon_{\text{eff}}-1)(1-\alpha+\alpha\varepsilon_{\text{eff}})}{1-\alpha}\right], \qquad \text{Var}_\omega(\varepsilon) = \omega_R\omega_P(\varepsilon_P-\varepsilon_R)^2. \tag{D.21}$$

The second term in brackets is the usual curvature of a monopolist with decreasing returns. The first is the expenditure-weighted variance of the elasticity across the firm's customers, and it works against concavity: a price increase loses mostly elastic poor customers, so the elasticity of the remaining demand falls as the price rises. With a common elasticity the variance is zero and $\varpi''(0) < 0$.

The same formula holds at any critical point x^* of ϖ, up to a positive factor, once ω_h is replaced by the expenditure shares at that price, $\omega_h(x^*) \propto \omega_h e^{-\varepsilon_h x^*}$, and ε_{eff} by their weighted elasticity: $\varpi(x^*+x')$ has the form (D.20) in x', and $\varpi'(x^*) = 0$ plays the

role of the steady-state normalization. The variance is at most $(\varepsilon_P - \varepsilon_R)^2/4$, and the second term is increasing in the effective elasticity, which is at least ε_R. Condition (D.1) therefore makes the bracket negative at every critical point, whatever the expenditure shares there. Every critical point is then a strict local maximum, so ϖ has at most one. Since $\varpi(x) \to -\infty$ as $x \to -\infty$ and $\varpi(x) \to 0 < \varpi(0)$ as $x \to \infty$, it has exactly one, $x = 0$, which is the global maximum of static profit. The Rotemberg cost is non-negative and zero on a constant price path, so the constant symmetric price then also maximizes the discounted objective in the zero-inflation steady state, which is the maintained pricing assumption that Section 2 states.

Condition (D.1) is sufficient, not necessary, and the local condition $\varpi''(0) < 0$ alone does not ensure a global maximum. At the baseline calibration, $\mathrm{Var}_\omega(\varepsilon) = 2.49$ against 34.2 for the second term, so $\varpi''(0) = -8.07$, and (D.1) reads $12.96 < 60$. The condition does not involve λ or R, so at the baseline elasticities it holds on the whole fiscal branch and at every (λ, R), including the supremum in Panel C of Table C.5. Because the maintained assumption is global optimality itself, we also verify it directly, locating every critical point of (D.20) and comparing profits. The symmetric price is the global maximizer at every calibration reported in the paper: the baseline; the whole fiscal branch, in TE-TANK and in both homogeneous benchmarks, where (D.1) holds trivially because the elasticity gap is zero; every point of the fixed-ratio and fixed-gap paths and every row of Table C.5 but one; and every admissible (λ, R) at the baseline elasticities. At the high-markup end of the fixed-gap path (D.1) fails, but the symmetric price remains the global maximizer there. The exception is the 10 percent point of Panel C, where profit has a second, higher peak; Appendix C.2 rules out that parameterization for this reason, among others. When global optimality fails, the root of Proposition 1(i) is not an equilibrium.

D.6 Steady State

The two-equation fixed point (8)–(9) has a closed-form characterization. Substituting (9) into (8) reduces it to a single quadratic in $\bar{H}$,

$$\mathsf{A}\,\bar{H}^2 + \big[\mathsf{B} - \mathsf{A}(\mathsf{C} - \mathsf{D})\big]\bar{H} - \big[\mathsf{B}(\mathsf{C} - \mathsf{D}) + \mathsf{D}\big] = 0, \tag{D.22}$$

which determines $\bar{H}$ and, through (9), $\bar{\mathcal{M}}$. Once these two objects are known, output follows recursively from labor supply and production, as derived below. Holding primitives fixed, both endogenous objects lower steady-state output: a higher markup depresses labor demand, while a higher $\bar{H}$ raises rich consumption and hence depresses labor supply through the Ricardian intratemporal condition. Proposition 1 shows that the steady state exists and is unique on the whole fiscal square.

From the markup definition in (D.25), which Appendix D.8 derives below, evaluated at the steady state with $A = 1$, $C^R = HC$, and market clearing $C = Y$, the steady-state markup is

$$\mathcal{M} = (1-\alpha)(HY)^{-\sigma} N^{-(\alpha+\varphi)}.$$

Substituting the steady-state production function $N = Y^{1/(1-\alpha)}$ and solving for Y yields

$$\bar{Y} = \left(\frac{1-\alpha}{\bar{\mathcal{M}}\bar{H}^{\sigma}}\right)^{\frac{1}{\frac{\alpha+\varphi}{1-\alpha}+\sigma}} = \left(\frac{1-\alpha}{\bar{\mathcal{M}}\bar{H}^{\sigma}}\right)^{\frac{1-\alpha}{\alpha+\varphi+\sigma(1-\alpha)}}. \tag{D.23}$$

D.7 Dynamic IS Curve

Start from the rich household's Euler equation (D.6), substitute $C_t^R = H_t C_t$, and use that the price index is common to both household types ($\Pi_{t+1}^R = \Pi_{t+1}$):

$$1 = \mathbb{E}_t\left[\beta\left(\frac{H_{t+1}C_{t+1}}{H_t C_t}\right)^{-\sigma}\frac{Z_{t+1}}{Z_t}\frac{1+i_t}{\Pi_{t+1}}\right].$$

Taking logs and log-linearizing around the deterministic steady state (a standard first-order Taylor expansion of the Euler equation, as in the textbook derivation of the New Keynesian IS curve), using the preference-shock process $z_t = \rho_z z_{t-1}$, and expressing variables as deviations from steady state ($\hat{i}_t = i_t - \bar{i}$, $\hat{\pi}_t = \pi_t - \bar{\pi}$) gives

$$y_t = \mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\right) + \mathbb{E}_t[\Delta\hat{h}_{t+1}] + \frac{1}{\sigma}(1-\rho_z)z_t.$$

Subtracting natural output y_t^n from both sides and defining the output gap $\tilde{y}_t \equiv y_t - y_t^n$ yields the pre-substitution dynamic IS curve,

$$\tilde{y}_t = \mathbb{E}_t[\tilde{y}_{t+1}] - \frac{1}{\sigma}\left(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\right) + \mathbb{E}_t[\Delta\hat{h}_{t+1}] + \frac{1}{\sigma}(1-\rho_z)z_t + \mathbb{E}_t[\Delta\hat{y}_{t+1}^n], \tag{D.24}$$

which contains $\mathbb{E}_t[\Delta\hat{h}_{t+1}]$ explicitly, as Section 2.5 notes; substituting (10) gives the reduced form (11).

D.8 New Keynesian Phillips Curve

This subsection log-linearizes the pricing first-order condition and writes it in terms of the output gap and the heterogeneity index. We start with the first-order condition (D.13) rewritten as:

$$(\Pi_t - 1)\Pi_t = \sum_h \frac{s_h}{\zeta} H_t^h (1 - \varepsilon_h + MC_t\varepsilon_h) + \mathbb{E}_t\left[\Lambda_{t+1,t}(\Pi_{t+1}-1)\Pi_{t+1}\frac{Y_{t+1}}{Y_t}\right]$$

and log-linearize around the zero-inflation steady state. A standard first-order expansion (dropping products of deviations) gives $(\Pi_t - 1)\Pi_t \approx \pi_t \equiv \ln \Pi_t$ on the left, collapses the forward-looking term to $\beta \mathbb{E}_t\{\pi_{t+1}\}$, and, using $H_t^h \approx H^h(1 + \hat{h}_t^h)$ and $MC_t \approx MC(1 + \hat{mc}_t)$, linearizes the heterogeneity-and-marginal-cost term to the model-specific form below.

Combining terms. Putting all linearized terms together:

$$\pi_t \approx \sum_h \frac{s_h}{\zeta} \varepsilon_h \left(\frac{1}{\varepsilon_h} - 1 + MC \right) H^h \hat{h}_t^h + \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h MC \hat{mc}_t + \beta \mathbb{E}_t \{\pi_{t+1}\}$$

Using the steady-state result $MC = \frac{\sum_h s_h H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)}{\sum_h s_h \varepsilon_h H^h}$ from (D.14), we can simplify:

$$\begin{aligned} \pi_t &= \sum_h \frac{s_h}{\zeta} \varepsilon_h \left(\frac{1}{\varepsilon_h} - 1 + MC \right) H^h \hat{h}_t^h + \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h \frac{\sum_h s_h H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)}{\sum_h s_h \varepsilon_h H^h} \hat{mc}_t + \beta \mathbb{E}_t \{\pi_{t+1}\} \\ &= \sum_h \frac{s_h}{\zeta} \varepsilon_h \left(\frac{1}{\varepsilon_h} - 1 + MC \right) H^h \hat{h}_t^h + \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right) \hat{mc}_t + \beta \mathbb{E}_t \{\pi_{t+1}\} \end{aligned}$$

Defining the coefficients:

$$\delta_1^P = \frac{s_P}{\zeta} \varepsilon_P \left(\frac{1}{\varepsilon_P} - 1 + MC \right) H^P, \quad \delta_1^R = \frac{s_R}{\zeta} \varepsilon_R \left(\frac{1}{\varepsilon_R} - 1 + MC \right) H^R,$$
$$\delta_2 = \sum_h \frac{s_h}{\zeta} H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)$$

The linearized Phillips Curve becomes:

$$\pi_t = \beta \mathbb{E}_t \{\pi_{t+1}\} + \delta_1^P \hat{h}_t^P + \delta_1^R \hat{h}_t^R + \delta_2 \hat{mc}_t$$

Relating markup to marginal cost. From the markup definition $\mathcal{M}_t = \frac{P_t}{MC_t^{nom}} = \frac{P_t}{MC_t \cdot P_t} = \frac{1}{MC_t}$, we have $\mu_t = -mc_t$ in logs. Since $\hat{mc}_t = mc_t - mc$ and $\hat{\mu}_t = \mu_t - \mu$, and given that $\mu = -mc$, it follows that $\hat{\mu}_t = -\hat{mc}_t$. Substituting:

$$\pi_t = \beta \mathbb{E}_t \{\pi_{t+1}\} + \delta_1^P \hat{h}_t^P + \delta_1^R \hat{h}_t^R - \delta_2 \hat{\mu}_t$$

Expressing markup in terms of output gap. To arrive at the NKPC, we replace $\hat{\mu}_t$ with the output gap $\tilde{y}_t$. Starting with the definition of the gross nominal markup:

$$\mathcal{M}_t = \frac{P_t}{MC_t^{nom}} = \frac{P_t}{\frac{W_t}{\partial Y_t / \partial N_t}} = P_t \frac{\partial Y_t / \partial N_t}{W_t} = \frac{(1 - \alpha) A_t N_t^{-\alpha} P_t}{W_t}.$$

From the Ricardian household's intratemporal condition (D.5), using $P_t^R = P_t$ in symmetric equilibrium and $N_t^R = N_t$ from labor market clearing, $W_t/P_t = (C_t^R)^\sigma N_t^\varphi$, so

$$\begin{aligned} \mathcal{M}_t &= \frac{(1-\alpha)A_t}{(C_t^R)^\sigma N_t^\varphi} N_t^{-\alpha} = (1-\alpha)A_t (C_t^R)^{-\sigma} N_t^{-\alpha-\varphi}, \\ \mu_t &= \ln(1-\alpha) + a_t - \sigma c_t^R - (\alpha+\varphi) n_t. \end{aligned} \tag{D.25}$$

Using $C_t^R = H_t C_t$, market clearing $C_t = Y_t$ (Rotemberg costs are second-order), and the production function $Y_t = A_t N_t^{1-\alpha}$ (so $n_t = (y_t - a_t)/(1-\alpha)$) to substitute out c_t^R and n_t, and taking log-deviations from steady state gives

$$\hat{\mu}_t = \left(1 + \frac{\alpha+\varphi}{1-\alpha}\right) a_t - \sigma \hat{h}_t - \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right) \hat{y}_t.$$

Deriving the natural rate of output. With flexible prices, $\mathcal{M}_t = \mathcal{M}_t^n(H_t)$ for all t. Here $\mathcal{M}_t^n \neq \mathcal{M}_t^e = 1$. This is also the case in the standard TANK without heterogeneous elasticities. However, in the TANK the difference $\mathcal{M}^n - \mathcal{M}^e = \frac{\varepsilon}{\varepsilon-1} - 1$ is constant, whereas in our model $\mathcal{M}_t^n(H_t)$ is in principle state-dependent. Yet the heterogeneity index in the flex-price equilibrium does not respond to technology or preference shocks: from equation (D.11), H_t^n is determined by an implicit equation involving only $\mathcal{M}_t^n(H_t^n)$ and the fiscal parameters (δ, τ), with no dependence on A_t or Z_t. Since $H_t^n = H$ for all t, the flex-price markup remains at its steady-state value ($\hat{\mu}_t^n = 0$), and divine coincidence is preserved. The time-varying markup channel would, however, generate a cost-push term and break divine coincidence if the model included shocks that directly affect the demand composition (e.g., stochastic redistribution or heterogeneous preference shocks).

In the flex-price equilibrium, the markup equals the desired markup $\hat{\mu}_t = \hat{\mu}_t^n$, where $\hat{\mu}_t^n$ denotes the log-deviation of the flexible-price markup (D.15) from its steady-state value. In general, log-linearizing the flex-price markup around H with $\Phi \equiv (1-\lambda)(\varepsilon_R - \varepsilon_P)$ gives:

$$\hat{\mu}_t^n = \frac{-\Phi H}{(\Phi H + \varepsilon_P)(\Phi H + \varepsilon_P - 1)} \hat{h}_t$$

Since $H_t^n = H$ for all t (as argued above), we have $\hat{h}_t^n = 0$ and therefore $\hat{\mu}_t^n = 0$. Setting $\hat{\mu}_t = \hat{\mu}_t^n = 0$ in the markup expression:

$$0 = \left(1 + \frac{\alpha+\varphi}{1-\alpha}\right) a_t - \left(\frac{\alpha+\varphi}{1-\alpha} + \sigma\right) \hat{y}_t^n$$

Subtracting this from the sticky-price markup expression:

$$\begin{aligned}\hat{\mu}_t &= -\sigma\hat{h}_t - \left(\frac{\alpha+\varphi}{1-\alpha}+\sigma\right)(\hat{y}_t - \hat{y}_t^n) \\ &= -\sigma\hat{h}_t - \left(\frac{\alpha+\varphi}{1-\alpha}+\sigma\right)\tilde{y}_t \end{aligned} \tag{D.26}$$

Since $\hat{mc}_t = -\hat{\mu}_t$, $\hat{mc}_t = \sigma\hat{h}_t + \left(\frac{\alpha+\varphi}{1-\alpha}+\sigma\right)\tilde{y}_t$, so the NKPC is $\pi_t = \beta\mathbb{E}_t\{\pi_{t+1}\} + \delta_1^P\hat{h}_t^P + \delta_1^R\hat{h}_t^R + \delta_2[\sigma\hat{h}_t + (\frac{\alpha+\varphi}{1-\alpha}+\sigma)\tilde{y}_t]$.

Expressing in terms of aggregate heterogeneity. We relate $\hat{h}_t^P$ to $\hat{h}_t^R$ using market clearing, $\lambda H_t^P + (1-\lambda)H_t^R = 1$: log-linearizing around the steady state (where $\lambda H^P + (1-\lambda)H^R = 1$) gives $\lambda H^P\hat{h}_t^P + (1-\lambda)H^R\hat{h}_t^R = 0$, so, using $\lambda H^P = 1-(1-\lambda)H^R$,

$$\hat{h}_t^P = \frac{(\lambda-1)H^R}{1+(\lambda-1)H^R}\hat{h}_t^R.$$

Define $\hat{h}_t^R \equiv \hat{h}_t$ and $H^R \equiv H$ as the aggregate heterogeneity index. Substituting, and using $\delta_1^P = \frac{1}{\zeta}\varepsilon_P(\frac{1}{\varepsilon_P} - 1 + MC)(1-(1-\lambda)H)$ and $\delta_1^R = \frac{1-\lambda}{\zeta}\varepsilon_R(\frac{1}{\varepsilon_R} - 1 + MC)H$ (both immediate from the definitions of δ_1^P, δ_1^R and $H^P = [1-(1-\lambda)H]/\lambda$), the coefficient on $\hat{h}_t$ collapses, after simplification, to $\delta_1^P\frac{(\lambda-1)H}{1+(\lambda-1)H} + \delta_1^R = \frac{(1-\lambda)H}{\zeta}(1-MC)(\varepsilon_P - \varepsilon_R)$. Adding the $\delta_2\sigma$ term from marginal cost, the NKPC with heterogeneity is

$$\pi_t = \beta\mathbb{E}_t\{\pi_{t+1}\} + \kappa_h\hat{h}_t + \kappa_y\tilde{y}_t + u_t,$$

where

$$\kappa_h = \underbrace{\frac{(1-\lambda)H}{\zeta}\left((1-MC)(\varepsilon_P-\varepsilon_R)\right)}_{\text{from firm FOC heterogeneity}} + \underbrace{\delta_2\sigma}_{\text{from Ricardian consumption in markup}},$$

$$\kappa_y = \delta_2\left(\frac{\alpha+\varphi}{1-\alpha}+\sigma\right),$$

and, substituting $H^P = [1-(1-\lambda)H]/\lambda$ into the definition of δ_2 and simplifying,

$$\delta_2 = \sum_h \frac{s_h}{\zeta}H^h\varepsilon_h\left(1-\frac{1}{\varepsilon_h}\right) = \frac{1}{\zeta}\left[H(1-\lambda)(\varepsilon_R-\varepsilon_P) + \varepsilon_P - 1\right].$$

The cost-push disturbance u_t is appended as an additive wedge to this linearized optimality condition, normalized to enter with a unit coefficient: a disturbance that raises real marginal cost by $\hat{u}_t$ independently of the output gap appears as $u_t = \delta_2\hat{u}_t$. The same loading applies to an exogenous log desired-markup shifter q_t. Write the desired markup as $\hat{\mu}_t^n = \omega\hat{h}_t + q_t$, where ω is the elasticity in the display above. The pricing

gap is then $-\delta_2(\hat{\mu}_t - \hat{\mu}_t^n)$, so q_t enters as $u_t = \delta_2 q_t$. The coefficient $\kappa_h - \delta_2\sigma = \delta_2\omega$ applies instead to a unit change in the heterogeneity index and is not the loading on a shifter already expressed in log-markup units. Thus equal real-marginal-cost and equal desired-markup shifters are algebraically equivalent normalizations, including in homogeneous TANK, where $\delta_2 = (\varepsilon - 1)/\zeta > 0$. This is equation (12) in the main text.

Ricardian wage channel. The term $\delta_2\sigma$ in κ_h is a Ricardian wage channel, present even when both types share one elasticity. Wages are flexible, and only rich households are on their labor-supply curve. Section 2.1 makes hours common across types and rations hand-to-mouth households at them, so the wage is the one that makes rich households willing to supply N_t. It therefore clears the Ricardian intratemporal condition

$$\frac{W_t}{P_t} = (C_t^R)^\sigma N_t^\varphi. \tag{D.27}$$

The wage is thus priced off the rich household's marginal rate of substitution, not off average consumption, and real marginal cost is that MRS relative to the marginal product of labor. Since $C_t^R = H_t C_t$, the distribution enters price setting through the wage. Log-linearizing, as in (D.26),

$$\widehat{mc}_t = \sigma\hat{h}_t + G\tilde{y}_t, \qquad G \equiv \frac{\varphi + \alpha}{1 - \alpha} + \sigma. \tag{D.28}$$

The $G\tilde{y}_t$ term is the usual RANK loading: more hours, a lower marginal product, and higher consumption. The extra $\sigma\hat{h}_t$ term is the TANK piece. Because H_t is a ratio, a higher $\hat{h}_t$ at a given level of aggregate consumption raises rich consumption and lowers hand-to-mouth consumption; the richer unconstrained household then has lower marginal utility of consumption and requires a higher real wage to supply the same hours. Substituting $\hat{h}_t = -\chi_y\tilde{y}_t$ from (10), a boom compresses inequality, so C_t^R rises by less than aggregate consumption, the wage that clears (D.27) rises by less, and inflation responds less to output. After this substitution the route $\delta_2\sigma$ becomes the wage factor $1/(1 + \Gamma\sigma)$ of (15).

Rotemberg versus Calvo. Unlike its Calvo counterpart, δ_2 carries no decreasing-returns correction $(1 - \alpha)/(1 - \alpha + \alpha\varepsilon)$: that factor arises because a resetting Calvo firm's own marginal cost differs from the average, whereas under Rotemberg pricing all firms stay symmetric and this wedge never opens (Ascari and Rossi, 2012).

D.9 Dividends

Linearization of Dividends. The aggregate accounting identity $Y_t \Delta^p(\Pi_t) = \frac{W_t}{P_t} N_t + D_t$ (output net of price dispersion equals real labor income plus dividends), together with $W_t N_t / P_t = \frac{1-\alpha}{\mathcal{M}_t} Y_t$, gives $D_t = Y_t[\Delta^p(\Pi_t) - \frac{1-\alpha}{\mathcal{M}_t}]$, so in steady state ($\Pi = \Delta^p(1) = 1$), $D = Y(\mathcal{M} - (1-\alpha))/\mathcal{M}$. Since price dispersion is minimized at zero inflation, $\partial D / \partial \Pi|_{\Pi=1} = 0$, so a first-order Taylor expansion of the deviation $D_t - D$ around the steady state, normalized by D and using $1 - \frac{1-\alpha}{\mathcal{M}} = D/Y$, simplifies to

$$\hat{d}_t = \hat{y}_t + \frac{1-\alpha}{\mathcal{M} - (1-\alpha)} \hat{\mu}_t. \tag{D.29}$$

D.10 Proofs of the Lemma and Propositions

The proofs below establish Lemma 1 and Propositions 1, 2, 3, and 4; Appendix D.11 states and proves Corollary 5, which builds on all five. Appendices D.4, D.7, D.8, and D.5.1 supply the derivations that the proofs use. Throughout, $\mathsf{A} = (1-\lambda)(\varepsilon_R - \varepsilon_P)$, $\mathsf{B} = \varepsilon_P$, $\mathsf{C} = \frac{1+\delta\lambda(\tau-1)}{1-\lambda}$, and $\mathsf{D} = \frac{(1-\alpha)\lambda(1+\delta(\tau-1))}{1-\lambda}$, as in Proposition 1.

D.10.1 Fiscal Sufficient Statistic (Lemma 1)

Proof of Lemma 1. After eliminating gross transfer flows and equity values, the fiscal parameters enter the reduced equilibrium through the following combinations, each a function of $R = \delta(1-\tau)$ through $1 + \delta(\tau - 1) = 1 - R$:

$$\begin{aligned}
\mathsf{C} &= \frac{1 + \delta\lambda(\tau-1)}{1-\lambda} = \frac{1-\lambda R}{1-\lambda}, \\
\mathsf{D} &= \frac{(1-\alpha)\lambda\big(1+\delta(\tau-1)\big)}{1-\lambda} = \frac{(1-\alpha)\lambda(1-R)}{1-\lambda}, \\
\bar{H} - 1 &= \frac{\lambda(1-R)}{1-\lambda}\left(1 - \frac{1-\alpha}{\bar{\mathcal{M}}}\right), \\
\Gamma &= \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}}.
\end{aligned}$$

The eliminated objects do not add a second fiscal sufficient statistic. Consider equity first. Total equity supply is one and only rich households hold shares, so market clearing implies

$$F_t^R = F_{t-1}^R = \frac{1}{1-\lambda}.$$

The purchase and resale-value terms in (D.3) therefore cancel period by period:

$$Q_t F_{t-1}^R - Q_t F_t^R = Q_t\big(F_{t-1}^R - F_t^R\big) = 0.$$

Although the equity Euler equation implies

$$\bar{Q} = \frac{\beta(1-\delta)}{1-\beta}\bar{D},$$

so that the equity valuation depends separately on δ, this valuation does not enter the household's flow resources or the reduced equilibrium.

Second, the transfer rule (D.2) can be written as

$$T_t^P = (1-\tau)\delta D_t = RD_t,$$
$$T_t^R = \frac{\delta(1-\lambda+\tau\lambda)}{1-\lambda}D_t = \frac{\delta-\lambda R}{1-\lambda}D_t.$$

Thus T_t^R itself depends on δ separately at fixed R, but rich households receive it together with after-tax dividend income. Their combined profit-related resources are

$$\begin{aligned}(1-\delta)D_t F_{t-1}^R + T_t^R &= \frac{(1-\delta)+\delta-\lambda R}{1-\lambda}D_t \\ &= \frac{1-\lambda R}{1-\lambda}D_t = \mathsf{C}D_t.\end{aligned}$$

Hence changing (δ,τ) while holding R fixed changes the composition of rich households' income between dividends and transfers, as well as the equity valuation, but not either household type's disposable resources.

The quadratic primitives A and B contain no fiscal parameters; hence $\bar{H}$, the root of (D.22), depends on (δ,τ) only through (C,D), that is, through R. Then $\bar{\mathcal{M}} = \varepsilon_{\text{eff}}(\bar{H})/(\varepsilon_{\text{eff}}(\bar{H})-1)$; the coefficients δ_2, κ_y, and κ_h (functions of $\bar{H}$ and $\bar{\mathcal{M}}$ only); Γ (an explicit factor $1-R$ times $1/(\bar{\mathcal{M}}\bar{H})$); $\chi_y = \Gamma\left(\frac{\varphi+\alpha}{1-\alpha}+\sigma\right)/(1+\Gamma\sigma)$; and $\kappa = \kappa_y - \chi_y\kappa_h$ all inherit dependence on R alone. No other occurrence of δ or τ appears in the derivations of Appendix D.4, Appendix D.8, or the dynamic IS block of Appendix D.7. Therefore fiscal pairs with the same R yield the same steady-state allocation and first-order dynamics. Finally, $\bar{H}=1 \iff 1-R=0 \iff R=1$, and $(\delta,\tau)=(1,0)$ gives $R=1$; the cyclicality derivation below establishes $\operatorname{sign}(\chi_y)=\operatorname{sign}(\Gamma)=\operatorname{sign}(1-R)$. □

D.10.2 Existence, Uniqueness, and Fiscal Comparative Statics (Proposition 1)

Proof of Proposition 1. (i) The fixed point $\bar{H} = \mathsf{C} - \mathsf{D}/\bar{\mathcal{M}}$ is the steady-state heterogeneity index of Appendix D.4 evaluated at $\Delta^p(\bar{\Pi})=1$, that is, equation (8); the flex-price markup $\bar{\mathcal{M}} = (\mathsf{A}\bar{H}+\mathsf{B})/(\mathsf{A}\bar{H}+\mathsf{B}-1)$ is equation (9) written with (A,B), and Appendix D.5.1 derives it. Substituting $1/\bar{\mathcal{M}} = (\mathsf{A}\bar{H}+\mathsf{B}-1)/(\mathsf{A}\bar{H}+\mathsf{B})$ and clearing the

denominator:

$$(\bar{H} - \mathsf{C})(\mathsf{A}\bar{H} + \mathsf{B}) = -\mathsf{D}(\mathsf{A}\bar{H} + \mathsf{B} - 1)$$
$$\Rightarrow \quad \mathsf{A}\bar{H}^2 + \big[\mathsf{B} - \mathsf{A}(\mathsf{C} - \mathsf{D})\big]\bar{H} - \big[\mathsf{B}(\mathsf{C} - \mathsf{D}) + \mathsf{D}\big] = 0,$$

which is the quadratic (D.22). *Existence and uniqueness on the fiscal square.* Let $e(H) \equiv \mathsf{A}H + \mathsf{B} = \varepsilon_{\text{eff}}(H)$ and

$$H_{\max} \equiv \frac{\varepsilon_P - 1}{(1-\lambda)(\varepsilon_P - \varepsilon_R)},$$

so that $e(H) > 1$, equivalently $\mathcal{M}(H) > 1$, exactly on $H < H_{\max}$. The steady state is a zero of

$$q(H; R) \equiv \mathsf{C}(R) - \mathsf{D}(R) + \frac{\mathsf{D}(R)}{e(H)} - H.$$

Using $\mathsf{C} - 1 = \frac{\lambda(1-R)}{1-\lambda}$ and $\mathsf{D} = (1-\alpha)(\mathsf{C}-1)$, for $R < 1$,

$$q(1) = (\mathsf{C} - 1)\left[\alpha + \frac{1-\alpha}{\bar{\varepsilon}}\right] > 0,$$

since $e(1) = \bar{\varepsilon} > 1$. Throughout we maintain $\varepsilon_R > 1$. A CES demand elasticity below one is not a sensible restriction for ordinary consumption goods: the desired markup $\varepsilon/(\varepsilon - 1)$ is then negative and the monopolist's problem has no interior solution, profit rising without bound as the price rises. Because $\varepsilon_P > \varepsilon_R$, the restriction binds only through the rich elasticity. Moreover $H_{\max} - \frac{1}{1-\lambda} = \frac{\varepsilon_R - 1}{(1-\lambda)(\varepsilon_P - \varepsilon_R)} > 0$ under it, while $\mathsf{C} \le \frac{1}{1-\lambda}$, so $\lim_{H \uparrow H_{\max}} q(H; R) = \mathsf{C} - H_{\max} < 0$. Finally $q_{HH}(H; R) = 2\mathsf{D}\mathsf{A}^2/e(H)^3 > 0$: q is convex, positive at $H = 1$, and negative at the upper boundary, so it crosses zero exactly once on $(1, H_{\max})$. The crossing is strict: if $q_H(\bar{H}; R) \ge 0$, convexity would imply $q(H; R) \ge 0$ for every $H > \bar{H}$, contradicting its negative limit at $H_{\max}$. Thus $q_H(\bar{H}; R) < 0$. The quadratic's second root lies beyond $H_{\max}$, where $\mathcal{M} < 1$ and, because $H_{\max} > \frac{1}{1-\lambda}$, poor-household consumption is negative (the quadratic equals $-e(H)q(H; R)$; it is positive at $H_{\max}$, where $q < 0$ and $e = 1$, and has negative leading coefficient A, so its remaining root exceeds $H_{\max}$). At the root, $\bar{H} = \mathsf{C} - \mathsf{D} + \mathsf{D}/e(\bar{H}) < \mathsf{C} \le \frac{1}{1-\lambda}$, so the poor household's consumption relative to the average, $\bar{H}^P = [1 - (1-\lambda)\bar{H}]/\lambda$, is strictly positive. Hence for every $R \in [0, 1)$ there is exactly one admissible steady state, with $1 < \bar{H} < \mathsf{C}$. At $R = 1$, $\mathsf{C} = 1$ and $\mathsf{D} = 0$, so the quadratic factors as $(H-1)(\mathsf{A}H + \mathsf{B}) = 0$. Under $\varepsilon_P > \varepsilon_R$, its second root is

$$-\frac{\mathsf{B}}{\mathsf{A}} = \frac{\varepsilon_P}{(1-\lambda)(\varepsilon_P - \varepsilon_R)} > \frac{1}{1-\lambda},$$

which implies negative poor-household consumption; hence $\bar{H} = 1$ is the unique feasible root. In the homogeneous TANK case $\mathsf{A} = 0$, the system is linear and triv-

ially unique. Existence and uniqueness therefore hold on the entire fiscal square $(\delta, \tau) \in [0,1]^2$. The root solves the pricing first-order condition. By the maintained pricing assumption, the symmetric price it implies is each firm's global profit maximizer, so the root is a steady state and not only a stationary point of profit. Appendix D.5.2 shows that condition (D.1) guarantees this assumption.

(ii) Differentiate the fixed-point condition $q(\bar{H}; R) = 0$. Its partial derivatives are

$$\begin{aligned} q_R(\bar{H}; R) &= -\frac{\lambda}{1-\lambda} + \frac{(1-\alpha)\lambda}{1-\lambda} - \frac{(1-\alpha)\lambda/(1-\lambda)}{e(\bar{H})} \\ &= -\frac{\lambda}{1-\lambda}\left[\alpha + \frac{1-\alpha}{e(\bar{H})}\right] < 0, \\ q_H(\bar{H}; R) &= -\left[1 + \frac{\mathsf{DA}}{e(\bar{H})^2}\right] < 0, \end{aligned}$$

where the second inequality follows from the strict downcrossing established in part (i). The implicit-function theorem therefore gives

$$\frac{\mathrm{d}\bar{H}}{\mathrm{d}R} = -\frac{q_R}{q_H} = -\frac{\lambda}{1-\lambda} \cdot \frac{\alpha + (1-\alpha)/e(\bar{H})}{1 + \mathsf{DA}/e(\bar{H})^2} < 0.$$

This derivative extends continuously to $R = 1$ from the left. For the markup, $\bar{\mathcal{M}} = e/(e-1)$ implies

$$\frac{\mathrm{d}\bar{\mathcal{M}}}{\mathrm{d}\bar{H}} = -\frac{\mathsf{A}}{[e(\bar{H})-1]^2} = \frac{(1-\lambda)(\varepsilon_P - \varepsilon_R)}{[e(\bar{H})-1]^2} > 0.$$

Combining the two derivatives yields

$$\frac{\mathrm{d}\bar{\mathcal{M}}}{\mathrm{d}R} = -\frac{\lambda(\varepsilon_P - \varepsilon_R)\left[\alpha + (1-\alpha)/e(\bar{H})\right]}{[e(\bar{H})-1]^2\ [1 + \mathsf{DA}/e(\bar{H})^2]} < 0.$$

Thus redistribution lowers both steady-state inequality and the desired markup. Using $\mathsf{D} = \Gamma \bar{\mathcal{M}} \bar{H}$ and the definition of η_μ in (13), the feedback denominator satisfies

$$1 + \frac{\mathsf{DA}}{e(\bar{H})^2} = 1 - \Gamma \eta_\mu > 0. \tag{D.30}$$

Hence the endogenous-markup feedback amplifies the direct redistribution effect by $(1 - \Gamma\eta_\mu)^{-1}$. □

Benchmark nesting. Subtracting one from (8) gives

$$\bar{H} - 1 = \frac{\lambda(1-R)}{1-\lambda}\left(1 - \frac{1-\alpha}{\bar{\mathcal{M}}}\right).$$

Because $\bar{\mathcal{M}} > 1 > 1 - \alpha$, the second factor is strictly positive, so $\bar{H} = 1 \iff R = 1$. On the fiscal square, $R = 1$ requires $(\delta, \tau) = (1, 0)$. At this point $\varepsilon_{\text{eff}}(1) = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P = \bar{\varepsilon}$ and $\bar{\mathcal{M}} = \bar{\varepsilon}/(\bar{\varepsilon} - 1)$. Moreover,

$$\Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\bar{\mathcal{M}}\bar{H}}, \qquad \chi_y = \frac{\Gamma\left(\frac{\varphi+\alpha}{1-\alpha} + \sigma\right)}{1 + \Gamma\sigma},$$

so the RANK nesting holds:

$$\bar{H} = 1 \iff R = 1 \iff \Gamma = 0 \iff \chi_y = 0 \iff \hat{h}_t \equiv 0 \quad \text{along every first-order equilibrium path.} \tag{D.31}$$

The final equivalence concerns the loading, not a single realized path: since $\hat{h}_t = -\chi_y \tilde{y}_t$, a path with a closed output gap has $\hat{h}_t \equiv 0$ even when $\chi_y \neq 0$, whereas $\chi_y = 0$ sets $\hat{h}_t \equiv 0$ under every admissible disturbance. The elasticity-gap term in the NKPC then multiplies $\hat{h}_t = 0$, while the DIS curve loses its heterogeneity term; every structural equation reduces to its RANK($\bar{\varepsilon}$) counterpart to first order. In the alternative homogeneous-elasticity benchmark $\varepsilon_R = \varepsilon_P = \varepsilon$, $\mathsf{A} = 0$ and the steady-state system is linear, with $\bar{H} = \mathsf{C} - \mathsf{D}(\varepsilon - 1)/\varepsilon$ and fixed $\bar{\mathcal{M}} = \varepsilon/(\varepsilon - 1)$. Thus standard TANK can retain consumption inequality, but its desired markup does not respond to it.

D.10.3 Endogenous Flattening of the Phillips Curve (Proposition 2)

Proof of Proposition 2. Step 1: part (i), the exact cancellation. The reduced-form slope in (14) is $\kappa = \kappa_y - \kappa_h \chi_y$, with $\kappa_y = \delta_2 G$ and $\kappa_h = \delta_2(\sigma + \eta_\mu)$ from (13), and $\chi_y = \Gamma G/(1 + \Gamma\sigma)$ from Appendix D.4. Substituting and putting both terms over the common denominator $1 + \Gamma\sigma$,

$$\begin{aligned}\kappa &= \delta_2 G - \delta_2(\sigma + \eta_\mu)\frac{\Gamma G}{1 + \Gamma\sigma} = \delta_2 G \frac{(1 + \Gamma\sigma) - \Gamma(\sigma + \eta_\mu)}{1 + \Gamma\sigma} \\ &= \delta_2 G \frac{1 + \Gamma\sigma - \Gamma\sigma - \Gamma\eta_\mu}{1 + \Gamma\sigma} = \delta_2 G \frac{1 - \Gamma\eta_\mu}{1 + \Gamma\sigma} = \kappa_y \frac{1 - \Gamma\eta_\mu}{1 + \Gamma\sigma},\end{aligned}$$

which is (15). The two $\Gamma\sigma$ terms cancel exactly: the Ricardian wage route of κ_h, the term $\delta_2\sigma$, survives only through the denominator $1 + \Gamma\sigma$, and the entire numerator correction is the desired-markup elasticity η_μ.

The benchmark slope is strictly positive, $\kappa_y = \delta_2 G > 0$, because both factors are. First, $G = \frac{\varphi+\alpha}{1-\alpha} + \sigma > 0$, since $\varphi \geq 0$, $\alpha \in [0, 1)$, and $\sigma > 0$. Second, by Appendix D.8,

$$\delta_2 = \frac{(1-\lambda)\bar{H}(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1}{\zeta} = \frac{\varepsilon_{\text{eff}}(\bar{H}) - 1}{\zeta} > 0,$$

because $\zeta > 0$ and $\varepsilon_{\text{eff}}(\bar{H}) = e(\bar{H}) > 1$ at the admissible steady state: for $R < 1$, Proposition 1(i) gives $\bar{\mathcal{M}} = e(\bar{H})/[e(\bar{H}) - 1] > 1$, equivalently $e(\bar{H}) > 1$; at $R = 1$ it gives $\bar{H} = 1$, hence $e(1) = \bar{\varepsilon} = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P > 1$ under $\varepsilon_P \geq \varepsilon_R > 1$.

Step 2: part (ii), the two bounds on the numerator. For the strict lower bound, use $\mathsf{D} = \Gamma\bar{\mathcal{M}}\bar{H}$, $\bar{\mathcal{M}} = e(\bar{H})/[e(\bar{H}) - 1]$, and $\eta_\mu = -\mathsf{A}\bar{H}/\big[e(\bar{H})(e(\bar{H}) - 1)\big]$ from (13) to write

$$\Gamma\eta_\mu = \frac{\mathsf{D}}{\bar{\mathcal{M}}\bar{H}} \cdot \frac{-\mathsf{A}\bar{H}}{e(\bar{H})\big[e(\bar{H}) - 1\big]} = -\frac{\mathsf{DA}}{e(\bar{H})^2},$$

so that $1 - \Gamma\eta_\mu = 1 + \mathsf{DA}/e(\bar{H})^2 = -q_H(\bar{H}; R)$. This is exactly (D.30), whose strict positivity follows from the strict downcrossing $q_H(\bar{H}; R) < 0$ established in the proof of Proposition 1(i). In the degenerate cases the bound is immediate: $R = 1$ gives $\mathsf{D} = 0$ and $\varepsilon_P = \varepsilon_R$ gives $\mathsf{A} = 0$, so $\Gamma\eta_\mu = 0$ in either case.

For the upper bound, both factors of $\Gamma\eta_\mu$ are non-negative on the fiscal square. Since $R = \delta(1-\tau) \in [0,1]$, $\alpha \in [0,1)$, $\lambda \in (0,1)$, and $\bar{\mathcal{M}}\bar{H} > 0$,

$$\Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}} \;\geq\; 0,$$

with equality if and only if $R = 1$. And $\mathsf{A} = (1-\lambda)(\varepsilon_R - \varepsilon_P) \leq 0$ under $\varepsilon_P \geq \varepsilon_R$, while $e(\bar{H}) > 1$ by Step 1, so

$$\eta_\mu = \frac{(1-\lambda)\bar{H}(\varepsilon_P - \varepsilon_R)}{e(\bar{H})\big[e(\bar{H}) - 1\big]} \;\geq\; 0,$$

with equality if and only if $\varepsilon_P = \varepsilon_R$. Hence $\Gamma\eta_\mu \geq 0$ and therefore $1 - \Gamma\eta_\mu \leq 1$, which together with the lower bound gives $0 < 1 - \Gamma\eta_\mu \leq 1$. Because $\Gamma \geq 0$ and $\sigma > 0$ also give $1 + \Gamma\sigma \geq 1$, Step 1 yields

$$0 < \kappa = \kappa_y \frac{1 - \Gamma\eta_\mu}{1 + \Gamma\sigma} \leq \kappa_y,$$

that is, $\kappa \in (0, \kappa_y]$: the reduced-form Phillips-curve slope is strictly positive at every point of the fiscal square.

Step 3: part (iii), when the flattening is strict. Write the comparison in level form, $\kappa - \kappa_y = -\kappa_h\chi_y$. Step 1 gives $\delta_2 > 0$ and Step 2 gives $\eta_\mu \geq 0$, so with $\sigma > 0$,

$$\kappa_h = \delta_2\big(\sigma + \eta_\mu\big) > 0,$$

and therefore $\operatorname{sign}(\kappa - \kappa_y) = -\operatorname{sign}(\chi_y)$, that is, $\kappa < \kappa_y \iff \chi_y > 0$. Next, $\chi_y = \Gamma G/(1 + \Gamma\sigma)$ with $G > 0$ by Step 1 and $1 + \Gamma\sigma \geq 1 > 0$ by Step 2, so $\operatorname{sign}(\chi_y) = \operatorname{sign}(\Gamma)$; and the remaining factors of Γ in the display above are strictly positive, so $\operatorname{sign}(\Gamma) = \operatorname{sign}(1 - R)$. Since $R \leq 1$ on the fiscal square, $\chi_y > 0 \iff R < 1$, and equality $\kappa = \kappa_y$ holds exactly at $R = 1$, where $\Gamma = \chi_y = 0$ and both correction factors

of part (i) equal one.

Finally, fix $R < 1$, so $\Gamma > 0$. Then $1 - \Gamma\eta_\mu < 1 \iff \Gamma\eta_\mu > 0 \iff \eta_\mu > 0$, and by the expression for η_μ in Step 2, whose remaining factors $(1-\lambda)\bar{H}$ and $e(\bar{H})[e(\bar{H})-1]$ are strictly positive, $\eta_\mu > 0 \iff \varepsilon_P > \varepsilon_R$. The desired-markup correction is thus strictly below one precisely when the two elasticities differ; when $\varepsilon_P = \varepsilon_R$ the flattening $\kappa < \kappa_y$ is entirely the Ricardian wage channel $1/(1+\Gamma\sigma)$. □

D.10.4 Elasticity Dispersion and IS Amplification (Proposition 3)

IS semi-elasticity and admissible region. The DIS coefficient $1/[\sigma(1-\chi_y)]$ follows from substituting (10) into the dynamic IS curve of Appendix D.7: its $\mathbb{E}_t\Delta\hat{h}_{t+1}$ term contributes $-\chi_y\mathbb{E}_t\Delta\tilde{y}_{t+1}$, whose contemporaneous piece scales $\tilde{y}_t$ by $(1-\chi_y)^{-1}$, yielding (11). The amplification factor relative to RANK is $1/(1-\chi_y)$ since RANK has $\chi_y = 0$. Finiteness and positivity: $\chi_y < 1 \iff \Gamma\left(\frac{\varphi+\alpha}{1-\alpha} + \sigma\right) < 1 + \Gamma\sigma \iff \Gamma\frac{\varphi+\alpha}{1-\alpha} < 1$; with $\chi_y > 0$ under the baseline, this gives $\chi_y \in (0,1)$. The maintained assumptions do not imply $\chi_y < 1$. Because $\bar{\mathcal{M}}\bar{H} > 1$ on the admissible branch, $\lambda(1-R)(\varphi+\alpha) \leq 1-\lambda$ is a sufficient condition in primitives; at the baseline φ and α and at $R = 0$ it reads $\lambda \leq 3/7$, so it holds on the whole fiscal branch at $\lambda = 0.40$, whereas at $\lambda = 0.8$ and $R = 0$ the loading is $\chi_y = 1.24$, the IS curve has the wrong sign, and the equilibrium is indeterminate under the calibrated Taylor rule in TE-TANK and in both homogeneous benchmarks. Outside $\chi_y < 1$ the Taylor principle no longer guarantees determinacy, although it does not fail everywhere there.

Proof of Proposition 3. The TANK/TE ordering: both economies share (C, D), since (λ, δ, τ) are common, and each steady state solves $H = \mathsf{C} - \mathsf{D}/\mathcal{M}(H)$. For $R < 1$ the TANK fixed point is $\bar{H}^{\text{TANK}} = \mathsf{C} - \mathsf{D}/\mathcal{M}(\bar{\varepsilon}) > 1$. For any $H > 1$, the TE effective elasticity is $\varepsilon^{\text{TE}}_{\text{eff}}(H) = \bar{\varepsilon} + \mathsf{A}(H-1) < \bar{\varepsilon}$ because $\mathsf{A} = (1-\lambda)(\varepsilon_R - \varepsilon_P) < 0$, and $\mathcal{M}(\cdot)$ is decreasing in the elasticity, so $\mathcal{M}^{\text{TE}}(H) > \mathcal{M}(\bar{\varepsilon})$. Hence the TE fixed-point residual $q(H) = \mathsf{C} - \mathsf{D}/\mathcal{M}^{\text{TE}}(H) - H$ of the proof of Proposition 1(i) (the same q, since $1/\mathcal{M}(H) = 1 - 1/e(H)$) is strictly positive at $\bar{H}^{\text{TANK}}$:

$$q\left(\bar{H}^{\text{TANK}}\right) = \mathsf{D}\left[\frac{1}{\mathcal{M}(\bar{\varepsilon})} - \frac{1}{\mathcal{M}^{\text{TE}}(\bar{H}^{\text{TANK}})}\right] > 0.$$

Since q is convex, positive at $H = 1$, and crosses zero exactly once on the admissible band (proof of Proposition 1(i)), and since $\bar{H}^{\text{TANK}} < \mathsf{C} \leq \frac{1}{1-\lambda} < H_{\max}$ places $\bar{H}^{\text{TANK}}$ inside that band, $q > 0$ exactly to the left of the TE root, so $\bar{H}^{\text{TE}} > \bar{H}^{\text{TANK}}$. With $\bar{H}^{\text{TE}} > 1$, also $\bar{\mathcal{M}}^{\text{TE}} = \mathcal{M}^{\text{TE}}(\bar{H}^{\text{TE}}) > \mathcal{M}(\bar{\varepsilon}) = \bar{\mathcal{M}}^{\text{TANK}}$. Both factors of $\bar{\mathcal{M}}\bar{H}$ are

therefore strictly larger in TE-TANK, so

$$\bar{\mathcal{M}}^{\text{TE}}\bar{H}^{\text{TE}} > \bar{\mathcal{M}}^{\text{TANK}}\bar{H}^{\text{TANK}} \;\Rightarrow\; \Gamma^{\text{TE}} < \Gamma^{\text{TANK}} \;\Rightarrow\; \chi_y^{\text{TE}} < \chi_y^{\text{TANK}}.$$

□

This is the one comparison in the paper made at the mean-matched benchmark rather than at ε^*, and it is a markup-level result. At the baseline calibration, $1/(1-\chi_y)$ orders 1.000 (RANK) $<$ 1.903 (TE-TANK) $<$ 1.973 (mean-matched TANK, $\bar{\varepsilon} = 4.44$). Elasticity dispersion raises $\bar{\mathcal{M}}$ by more than it raises $\bar{H}$, so $\bar{\mathcal{M}}\bar{H}$ rises, Γ falls, and χ_y falls to 0.4746. A higher steady-state markup is a lower labor share: hand-to-mouth income is a smaller part of aggregate income, so a one percent output gap moves the heterogeneity index by less and the New Keynesian cross is weaker. Dispersion acts here only through the point around which the system is linearized. The markup-matched TANK(ε^*) shares TE-TANK's $\bar{\mathcal{M}}\bar{H}$ and hence its χ_y, so the two amplification factors coincide: the proposition has no dispersion-pure content.

D.10.5 Cost-Push Pass-Through (Proposition 4)

Cyclicality and flattening. Appendix D.4 derives the relation $\hat{h}_t = -\chi_y \tilde{y}_t$. Write $\Gamma = \Gamma_0\big(1+\delta(\tau-1)\big)$ with

$$\Gamma_0 = \frac{(1-\alpha)\lambda}{\bar{\mathcal{M}}\bar{H}(1-\lambda)} > 0$$

on the admissible region ($\bar{\mathcal{M}} > 1$, $\bar{H} > 0$), so $\operatorname{sign}(\Gamma) = \operatorname{sign}\big(1+\delta(\tau-1)\big)$. Since $\chi_y = \Gamma\big(\frac{\varphi+\alpha}{1-\alpha}+\sigma\big)/(1+\Gamma\sigma)$ with a strictly positive numerator factor and $1+\Gamma\sigma > 0$, $\operatorname{sign}(\chi_y) = \operatorname{sign}(\Gamma)$. At $\delta = 1$, $1+\delta(\tau-1) = \tau$.

Appendix D.8 derives the reduced form $\kappa = \kappa_y - \chi_y\kappa_h$, the decomposition of κ_h, and the loading δ_2. Signs: $\overline{MC} = 1/\bar{\mathcal{M}} < 1$ since $\bar{\mathcal{M}} > 1$, so $1 - \overline{MC} > 0$; $\varepsilon_P - \varepsilon_R \geq 0$ under the maintained assumptions; and the second piece $\delta_2\sigma > 0$ because $\delta_2 = \big[(1-\lambda)\bar{H}(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1\big]/\zeta = (\varepsilon_{\text{eff}}(\bar{H}) - 1)/\zeta > 0$. Therefore $\kappa_h > 0$. Since $\kappa_y - \kappa = \chi_y\kappa_h$ and $\kappa_h > 0$, $\kappa < \kappa_y \iff \chi_y > 0 \iff R < 1$.

Benchmark. κ_y is the slope obtained by holding $(\bar{H}, \bar{\mathcal{M}}, \delta_2)$ at their TE-TANK values and switching off the $-\chi_y\kappa_h$ feedback, equivalently by imposing $\hat{h}_t \equiv 0$ in the structural NKPC (12). This is the same-steady-state, feedback-off comparison. It is not the slope of the homogeneous-elasticity economy, which additionally resets δ_2, κ_y, and κ_h to their $\bar{\varepsilon}$ values.

Proof of Proposition 4. With u_t the only disturbance, $z_t = a_t = v_t = 0$ implies $r_t^n = 0$ and $\hat{y}_t = \tilde{y}_t$, so the equilibrium reduces to (14), (11), and (7) with forcing process $u_t = \rho_u u_{t-1} + \varepsilon_t^u$. Guess $\pi_t = \Psi_\pi u_t$ and $\tilde{y}_t = \Psi_y u_t$, so $\mathbb{E}_t[\pi_{t+1}] = \rho_u\Psi_\pi u_t$ and

$\mathbb{E}_t[\tilde{y}_{t+1}] = \rho_u \Psi_y u_t$. Matching coefficients in (14) gives $(1 - \beta\rho_u)\Psi_\pi = \kappa\Psi_y + 1$; substituting the Taylor rule into (11) gives $\Psi_y\left[\tilde{\sigma}(1-\rho_u) + \phi_y\right] = -(\phi_\pi - \rho_u)\Psi_\pi$, that is, $\Psi_y = -\Theta\Psi_\pi$ with $\Theta = (\phi_\pi - \rho_u)/\left[\tilde{\sigma}(1-\rho_u) + \phi_y\right] > 0$. Combining the two conditions yields $\Psi_\pi = \left[(1-\beta\rho_u) + \kappa\Theta\right]^{-1} \equiv 1/D_\pi > 0$. The standard determinacy condition is $\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0$ (Bullard and Mitra, 2002), which is necessary and sufficient when $\phi_\pi, \phi_y \geq 0$ and $\tilde{\sigma} > 0$ (that is, $\chi_y < 1$). Under $\phi_\pi > 1$, $\phi_y \geq 0$, $\beta < 1$, and $\kappa > 0$ it holds, so this is the unique bounded solution. Given $\tilde{\sigma} > 0$, the condition involves $\tilde{\sigma}$ nowhere else, so within the admissible region $\chi_y < 1$ the demand-composition feedback cannot threaten determinacy through χ_y: it holds at every point of the fiscal branch $R \in [0,1]$ at the baseline calibration, as does the maintained restriction $0 \leq \chi_y < 1$, whose maximum along the branch is 0.584. Differentiating,

$$\frac{\partial \Psi_\pi}{\partial \kappa} = -\Theta\,\Psi_\pi^2 < 0, \qquad \frac{\partial \Psi_\pi}{\partial \tilde{\sigma}} = \frac{(1-\rho_u)\,\kappa\,(\phi_\pi - \rho_u)}{\left[\tilde{\sigma}(1-\rho_u) + \phi_y\right]^2}\,\Psi_\pi^2 > 0.$$

For the ratio, $\Psi_\pi^{\mathrm{TE}}/\Psi_\pi^{\mathrm{TANK}} = D_\pi^{\mathrm{TANK}}/D_\pi^{\mathrm{TE}} > 1$ if and only if $\kappa^{\mathrm{TE}}\Theta^{\mathrm{TE}} < \kappa^{\mathrm{TANK}}\Theta^{\mathrm{TANK}}$. The sufficient conditions stated after the proposition follow because Θ is increasing in χ_y through $\tilde{\sigma} = \sigma(1-\chi_y)$. Finally, $\Psi_y = -\Theta\Psi_\pi < 0$, and dividing the TE and TANK output responses gives the stated contraction ratio. □

Quantitative magnitudes. At the baseline calibration, the feedback removes 19.5 percent of κ_y. Moving from κ_y to κ at fixed $\tilde{\sigma}$ raises cost-push pass-through by 16.6 percent. Table C.6 reports the pass-through and output ratios across four homogeneous-elasticity benchmarks.

D.11 Comparative Statics

The markup-matched comparison isolates the elasticity-gap channel. Table D.1 shows that RANK, TANK, and TE-TANK share the steady-state markup at $\varepsilon^* = 3.93$. Moving from RANK to TANK raises χ_y from zero to 0.4746 and lowers κ from 0.0833 to 0.0701. Moving from TANK to TE-TANK changes only κ_h within the first-order policy block and lowers the slope further to 0.0671.

TABLE D.1: Structural Objects Across RANK, TANK, and TE-TANK

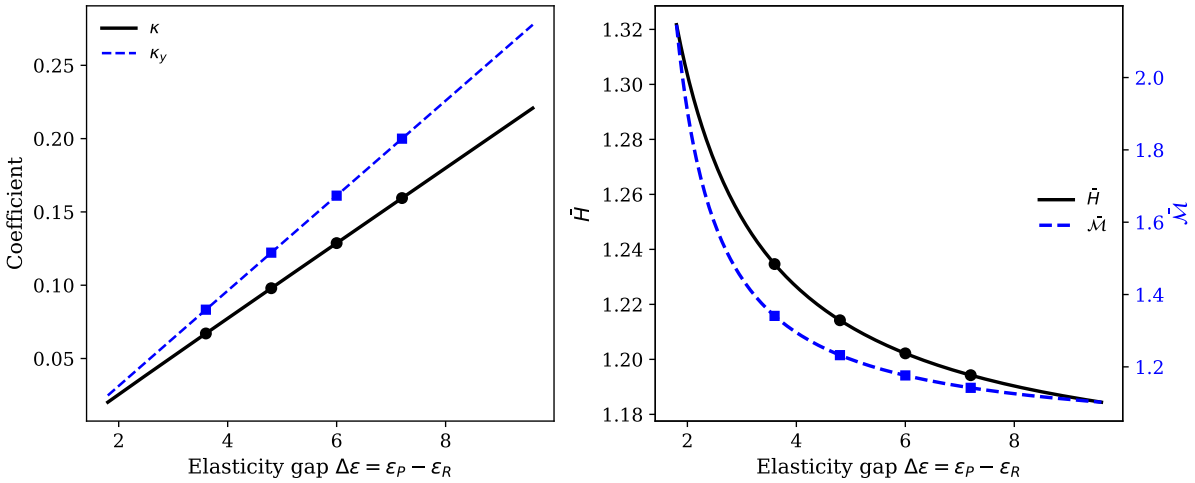
	RANK	TANK	TE-TANK
ε_R	3.93	3.93	3.00
ε_P	3.93	3.93	6.60
λ	0.00	0.40	0.40
τ	0.00	0.70	0.70
$\bar{H}$	1.0000	1.2347	1.2347
$\bar{\mathcal{M}}$	1.3409	1.3409	1.3409
Γ	0.0000	0.1879	0.1879
χ_y	0.0000	0.4746	0.4746
δ_2	0.0278	0.0278	0.0278
κ_y	0.0833	0.0833	0.0833
κ_h	0.0278	0.0278	0.0342
$\kappa = \kappa_y - \chi_y \kappa_h$	0.0833	0.0701	0.0671
$1/(1-\chi_y)$	1.0000	1.9032	1.9032

Notes: Structural steady-state and NKPC objects across three calibrations. $\kappa_h = \frac{(1-\lambda)\bar{H}}{\zeta}(1 - \overline{MC})(\varepsilon_P - \varepsilon_R) + \delta_2\sigma$ and $\chi_y = \Gamma G/(1+\Gamma\sigma)$ with $G = \frac{\varphi+\alpha}{1-\alpha} + \sigma$. Common parameters: $\sigma = \varphi = 1$, $\alpha = 1/3$, $\zeta = 105.63$, $\delta = 1$. The RANK and TANK columns are evaluated at the markup-matched elasticity $\varepsilon^* = 3.93$, so RANK(ε^*) and TANK(ε^*) have the same steady-state markup as TE-TANK, $\bar{\mathcal{M}} = 1.3409$ (Table C.6): all three columns share $\bar{\mathcal{M}}$, δ_2, κ_y; TANK and TE-TANK also share $\bar{H}$, Γ, χ_y, differing only in κ_h (hence κ).

Figure D.1 shows that both κ_y and κ rise along the fixed-ratio path, which varies the elasticity level while holding $\varepsilon_P/\varepsilon_R = 2.2$, and that the feedback remains a drag of about 20 percent on κ_y. The horizontal axis is the elasticity gap. The left panel plots the reduced-form slope and the feedback-off slope, and the right panel plots the heterogeneity index and the markup on separate axes. The elasticity gap and level rise together, so the sweep does not isolate either dimension. The heterogeneity index changes little, and the movement comes mainly from the pricing coefficients rather than from a large change in the consumption distribution.

Steady-state inequality and cost-push amplification. Higher steady-state inequality amplifies the markup response to a cost-push shock. The mechanism is compositional. With more steady-state inequality, a larger share of spending comes from rich households, whose demand is less price-elastic, so the demand curve firms face is less elastic and the desired markup is more sensitive to any further shift in the composition of demand. A cost-push shock produces exactly such a shift: it opens a negative output

FIGURE D.1: Comparative Statics along the Fixed-Ratio Elasticity Path



Notes: Phillips-curve slopes and steady-state objects as functions of $\Delta\varepsilon = \varepsilon_P - \varepsilon_R$, holding $\varepsilon_P/\varepsilon_R = 2.2$ fixed and all other parameters at baseline. Left: reduced-form slope κ (solid black) and feedback-off slope κ_y (dashed blue). Their distance is $\chi_y\kappa_h$. Right: heterogeneity index $\bar{H}$ (solid black, left axis) and markup $\bar{\mathcal{M}}$ (dashed blue, right axis). Markers trace $\Delta\varepsilon = 3.6$, 4.8, 6.0, and 7.2 along the same path. The leftmost marker is the baseline $(\varepsilon_R, \varepsilon_P) = (3, 6.6)$.

gap, inequality rises, spending tilts further toward the rich, and desired markups rise. Corollary 5 makes this comparative static exact: after the same shock, a more unequal economy shows a larger markup response. The sign, however, is not specific to the elasticity gap: homogeneous-elasticity TANK delivers it too, as the benchmark below establishes. What the elasticity gap contributes here is magnitude. Section 4.2 tests this prediction with cross-country local projections.

The comparative static runs along the admissible fiscal branch: the family of steady states traced out as the rebate parameter $R = \delta(1-\tau)$ moves over $[0,1]$, with all non-fiscal parameters held at baseline. At $R = 1$ profits are fully rebated as equal transfers and steady-state inequality vanishes; at $R = 0$ poor households receive no profit income and inequality is largest. The branch can be indexed by steady-state inequality itself, the heterogeneity index $\bar{H}$ (rich households' consumption relative to average consumption, equal to one under full equality), rather than by R. The *Benchmark nesting* display above, obtained by subtracting one from (8), gives $\bar{H} - 1 = \frac{\lambda(1-R)}{1-\lambda}\left(1 - \frac{1-\alpha}{\bar{\mathcal{M}}}\right)$, whose second factor is strictly positive because $\bar{\mathcal{M}} > 1 > 1-\alpha$ on the branch (Proposition 1(i)). Inverting is therefore legitimate:

$$1 - R = \frac{(1-\lambda)(\bar{H}-1)}{\lambda\left[1 - (1-\alpha)/\bar{\mathcal{M}}(\bar{H})\right]}. \tag{D.32}$$

Since $\mathrm{d}\bar{H}/\mathrm{d}R < 0$ (Proposition 1(ii)), $R \mapsto \bar{H}$ is a strictly decreasing bijection from $[0,1]$ onto $[1, \bar{H}(R{=}0)]$, with $\bar{H}(R{=}0) = 1.342$ at the baseline elasticities, and every object

below is a function of $\bar{H}$ alone along the branch. "Increasing in $\bar{H}$" therefore means increasing as redistribution weakens and steady-state inequality rises.

The corollary tracks, along the branch, the composite objects that Sections 2.5 and 2.6 define. In words: Γ is the strength of the demand-composition feedback, zero in the RANK limit; χ_y is the loading of inequality on the output gap in the distributional relation $\hat{h}_t = -\chi_y \tilde{y}_t$, so it measures how much inequality rises when a shock opens a negative gap; η_μ is the elasticity of the desired markup with respect to the heterogeneity index, that is, how strongly desired markups react to a shift in demand composition; δ_2 is the marginal-cost loading in the Phillips curve; κ is the reduced-form Phillips-curve slope; Θ converts inflation into the output contraction generated by the Taylor rule and the IS curve, through $\Psi_y = -\Theta\Psi_\pi$; and Ψ_π is cost-push pass-through, the impact response of inflation to a unit cost-push shock. Throughout, write $G = \frac{\varphi+\alpha}{1-\alpha} + \sigma$ for the coefficient linking real marginal cost to the output gap and $b = G - \sigma = \frac{\varphi+\alpha}{1-\alpha}$ for its labor-market part.

Two conditions on primitives organize the comparative statics:

$$(\varepsilon_R - 1)\big(1 - \alpha + \alpha\varepsilon_R\big) \;\geq\; \frac{\lambda}{1-\lambda}\big(\varepsilon_P - \varepsilon_R\big), \tag{D.33}$$

$$(1-\rho_u)\,\frac{\varphi+\alpha}{1-\alpha} \;\geq\; \phi_y. \tag{D.34}$$

Condition (D.33) keeps the rich elasticity far enough above one relative to the elasticity gap. Its left side rises with ε_R and vanishes as $\varepsilon_R \to 1$, the point where the desired markup diverges; its right side is the elasticity gap weighted by the relative population share of the poor. The condition rules out configurations in which the effective elasticity firms face comes close to one on the admissible branch; the remark after the proof makes this boundary explicit. Condition (D.34) caps the Taylor rule's output response ϕ_y at $(1-\rho_u)b$, the marginal-cost coefficient scaled by one minus the persistence of the cost-push shock; it rules out rules that respond so strongly to output that the demand-side term in the proof of part (iii) would work against the amplification. Both hold strictly at the baseline calibration, with room to spare: (D.33) reads $3.333 \geq 2.400$ and (D.34) reads $0.200 \geq 0.125$.

Corollary 5 (Steady-State Inequality and Cost-Push Amplification)**.** *Index the admissible fiscal branch by $\bar{H} \in \big[1, \bar{H}(R{=}0)\big]$ through (D.32), and maintain the cost-push conditions of Proposition 4 together with strict elasticity dispersion, $\varepsilon_P > \varepsilon_R$. Then:*

(i) Under (D.33), Γ, χ_y, η_μ, and Θ are strictly increasing in $\bar{H}$, while δ_2 and κ are strictly decreasing.

(ii) Under (D.33) alone, the impact output contraction $|\Psi_y| = \Theta\Psi_\pi$ is strictly increasing in $\bar{H}$.

(iii) Under (D.33) *and* (D.34)*, the impact markup response to the cost-push wedge,*

$$\frac{\partial \hat{\mu}_t}{\partial u_t} = \big[G - \sigma\chi_y\big]\Theta\Psi_\pi = \frac{G}{1+\Gamma\sigma}\,\Theta\Psi_\pi,$$

is strictly increasing in $\bar{H}$. Higher steady-state inequality therefore raises the markup response to a cost-push shock.

(iv) Numerical, at the baseline calibration only; not implied by (i)–(iii). *Inflation pass-through Ψ_π is strictly increasing in $\bar{H}$ along the branch, rising from* 2.739 *at $R = 1$ to* 3.212 *at $R = 0$, while κ falls from* 0.0977 *to* 0.0574 *and the markup response of (iii) rises by 27.5 percent.*

Proof of Corollary 5. Throughout, write $e \equiv e(\bar{H}) = \mathsf{A}\bar{H} + \mathsf{B} = \varepsilon_{\text{eff}}(\bar{H})$ for the effective demand elasticity firms face, the expenditure-weighted average of the two types' elasticities. Since $\mathsf{A} = -|\mathsf{A}| < 0$ with $|\mathsf{A}| = (1-\lambda)(\varepsilon_P - \varepsilon_R)$, e is strictly decreasing in $\bar{H}$: as inequality rises, spending shifts toward the low-elasticity rich and the average elasticity falls. Proposition 1(i) gives $1 \le \bar{H} < \mathsf{C} \le \frac{1}{1-\lambda}$ and $e > 1$, equivalently $\bar{\mathcal{M}} = e/(e-1) > 1$, everywhere on the branch.

Step 1: the feedback strength Γ is strictly increasing in $\bar{H}$; this is where (D.33) *does its work.* Substituting (D.32) into $\Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\bar{\mathcal{M}}\bar{H}}$ and using $\bar{\mathcal{M}} - (1-\alpha) = \frac{\alpha e + 1 - \alpha}{e-1}$ expresses Γ as a function of $\bar{H}$ alone,

$$\Gamma(\bar{H}) = \frac{(1-\alpha)(\bar{H}-1)}{\bar{H}\big[\bar{\mathcal{M}} - (1-\alpha)\big]} = \frac{(1-\alpha)(\bar{H}-1)(e-1)}{\bar{H}\big[\alpha e + 1 - \alpha\big]}.$$

Two forces compete: the factor $\bar{H} - 1$ rises with inequality, while the factor $e - 1$ falls with it. Log-differentiating and collecting the two e terms separates them,

$$\frac{\mathrm{d}\log\Gamma}{\mathrm{d}\bar{H}} = \frac{1}{\bar{H}(\bar{H}-1)} - \frac{|\mathsf{A}|}{(e-1)(\alpha e + 1 - \alpha)},$$

$$\text{so} \qquad \Gamma' > 0 \iff (e-1)(\alpha e + 1 - \alpha) > |\mathsf{A}|\,\bar{H}(\bar{H}-1).$$

Because e is strictly decreasing in $\bar{H}$, the left side of this inequality is strictly decreasing and the right side strictly increasing in $\bar{H}$; it therefore suffices to check it weakly at the right endpoint of the branch's closure. That endpoint is $\bar{H} = \frac{1}{1-\lambda}$, where $e\big(\frac{1}{1-\lambda}\big) = \frac{\mathsf{A}}{1-\lambda} + \mathsf{B} = \varepsilon_R$ exactly and $|\mathsf{A}|\frac{1}{1-\lambda}\big(\frac{1}{1-\lambda} - 1\big) = \frac{\lambda}{1-\lambda}(\varepsilon_P - \varepsilon_R)$, so the inequality evaluated there is precisely (D.33). Since $\bar{H} < \frac{1}{1-\lambda}$ strictly on the branch, the weak condition (D.33) delivers the strict inequality at every admissible $\bar{H}$, hence $\Gamma' > 0$. At the left endpoint $\Gamma(1) = 0$ and $\Gamma'(1) = \frac{(1-\alpha)(e(1)-1)}{\alpha e(1) + 1 - \alpha} > 0$.

Step 2: the remaining objects of part (i) follow by inspection once e falls and Γ rises with $\bar{H}$. By Appendix D.8, $\delta_2 = (e-1)/\zeta$, so $\delta_2 > 0$ and $\delta_2' < 0$. By (13), $\eta_\mu = |\mathsf{A}|\bar{H}/\big[e(e-1)\big]$

has a strictly increasing positive numerator and a strictly decreasing positive denominator, so $\eta_\mu > 0$ is strictly increasing: desired markups become more sensitive to demand composition. Since $\mathrm{d}\chi_y/\mathrm{d}\Gamma = G/(1+\Gamma\sigma)^2 > 0$, Step 1 makes χ_y strictly increasing; hence $\tilde{\sigma} = \sigma(1-\chi_y)$ is strictly decreasing and $\Theta = (\phi_\pi - \rho_u)/\big[\tilde{\sigma}(1-\rho_u) + \phi_y\big]$ strictly increasing: the demand side contracts more per unit of inflation. Finally, by (15), $\kappa = \delta_2 G \frac{1-\Gamma\eta_\mu}{1+\Gamma\sigma}$. Proposition 2(ii) already establishes $0 < 1-\Gamma\eta_\mu \leq 1$ analytically, in particular the strict positivity $1-\Gamma\eta_\mu = 1 + \mathrm{DA}/e^2 > 0$ of (D.30), which follows from the strict downcrossing of q. So κ is G times a product of three strictly positive factors, δ_2, $1-\Gamma\eta_\mu$, and $1/(1+\Gamma\sigma)$, of which the first is strictly decreasing and the other two are decreasing; therefore $\kappa' < 0$: higher steady-state inequality flattens the Phillips curve.

Step 3: part (ii), the output contraction deepens. By Proposition 4, $|\Psi_y| = \Theta\Psi_\pi = \Theta/\big[(1-\beta\rho_u) + \kappa\Theta\big] = \big[(1-\beta\rho_u)/\Theta + \kappa\big]^{-1}$. By part (i), Θ is strictly increasing and κ strictly decreasing, so both terms in the bracket fall, the bracket is strictly decreasing, and $|\Psi_y|$ is strictly increasing. Only (D.33) is used.

Step 4: part (iii), the markup response rises: it equals a fixed positive numerator divided by a denominator $N(\bar{H})$, and N falls. Combining (D.26) with the distributional relation (10) gives $\hat{\mu}_t = -\sigma\hat{h}_t - G\tilde{y}_t = -\big[G - \sigma\chi_y\big]\tilde{y}_t$, and Proposition 4 gives $\tilde{y}_t = -\Theta\Psi_\pi u_t$, which yields the response displayed in the statement. The second equality in that display is the exact identity

$$G - \sigma\chi_y = G - \frac{\sigma\Gamma G}{1+\Gamma\sigma} = \frac{G\big[(1+\Gamma\sigma) - \Gamma\sigma\big]}{1+\Gamma\sigma} = \frac{G}{1+\Gamma\sigma},$$

so the $\Gamma\sigma$ terms cancel exactly, as they do in (15). Substituting Θ and Ψ_π and clearing $1+\Gamma\sigma$ from numerator and denominator, using $(1+\Gamma\sigma)\kappa = \delta_2 G(1-\Gamma\eta_\mu)$ and $(1+\Gamma\sigma)\sigma(1-\chi_y) = \sigma(1-\Gamma b)$, the denominator collects all the $\bar{H}$ dependence:

$$\frac{\partial\hat{\mu}_t}{\partial u_t} = \frac{G(\phi_\pi - \rho_u)}{N(\bar{H})},$$

$$\begin{aligned} N = {} & (1-\beta\rho_u)\big[\sigma(1-\rho_u)(1-\Gamma b) + \phi_y(1+\Gamma\sigma)\big] \\ & + (\phi_\pi - \rho_u)G\delta_2(1-\Gamma\eta_\mu) > 0, \end{aligned}$$

with $N > 0$ because $D_\pi > 0$, $\tilde{\sigma}(1-\rho_u) + \phi_y > 0$, and $1+\Gamma\sigma > 0$, and with $\phi_\pi - \rho_u > 0$ under the maintained restrictions $\phi_\pi > 1 > \rho_u$. It remains to show $N' < 0$. Differentiating in $\bar{H}$, and noting that b, σ, ϕ_y, ϕ_π, ρ_u, and β are fixed along the branch,

$$N' = (1-\beta\rho_u)\,\sigma\,\Gamma'\big[\phi_y - (1-\rho_u)b\big] + (\phi_\pi - \rho_u)G\big[\delta_2'(1-\Gamma\eta_\mu) - \delta_2(\Gamma\eta_\mu)'\big].$$

The second bracket is the supply side, and under (D.33) it is strictly negative: $\delta_2' < 0$

and $1-\Gamma\eta_\mu > 0$ by Step 2, while $\Gamma \geq 0$ and $\eta_\mu > 0$ are both increasing with at least one strictly so, hence $(\Gamma\eta_\mu)' > 0$ and $\delta_2 > 0$. The first term is the demand side, and (D.34) is exactly what signs it: it gives $\phi_y - (1-\rho_u)b \leq 0$, and $\Gamma' > 0$ by Step 1, so the term is non-positive. Therefore $N' < 0$ and $G(\phi_\pi - \rho_u)/N$ is strictly increasing in $\bar{H}$.

Step 5: part (iv) is a numerical statement, not a theorem. By Proposition 4, pass-through is $\Psi_\pi = \left[(1-\beta\rho_u) + \kappa\Theta\right]^{-1}$, so it depends on $\bar{H}$ only through the product $\kappa\Theta$, and along the branch the two factors move in opposite directions. Part (i) gives $\mathrm{d}\kappa/\mathrm{d}\bar{H} < 0$: the Phillips curve flattens, each unit of output contraction buys less disinflation, and this force alone would raise Ψ_π through $\partial\Psi_\pi/\partial\kappa < 0$. But part (i) also gives $\mathrm{d}\tilde{\sigma}/\mathrm{d}\bar{H} < 0$, so Θ rises: the same policy rule generates a larger output contraction per unit of inflation, and this force alone would lower Ψ_π through $\partial\Psi_\pi/\partial\tilde{\sigma} > 0$. Conditions (D.33)–(D.34) sign the factors but not the product $\kappa\Theta$, so neither force dominates on paper; in particular, $\mathrm{d}\kappa/\mathrm{d}\bar{H} < 0$ does *not* by itself imply that pass-through rises. Part (iv) is accordingly verified numerically rather than proved: at the baseline calibration we evaluate the branch on a 2001-point grid in $R \in [0,1]$ and find Ψ_π strictly increasing in $\bar{H}$ at every grid step, with the endpoint values reported in the statement. Parts (i)–(iii) are unaffected, since none of them uses the monotonicity of Ψ_π. □

Parts (i)–(iii) are comparative statics across steady states on the fiscal branch, conditional on (D.33)–(D.34); they are not a global ordering over all parameterizations. The inequality in Step 1 genuinely fails as $e(\bar{H}) \downarrow 1$, where the desired markup $e/(e-1)$ diverges, and (D.33) is exactly the requirement that the admissible branch stay clear of that boundary. The magnitudes in part (iv) are calibration-specific.

Homogeneous-elasticity benchmark. Corollary 5 is stated for TE-TANK, but most of it survives the removal of the elasticity gap. The exception is pass-through. Setting $\varepsilon_R = \varepsilon_P = \varepsilon$ gives $\mathsf{A} = 0$, hence $e \equiv \varepsilon$, $\eta_\mu = 0$, and constant $\bar{\mathcal{M}} = \varepsilon/(\varepsilon-1)$ and $\delta_2 = (\varepsilon-1)/\zeta$ along the entire branch; only Γ still moves with $\bar{H}$. Three claims follow.

(a) The objects of part (i) keep their signs, and (D.33) is not needed. With $\bar{\mathcal{M}}$ constant, Step 1's expression reduces to $\Gamma(\bar{H}) = (1-\alpha)(\bar{H}-1)/\left[\bar{H}(\bar{\mathcal{M}} - (1-\alpha))\right]$, and since $\mathrm{d}\left[(\bar{H}-1)/\bar{H}\right]/\mathrm{d}\bar{H} = 1/\bar{H}^2 > 0$ with $\bar{\mathcal{M}} > 1 > 1-\alpha$, we get $\Gamma' > 0$ unconditionally. Condition (D.33) is used only to sign Γ' against the countervailing movement of e, which is now absent. Step 2 then gives $\chi_y' > 0$ and $\Theta' > 0$ as before, while $\kappa = \delta_2 G/(1+\Gamma\sigma)$ with δ_2 constant gives $\kappa' < 0$, and Step 3 applies verbatim, so $|\Psi_y|$ is strictly increasing.

(b) The markup comparative static is not specific to the elasticity gap. The response $\partial\hat{\mu}_t/\partial u_t = \left[G/(1+\Gamma\sigma)\right]\Theta\Psi_\pi$ of part (iii) has a prefactor $G/(1+\Gamma\sigma)$ free of η_μ; the elasticity gap enters through Γ and, inside Ψ_π, through δ_2 and $1-\Gamma\eta_\mu$ in κ. The slope route is the main reason the markup response rises by 27.5 percent along the branch

in TE-TANK against 2.7 percent in *markup-matched* TANK (Table 3). The sign of the response, however, does not depend on the gap. Concretely, in Step 4's expression for N' the substitutions $\delta_2' = 0$ and $\eta_\mu = 0$ annihilate the entire supply-side bracket, leaving

$$N' = (1 - \beta\rho_u)\,\sigma\,\Gamma'\big[\phi_y - (1 - \rho_u)b\big] \leq 0 \tag{D.35}$$

under (D.34), so $G(\phi_\pi - \rho_u)/N$ is again increasing: higher steady-state inequality raises the markup response to a cost-push shock in homogeneous-elasticity TANK too. The elasticity gap changes the magnitude, not the sign.

(c) Pass-through is where the two economies part. Because $\delta_2(1 - \Gamma\eta_\mu)$ is now constant, clearing $1 + \Gamma\sigma$ as in Step 4 gives the product $\kappa\Theta$ in closed form,

$$\kappa\Theta = \frac{\delta_2 G(\phi_\pi - \rho_u)}{\sigma(1 - \rho_u)(1 - \Gamma b) + \phi_y(1 + \Gamma\sigma)},$$

whose denominator has derivative $\sigma\Gamma'\big[\phi_y - (1 - \rho_u)b\big]$ in $\bar{H}$. Since $\Psi_\pi = \big[(1 - \beta\rho_u) + \kappa\Theta\big]^{-1}$,

$$\operatorname{sign}\left(\frac{\mathrm{d}\Psi_\pi}{\mathrm{d}\bar{H}}\right) = \operatorname{sign}\big[\phi_y - (1 - \rho_u)b\big], \tag{D.36}$$

so whenever (D.34) holds strictly, homogeneous-elasticity pass-through is strictly *decreasing* in steady-state inequality. This is the opposite of the TE-TANK sign in part (iv). The contrast is exact here only because the supply-side factor is constant; in TE-TANK that factor moves, which is precisely why Step 5 cannot resolve the sign on paper and part (iv) is numerical. Accordingly, the opposite signs are a property of the calibrated branch under the calibrated policy rule, not a theorem: the same bracket $\phi_y - (1 - \rho_u)b$ that (D.34) signs also governs (D.36), so a rule with ϕ_y above the threshold $(1 - \rho_u)b = 0.200$ would reverse the homogeneous sign, and at $\phi_y = 0$ both economies deliver a decreasing Ψ_π. At the calibrated $\phi_y = 0.125$ the threshold holds with room to spare. Holding the population mean at $\bar{\varepsilon} = 4.44$, the net change in TE-TANK pass-through between the endpoints $R = 1$ and $R = 0$ changes sign at an elasticity gap of $\varepsilon_P - \varepsilon_R = 1.19$ and is positive above it. This threshold concerns the endpoint change only. Above it, pass-through need not be monotone along the branch, because a positive endpoint difference does not imply a positive derivative at every R. At the baseline gap of 3.6, Step 5 shows that pass-through rises monotonically as redistribution falls on the numerical grid, so the baseline is not a knife-edge.

The pass-through sign in TE-TANK. In TE-TANK the supply-side factor moves along the branch, so the product $\kappa\Theta$ that governs Ψ_π has two moving parts. A lower rebate raises Γ and χ_y, so $\tilde{\sigma}$ falls and Θ rises: the flattening pushes Ψ_π up and the steeper

demand response pushes it down. Clearing $1 + \Gamma\sigma$ as in part (c), without holding the supply side fixed, gives

$$\kappa\Theta = \frac{\delta_2 G(1 - \Gamma\eta_\mu)(\phi_\pi - \rho_u)}{\sigma(1 - \rho_u)(1 - \Gamma b) + \phi_y(1 + \Gamma\sigma)}.$$

The Ricardian wage channel cancels from this product, and the demand side acts only through the denominator, as in part (c). What is new is the numerator. As the rebate falls, spending tilts toward the price-insensitive rich, and the effective elasticity firms face falls from 4.44 at $R = 1$ to 3.70 at $R = 0$. Both supply-side factors decline with it, the marginal-cost loading $\delta_2 = (\varepsilon_{\text{eff}}(\bar{H}) - 1)/\zeta$ and the desired-markup factor $1 - \Gamma\eta_\mu$, and neither has a counterpart in the denominator. In the additive split of the change in $\log(\kappa\Theta)$ along the branch, δ_2 contributes -0.242, the desired-markup factor -0.073, and the demand-side residual $+0.084$. The supply side is thus 3.7 times the residual in absolute value, so $\kappa\Theta$ falls and pass-through rises. Both supply-side terms come from one primitive, the endogenous effective elasticity, which moves if and only if $\varepsilon_P \neq \varepsilon_R$.

Table 3 in Section 3.4 reports the endpoint comparison for all three economies. Section 4.2 and Appendix B.5 take both comparative statics to the data.

D.12 Technology and Monetary Policy Shock Transmission

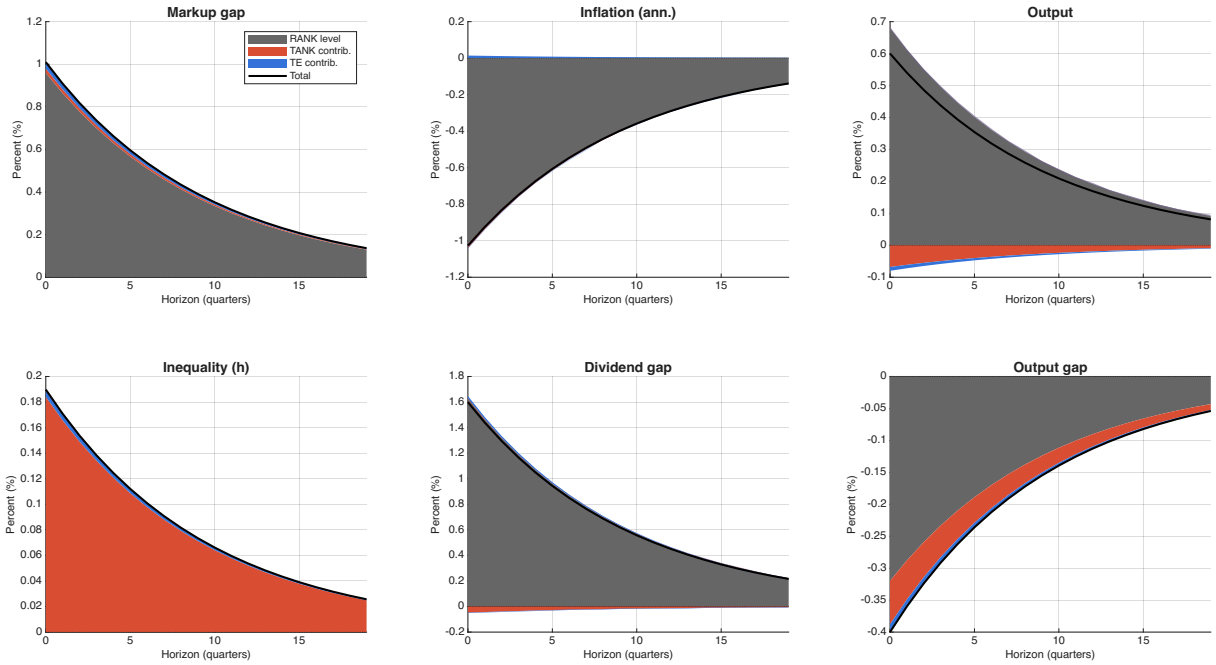
Figures D.2 and D.3 apply the band decomposition of Section 3.3 to a positive technology shock and to an expansionary monetary policy shock.[13] In Figure D.2, the gray and red bands reproduce the familiar New Keynesian and TANK shapes, with hand-to-mouth aggregation muting the output gain. The blue band adds inflation: inequality rises, firms raise desired markups, and because the shock is deflationary, the elasticity gap shrinks the fall in inflation. After the monetary policy shock, TE-TANK produces dynamics nearly identical to the markup-matched TANK($\varepsilon^* = 3.93$), so the blue elasticity-gap band is thin; both economies deviate substantially from the RANK corner, consistent with the literature documenting amplification from hand-to-mouth households.

The similarity between TE-TANK and TANK for demand shocks reflects a property of the transmission mechanism. Following the interest rate cut, rich households raise consumption. Because prices are sticky, this raises labor demand and the real wage for all households, including the poor, who consume their additional income. Firms accommodate the additional demand by hiring more labor.

As output rises, marginal cost rises under decreasing returns, which lowers markups through (D.26). The fall in markups dominates the offsetting rise in out-

[13] The preference shock case, not shown, yields no additional insights.

FIGURE D.2: Technology Shock: Channel Decomposition



Notes: Impulse responses to a one-standard-deviation positive technology shock. Section 3.3 defines the bands, homogeneous corners, and units. Panels as in Figure 2, plus output (top right) and the dividend gap (bottom middle). Inequality is identically zero in RANK, so the gray band is flat in the inequality panel.

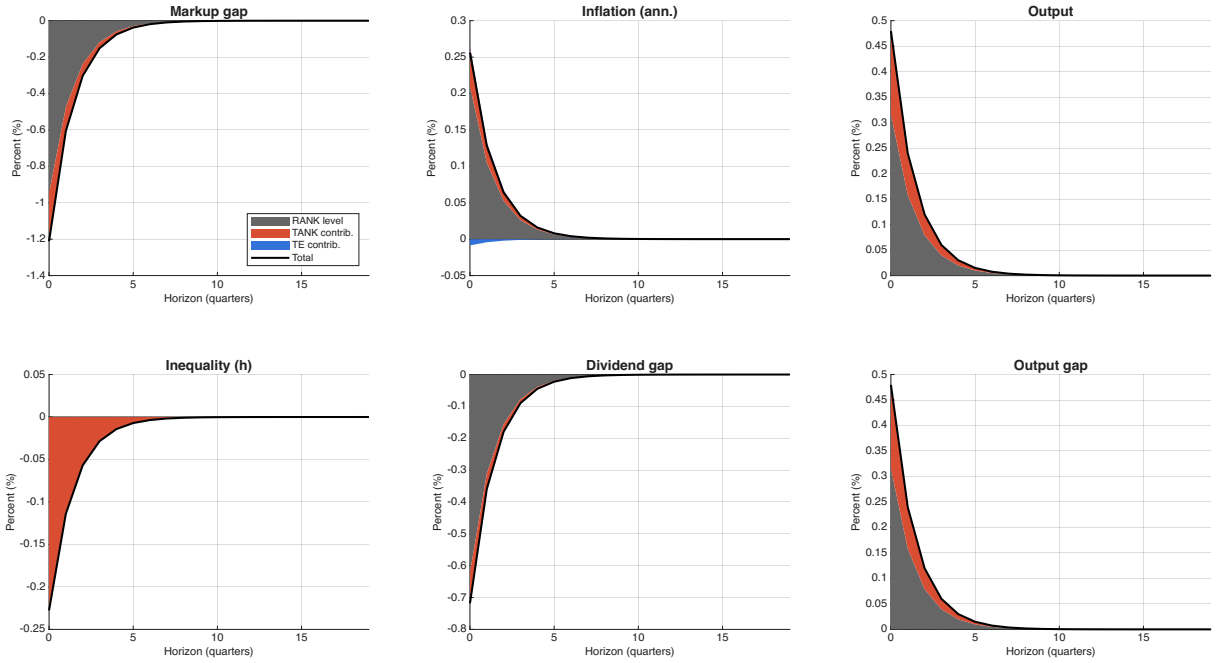
put, so dividends fall in (D.29) and transfers to both types decline. Poor households nevertheless gain, because their income is disproportionately labor income; firms must therefore produce still more, hiring more labor and pushing the real wage up further. This channel is absent in RANK, where the real effects of a demand shock are muted relative to TANK and TE-TANK.

The fall in the heterogeneity index leaves desired markups unchanged in TANK. In TE-TANK it lowers the desired markup, because the composition of demand shifts toward the more price-elastic poor households. Firms therefore raise prices by less in TE-TANK than in TANK, which damps inflation in TE-TANK relative to TANK. Total inflation still rises in both economies; the negative blue band in the inflation panel of Figure D.3 is the incremental contribution of the elasticity gap relative to TANK. The output gap rises by essentially the same amount in the two economies.

D.13 Fixed versus Consumption-Dependent Elasticities

The concern. The model fixes ε_R and ε_P within type. Auer et al. (2024), and the closest structural model (Mongey and Waugh, 2025), obtain the gap from *non-homothetic* preferences instead: a household's price elasticity depends on its own consumption, $\varepsilon_h = \varepsilon_h(C_h)$, and falls as the household consumes more, so elasticities move over the

FIGURE D.3: Monetary Policy Shock: Channel Decomposition



Notes: Impulse responses to a one-standard-deviation reduction in the nominal interest rate (expansionary shock). Section 3.3 defines the bands, homogeneous corners, and units. Panels as in Figure 2, plus output (top right) and the dividend gap (bottom middle). The red band amplifies output and inflation relative to RANK through the high marginal propensity to consume of hand-to-mouth households; inequality is zero in RANK by construction, so its response is carried entirely by the red and blue bands, and it falls because the expansion raises labor income while dividends fall. The blue band is negative in the inflation panel, the elasticity-gap damping discussed in the text.

business cycle. With such gradients, and against a markup-matched benchmark with the same average gradient, the desired-markup channel keeps its sign when the curvature index γ defined below is less than one and reverses when it exceeds one. The evidence does not identify γ. Across the assumed calibrations below, the pass-through gap ranges from 2.96 percent in the linear case to -0.60 percent when γ is fitted to the model's own consumption spread, against 3.11 percent with fixed elasticities.

Two forces. Let $\eta_h \equiv \mathrm{d}\varepsilon_h / \mathrm{d}\ln C_h < 0$ be the within-type gradient, in elasticity points per log point of own consumption, so that $\varepsilon_{h,t} = \bar{\varepsilon}_h + \eta_h \hat{c}_t^h$ to first order. The effective elasticity $\varepsilon_{\text{eff},t} = \sum_h s_h H_t^h \varepsilon_{h,t}$ then moves for two reasons,

$$\mathrm{d}\varepsilon_{\text{eff},t} = \underbrace{(\varepsilon_R - \varepsilon_P)(1-\lambda)\bar{H}\,\hat{h}_t}_{\text{composition}} + \underbrace{\omega_R \eta_R\,\hat{c}_t^R + \omega_P \eta_P\,\hat{c}_t^P}_{\text{within-type}}, \quad \omega_R = (1-\lambda)\bar{H}, \quad \omega_P = \lambda \bar{H}^P,$$

where ω_R and ω_P are the steady-state expenditure shares of the two types, $\bar{H}^P = [1 - (1-\lambda)\bar{H}]/\lambda$, and $\omega_R + \omega_P = 1$. The composition term is the paper's mechanism: spending shifts between the types while each ε_h stays fixed. The within-type term is what fixed elasticities switch off. Substituting $\hat{c}_t^R = \hat{y}_t + \hat{h}_t$ and $\hat{c}_t^P = \hat{y}_t - (\omega_R/\omega_P)\hat{h}_t$, which follow from $C_t^R = H_t C_t$, $\lambda H_t^P + (1-\lambda) H_t^R = 1$, and $C_t = Y_t$, gives

$$\mathrm{d}\varepsilon_{\text{eff},t} = (1-\lambda)\bar{H}\big[(\varepsilon_R - \varepsilon_P) + (\eta_R - \eta_P)\big]\hat{h}_t + \bar{\eta}\,\hat{y}_t, \qquad \bar{\eta} \equiv \omega_R \eta_R + \omega_P \eta_P.$$

The average gradient $\bar{\eta}$ multiplies aggregate output. Only the difference $\eta_R - \eta_P$ reaches the inequality term.

The average gradient. Linearizing the firm first-order condition (D.13) in ε_h adds $-[(1 - \overline{MC})/\zeta]\,\mathrm{d}\varepsilon_{\text{eff},t}$ to the Phillips curve, the route through which the composition term already enters. The $\bar{\eta}\hat{y}_t$ part adds $\Delta\kappa_y \equiv -(1 - \overline{MC})\bar{\eta}/\zeta > 0$ to κ_y and leaves κ_h unchanged: when output is high, every household is less price-sensitive, and the Phillips curve is steeper.

The difference in gradients. Define the curvature index $\gamma \equiv (\eta_R - \eta_P)/(\varepsilon_P - \varepsilon_R)$, the within-type gradient difference in units of the between-type elasticity gap. It is zero when the two types share a gradient and positive when the poor's elasticity falls more steeply with consumption than the rich's. The bracket in the last display equals $(\varepsilon_R - \varepsilon_P)(1-\gamma)$. When inequality rises, spending tilts toward the less price-sensitive rich, but the poor, now consuming less, become more price-sensitive, and for $\gamma > 0$ this pushes the average elasticity back up. The desired-markup route $\delta_2 \eta_\mu$ of κ_h therefore scales by $1 - \gamma$, while the wage route $\delta_2 \sigma$ is unchanged. With the subscript η marking

slopes under within-type gradients,

$$\kappa_\eta^{\text{TANK}^*} = \kappa^{\text{TANK}^*} + \Delta\kappa_y, \quad \kappa_\eta^{\text{TE}} = \kappa^{\text{TANK}^*}\left[1 - \Gamma\eta_\mu(1-\gamma)\right] + \Delta\kappa_y, \quad \kappa^{\text{TANK}^*} \equiv \frac{\kappa_y}{1+\Gamma\sigma},$$

$$\kappa_\eta^{\text{TE}} - \kappa_\eta^{\text{TANK}^*} = -\kappa^{\text{TANK}^*}\Gamma\eta_\mu(1-\gamma) = -\chi_y\delta_2\eta_\mu(1-\gamma),$$

$$\frac{\kappa_\eta^{\text{TANK}^*} - \kappa_\eta^{\text{TE}}}{\kappa_\eta^{\text{TANK}^*}} = \frac{\kappa^{\text{TANK}^*}}{\kappa^{\text{TANK}^*} + \Delta\kappa_y}\Gamma\eta_\mu(1-\gamma).$$

We assume that the comparison economy TANK(ε^*) shares the gradient $\bar{\eta}$ at $\varepsilon^* = \varepsilon_{\text{eff}}(\bar{H})$, so that only dispersion differs; a power law with a common γ and a linear form with a common slope both satisfy this. The common increment $\Delta\kappa_y$ then enters both slopes one-for-one and cancels from the level difference. It does not cancel from the ratio, because it also raises the denominator. A benchmark that instead keeps ε^* fixed changes the demand system itself; against it, the common increment outweighs the composition effect even at the income-based curvature below, so TE-TANK would be slightly steeper.

The level difference is proportional to $-(1-\gamma)$ and the ratio factor is positive. The channel therefore keeps its sign for $\gamma < 1$, vanishes at $\gamma = 1$, and reverses for $\gamma > 1$, whatever the size of $\Delta\kappa_y$. Pass-through falls with κ, and the gradients leave χ_y unchanged, so the markup-matched pass-through gap has the sign of the slope gap.

Calibrating γ. No study identifies γ separately from the cross-sectional gap. Auer et al. (2024) estimate how price sensitivity differs across households at a point in time, and Mongey and Waugh (2025) note that these estimates "cannot be used to determine the level of the elasticity of demand, only how it varies across individuals." Their own model makes a household's elasticity a function of its own resources. They discipline its curvature by indirect inference: they simulate the Auer et al. (2024) experiment inside the model and match its cross-sectional coefficient.[14] Non-homothetic demand systems make the price elasticity a function of the expenditure level by construction, from AIDS and QUAIDS (Deaton and Muellbauer, 1980; Banks et al., 1997) to non-homothetic CES across sectors (Comin et al., 2021). None of these studies estimates the within-household gradient the model needs. The gradient the calibration imports is cross-sectional, and carrying it inside the household is an assumption. We therefore assume functional forms, calibrate them to the same evidence, and report the implied γ; these values are assumptions, not bounds. In all cases the steady state is unchanged, since $\bar{\varepsilon}_R$ and $\bar{\varepsilon}_P$ stay at their baseline values.

[14] Mongey and Waugh (2025) calibrate a curvature parameter above one, but it is the curvature of the elasticity in variety-level quantity, not in a household's total consumption, and it is identified from the same cross-sectional coefficient, so it does not pin down γ.

(i) Linear, $\gamma = 0$. If ε is linear in log consumption, with the cross-sectional slope $-(\varepsilon_P - \varepsilon_R)/\ln 6 \approx -2.0$ points per log point, the two types share a gradient. The level difference is unchanged at $-\chi_y \delta_2 \eta_\mu$. The common increment still lowers the percentage gaps, the slope gap from 4.34 to 4.06 percent and the pass-through gap from 3.11 to 2.96 percent.

(ii)–(iv) Power law. A constant proportional gradient, $\varepsilon = AC^{-\gamma}$, gives $\gamma = \ln(\varepsilon_P/\varepsilon_R)/\ln S$, where $\varepsilon_P/\varepsilon_R = 2.2$ and S is the rich-to-poor spread over which that ratio is taken. A wider spread implies a smaller γ.

(ii) Income spread, $\gamma = 0.44$. The elasticity ratio is estimated across a 6-fold range of permanent income. The desired-markup route and the level difference keep $1 - \gamma = 56$ percent of their fixed-elasticity size. The slope gap falls to 2.30 percent and the pass-through gap to 1.65 percent, 53 percent of the fixed-elasticity values.

(iii) CEX consumption spread, $\gamma = 0.73$. The held-out CEX rich-to-poor consumption ratio of 2.93 measures the spread in consumption, the units in which γ is defined. The desired-markup route and the level difference keep $1 - \gamma = 27$ percent. The slope gap falls to 1.06 percent and the pass-through gap to 0.76 percent, 24 percent of the fixed-elasticity values.

(iv) The model's own spread, $\gamma = 1.22$. The steady-state ratio $\bar{C}^R/\bar{C}^P = \bar{H}^R/\bar{H}^P = 1.905$ gives $\gamma > 1$, and the channel reverses: the slope gap is -0.83 percent and the pass-through gap -0.60 percent. That ratio is neither a calibration target nor a data moment. It follows once λ and $R \equiv \delta(1-\tau)$ are set externally. Appendix C.2 discusses this choice. Any spread above 2.2 gives $\gamma < 1$, and both the CEX ratio and the widest spread the model can reach, 2.759 at $R = 0$, exceed it.

Within the power-law family, a single $\varepsilon(\cdot)$ with $\gamma < 1$ implies $\varepsilon_P/\varepsilon_R < 1.905$ at the model's types (1.33 at $\gamma = 0.44$) and cannot reproduce the calibrated ratio of 2.2, which would need $\gamma = 1.22$. Cases (ii) and (iii) therefore imply type-specific intercepts, $\varepsilon_h = A_h C_h^{-\gamma}$ with A_P/A_R between 1.37 and 1.66. The curvature index is still γ, so the scaling above is unchanged.

Natural allocation. Under flexible prices, with $\hat{h}_t^n = \Gamma \hat{\mu}_t^n$,

$$\hat{\mu}_t^n = b' \hat{y}_t^n, \quad b' \equiv \frac{-\bar{\eta}}{\varepsilon^*(\varepsilon^* - 1)\left[1 - \Gamma \eta_\mu (1-\gamma)\right]}, \quad \hat{y}_t^n = \frac{\Psi_a a_t}{G + b'(1 + \Gamma\sigma)}, \quad \Psi_a \equiv \frac{1+\varphi}{1-\alpha},$$
$$\pi_t - \beta \mathbb{E}_t \pi_{t+1} = \kappa_\eta^{\mathrm{TE}}(\hat{y}_t - \hat{y}_t^n) = \kappa_\eta^{\mathrm{TE}}(\hat{y}_t - \hat{y}_t^{\mathcal{M}}) + \Delta\kappa_y \hat{y}_t^{\mathcal{M}}, \qquad \hat{y}_t^{\mathcal{M}} \equiv \Psi_a a_t / G.$$

Fixed elasticities imply a constant flexible-price markup, so natural output equals constant-markup output $\hat{y}_t^{\mathcal{M}}$. With gradients, over cases (ii)–(iii), a one percent technology shock raises $\hat{\mu}_t^n$ by 0.14–0.23 percent and leaves $\hat{y}_t^n$ 5.7–9.1 percent below $\hat{y}_t^{\mathcal{M}}$. The natural-rate curve has no residual, so divine coincidence is unaffected; the wedge

$\Delta\kappa_y \hat{y}_t^{\mathcal{M}}$ appears only against constant-markup output, and only technology shocks move it.